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Questions

Q1.

Many predatory animals produce toxins to disable their prey. Some of these toxins inhibit nervous communication.

(a) Nerve impulses are transmitted along axons. Some axons are myelinated.

(i) Which type of cell produces myelin?

(1)

- ☐ **A** neurone
- ☐ **B** rod cell
- ☐ **C** Schwann cell
- ☐ **D** lymphocyte

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(ii) Which row correctly compares the conduction of an impulse along a myelinated neurone with conduction along a non-myelinated neurone?

(1)

	Compared with a non-myelinated neurone the speed of conduction in a myelinated neurone is	Because membrane depolarisation only takes place in the axon membrane
☐ **A**	faster	at nodes of Ranvier
☐ **B**	faster	in between nodes of Ranvier
☐ **C**	slower	at nodes of Ranvier
☐ **D**	slower	in between nodes of Ranvier

(b) Describe the role of ion transport in maintaining the resting potential of a neurone.

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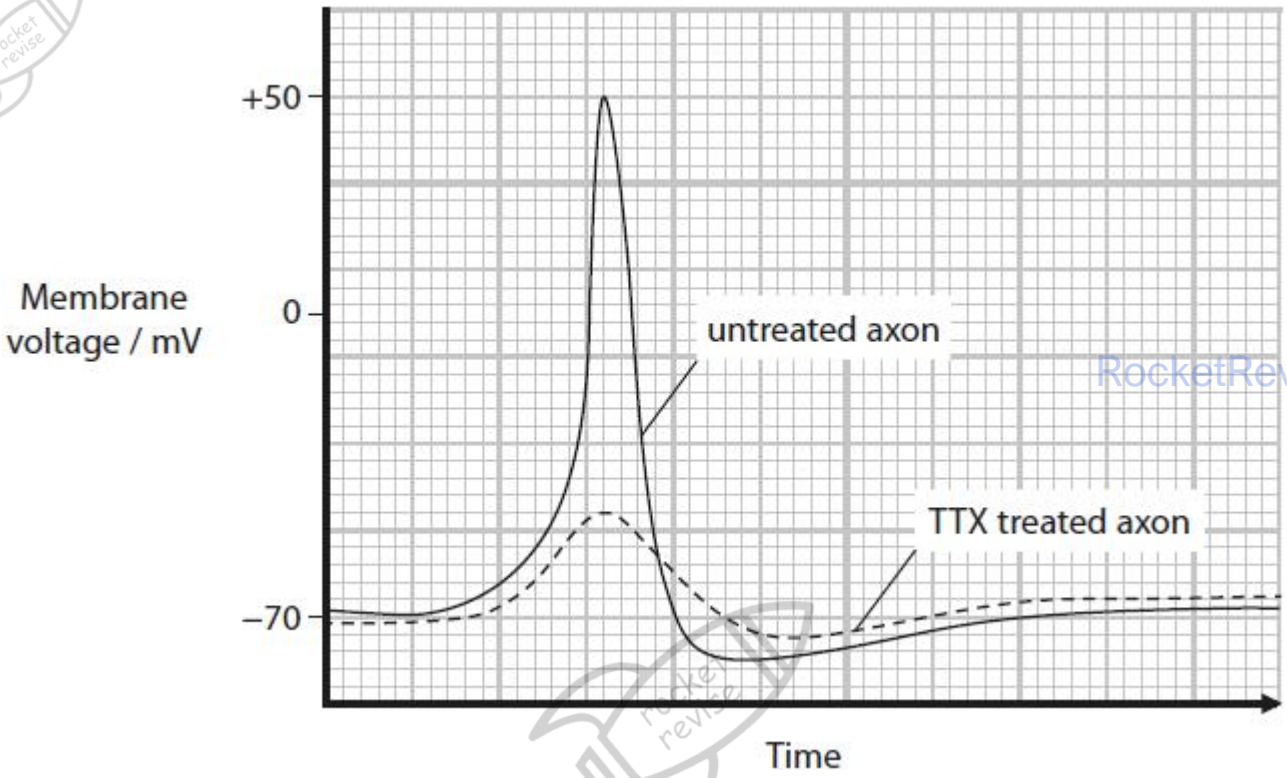
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(c) The pufferfish produces a neurotoxin called tetrodotoxin (TTX).

The graph shows the effect of TTX on the nerve impulse in an axon.



Deduce how TTX inhibits nerve impulses in an axon.

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(Total for question = 9 marks)

Q2.

The scientific document you have studied is adapted from an article in *National Geographic*: How personalized medicine is transforming your health care.

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Use the information from the scientific document and your own knowledge to answer the following questions.

(a) Explain what is meant by the phrase 'gene variants' (paragraph 4).

(2)

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(b) Explain why 'tumors ... riddled with different mutations' are good candidates for immunotherapies (paragraph 6).

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(c) Explain why a CT scan was used to show that Judy Perkins was free of tumours (paragraph 10 and Figure 1).

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(d) Checkpoints are a type of protein that prevent the activation of immune cells.

Explain why a checkpoint inhibitor was given with the lymphocytes in the immunotherapy used to treat Judy Perkins (paragraph 10).

(2)

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(e) Explain how 'a gene variant that produces a defective form of an enzyme' reduces the effectiveness of clopidogrel
(paragraph 16).

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(f) Researchers have used stem cells to produce spinal cord tissue from individuals with amyotrophic lateral sclerosis.

These functioning tissues contain motor neurones and blood vessels.
Suggest why blood vessels are required to form functioning spinal cord tissue (paragraph 31).

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(g) Suggest how 'a brew of growth factors and other proteins' can stimulate induced pluripotent stem cells to produce a functioning tissue (paragraph 33).

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(Total for question = 20 marks)

Q3.

Ecstasy (MDMA) is a recreational drug that can have serious toxic effects.

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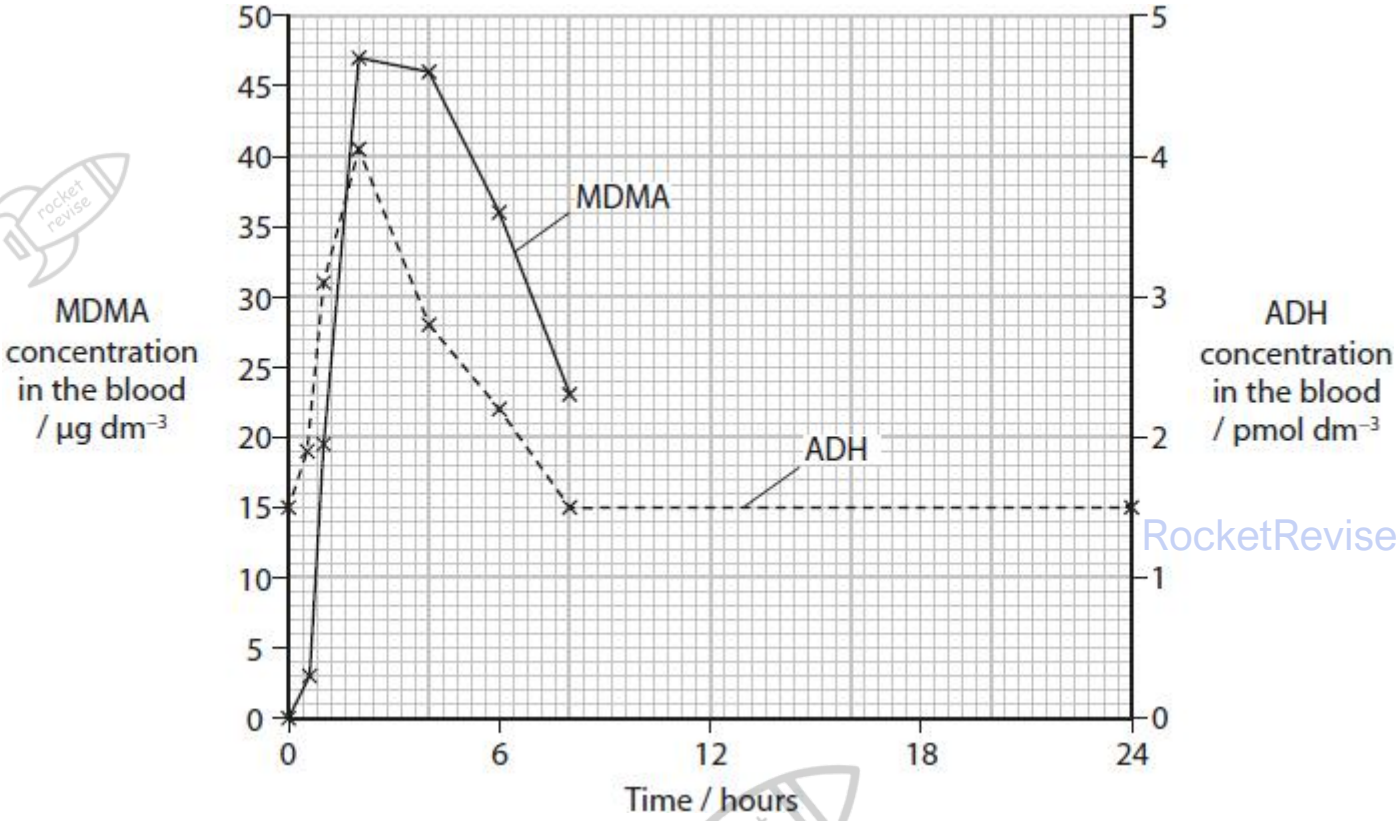
(a) The effect of MDMA on ADH concentration in blood has been studied.

In one study, each person in a group of volunteers was given MDMA. The MDMA and ADH concentrations in the blood were measured.

The graph shows the results of this study.

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(i) Estimate how long it takes the MDMA concentration in the blood to become 0.0 μg dm^{-3}.

(1)

Answer ..... hours

(ii) Calculate the rate of removal of ADH from the blood between 4 and 8 hours.

(2)

Answer .....

*(b) Taking MDMA causes a person to be very thirsty. This can lead to swelling of the brain, which can be fatal.

The table shows some laboratory results for one case in which a person suffered swelling of the brain.

Time since taking MDMA / hours	ADH concentration / pmol dm^{-3}	Sodium ion concentration in the blood / mmol dm^{-3}	Appearance of the brain in computed tomography (CT)
9	4.5	112	Swollen
96	1.2	131	Normal

Comment on the role of ADH in MDMA-induced brain swelling.
Use the information in the graph and table as well as your own knowledge to support your answer.

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(Total for question = 9 marks)

Q4.

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The eye is a sense organ specialised for detecting light stimuli.

(a) Pupils of the eye react to light.

The effects of different light intensities from two light sources have been investigated.
The table shows the results of this investigation.

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Light intensity / a.u. per m^2	Mean pupil diameter / mm	
	Incandescent light	LED light
0	25.5	25.5
75	24.0	22.0
150	24.0	18.5
200	23.0	17.5
300	21.5	16.0
650	17.5	12.0
1400	16.0	9.5

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(i) Describe the relationship between light intensity and pupil diameter.

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(ii) Calculate the amount of incandescent light that can pass through the pupil when the light intensity is 1400 a.u. per m^2.

Use a value of pi (π) = 3.14

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Answer .....

(iii) Describe how light entering the eye causes the pupil to respond.

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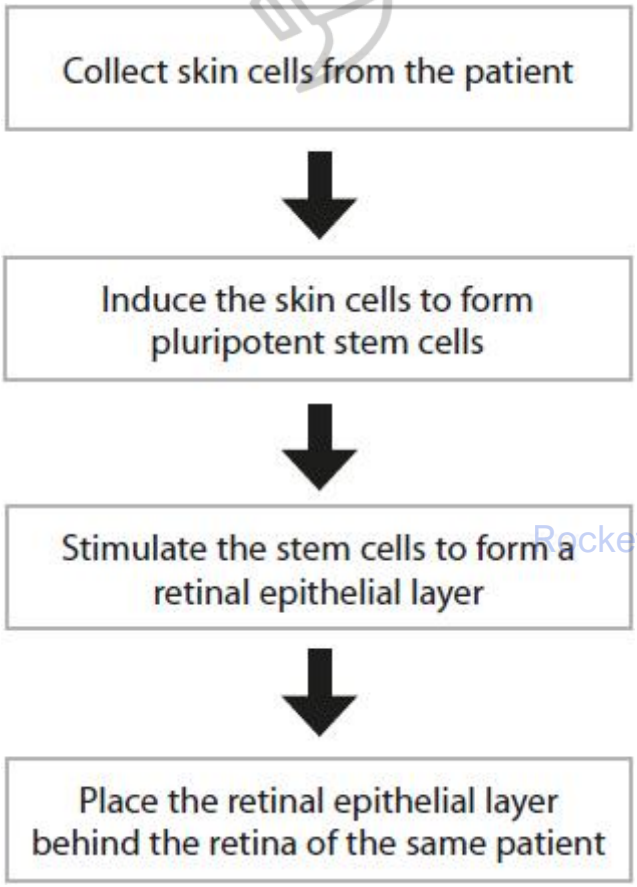
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(b) Age-related macular degeneration (AMD) is a common cause of blindness in older people.

In AMD the epithelial layer behind the retina is damaged.
A new treatment for AMD uses induced pluripotent stem cells. These cells are grown in a culture to produce a retinal
epithelial layer.
This retinal epithelial layer can then be used to replace the damaged layer of cells.
The procedure is outlined in the flow chart.



(i) State what is meant by the term **pluripotent stem cell**.

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(ii) The epithelial layer is checked to ensure that there are no stem cells present before it is placed in the eye of the patient.
Suggest why this check is carried out.

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(Total for question = 13 marks)

Q5.

Animals and plants can respond to light.

(a) The photograph shows the front of a human eye.



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(Source: © SCIENCE PHOTO LIBRARY)

(i) The size of the pupil changes when moving from dim to very bright light.
Explain how this change occurs.

(3)

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(ii) Describe the role of rhodopsin in causing changes in the polarisation of rod cells.

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*(b) Exposure to light controls the germination of seeds and the direction of growth of plants.

In an investigation the seeds of plantains were exposed to light.
The seeds were first exposed to red light and then to far-red light.
The percentage of seeds that germinated when exposed to different periods of red and far-red light was recorded.
The results are shown in the table.

Exposure to red light / min	Percentage germination (%)						
	Exposure to far-red light /seconds						
	1	5	10	15	30	60	960
32	100	20	14	6	4	6	10
16	92	30	40	28	30	33	20
8	100	62	50	57	44	47	56
4	100	67	60	57	40	37	92

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Comment on the effect of red and far-red light on the germination of these seeds and the processes that occur in their cells.
Use the information in the table to support your answer.

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(Total for question = 13 marks)

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Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

(a) The body length of the medicinal leech decreases when touched.

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(Source: © Mit Kapevski/Shutterstock)

The body length of the leech decreases less each time the leech is touched.

(i) What is the name of this effect?

(1)

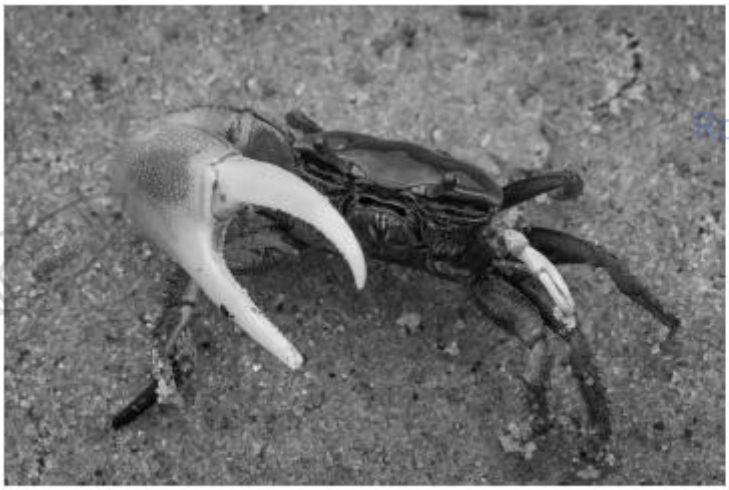
- ☒ **A** absorption
- ☒ **B** adaptation
- ☒ **C** habituation
- ☒ **D** sensitivity

(ii) Why is it an advantage for the leech to respond in this way?

(1)

- ☒ **A** it filters out what is important and what is not important to react to
- ☒ **B** it enables the animal to react more quickly to the stimulus
- ☒ **C** it keeps the habitat of the animal protected
- ☒ **D** it helps the animal find food more easily

(b) The photograph shows a fiddler crab.



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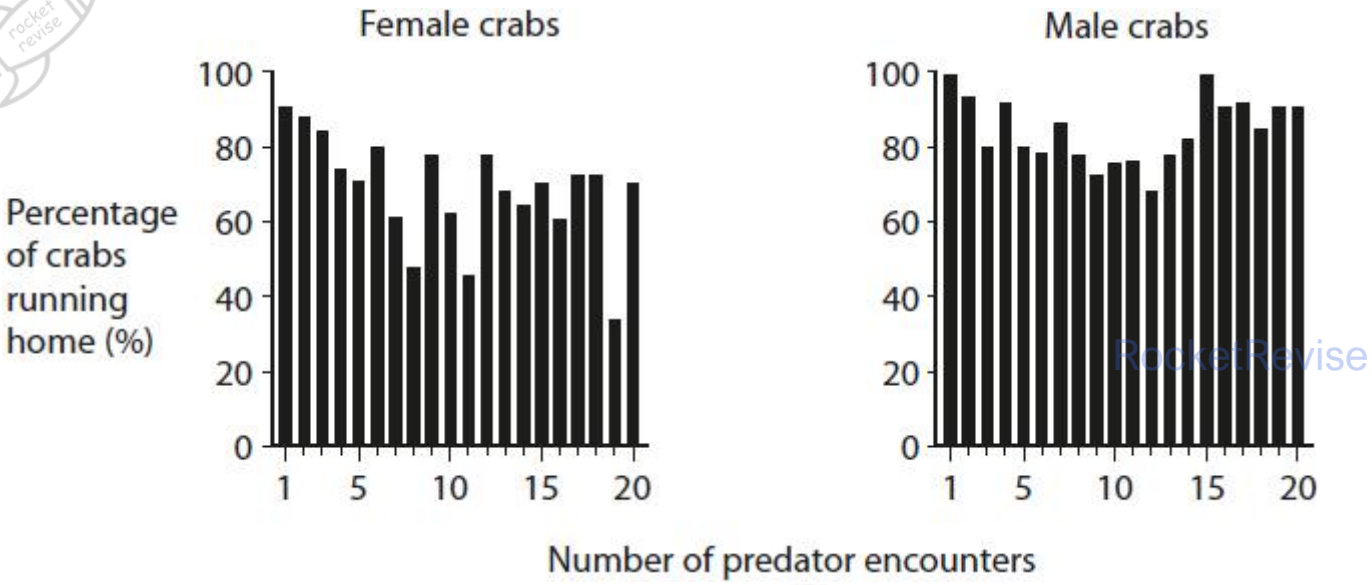
(Source: © Jay Gao/Shutterstock)

In an experiment, the fiddler crab was used to investigate responses to a predator. The responses by females and males were compared.

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A predator approached a group of female crabs from one direction.
The percentage of crabs running home was recorded. This was repeated 20 times.
This experiment was repeated with a group of male crabs.
The graphs show the percentage of crabs running home for both groups of crabs against the total number of times each group of crabs encountered a predator.



(i) Comment on the results of this experiment.

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(ii) Which statistical test would be used to analyse the effect of the increase in the number of predator encounters on the percentage of crabs running home?

(1)

- ☒ **A** correlation coefficient
- ☒ **B** Hardy-Weinberg
- ☒ **C** index of diversity
- ☒ **D** Student's t-test

(Total for question = 6 marks)

Q7.

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Many neurones in the human nervous system are myelinated.

(a) A myelinated axon conducts impulses faster than a non-myelinated axon of the same diameter.

Explain this difference.

(3)

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(b) Polyneuropathy is a disorder that damages the myelin sheath of neurones throughout the body.

One symptom is muscle weakness. Muscle weakness is the reduced strength in one or more muscles.
Suggest how damage to the myelin sheaths of neurones can lead to muscle weakness.

(2)

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(c) Dementia is a condition associated with the ongoing decline of brain functioning. There are many types of dementia.

The relationship between myelin in brain tissue and types of dementia has been investigated.
The mean quantity of myelin in samples of brain tissue from groups of people with types of dementia and from a control group was measured.
The table shows the results of this investigation.

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Group	Mean quantity of myelin in a brain tissue sample / a.u.	Standard deviation
Control (no dementia)	52	± 3.2
Patients with vascular dementia	25	± 5.9
Patients with Alzheimer's dementia	32	± 4.1
Patients with Lewy Body dementia	42	± 5.0

A student concluded that there was a relationship between the quantity of myelin in the brain of a person and whether or not they had dementia.
Comment on the validity of this conclusion.

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(Total for question = 7 marks)

Q8.

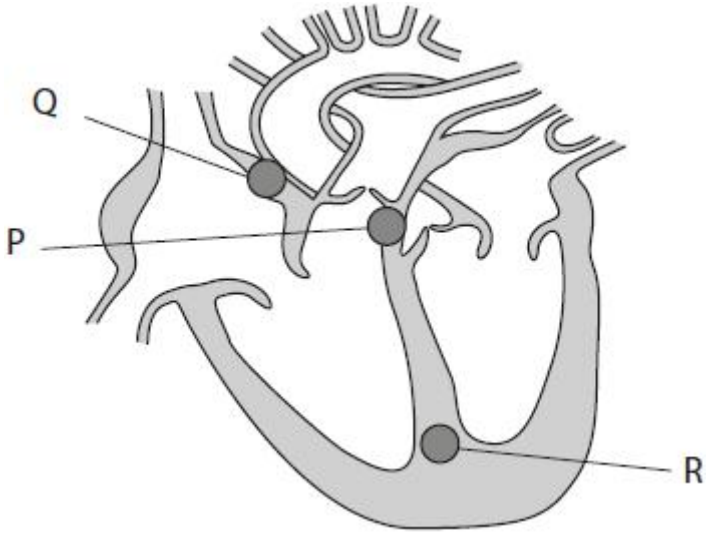
Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

In humans the sinoatrial node, atrioventricular node and bundle of His are involved in regulation and coordination of the cardiac cycle.

(a) The diagram shows a human heart.

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(i) Which **one** of the following statements is correct?

(1)

- ☒ **A** Q is the sinoatrial node (SAN) and R is the atrioventricular node (AVN)
- ☒ **B** Q is the sinoatrial node (SAN) and P is the atrioventricular node (AVN)
- ☒ **C** Q is the bundle of His and R is the sinoatrial node (SAN)
- ☒ **D** R is the sinoatrial node (SAN) and P is the Purkinje fibres

(ii) The electrical impulse that initiates the cardiac cycle begins without the need for a nerve impulse. What is the term for this process?

(1)

- ☒ **A** diastole
- ☒ **B** myogenic
- ☒ **C** polarisation
- ☒ **D** systole

(b) A football player undertook regular exercise as part of a training programme.

During exercise there is a change in the duration of the cardiac cycle.

(i) Explain why there is a change in the cardiac cycle during this exercise.

(3)

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(ii) The table shows the effect of the training programme on the cardiac output and resting heart rate of this football player.

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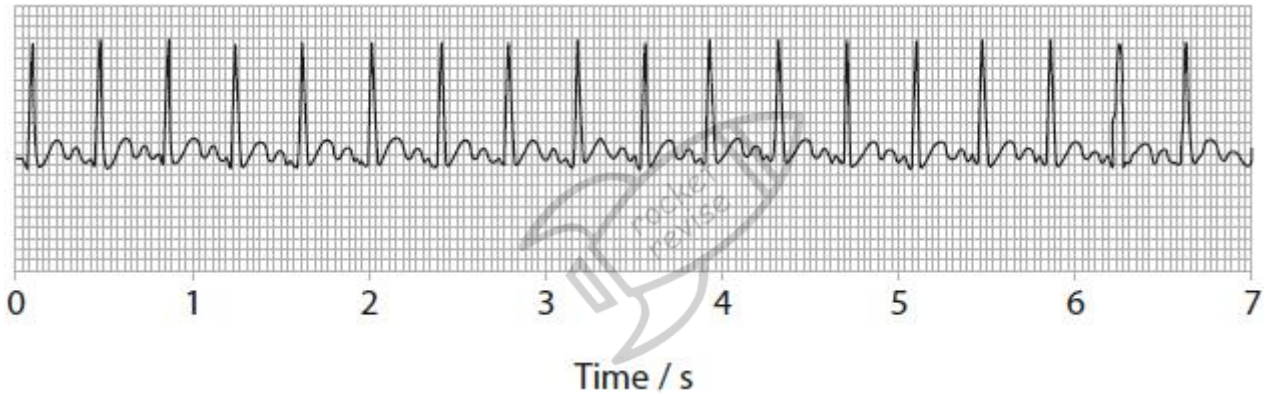
When measurements were made	Cardiac output / $cm^3 min^{-1}$	Resting heart rate / bpm
Before training	4500	72
After training	4500	51

Calculate the stroke volume of this football player, after training.

(1)

Answer ..... cm^3

(iii) After training, the footballer had an ECG.
This ECG trace is shown in the diagram.

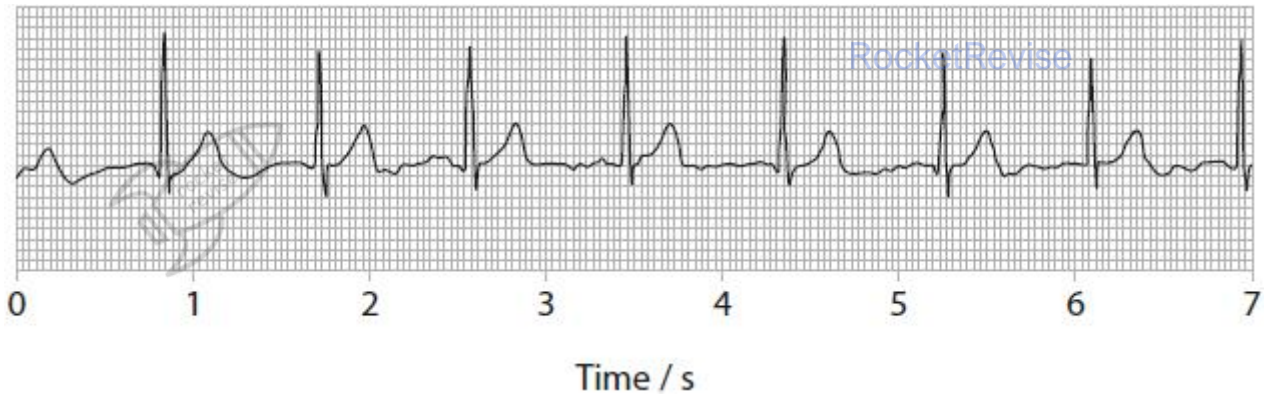


Calculate the heart rate of the footballer using the ECG in this diagram.

(1)

Answer .....

(iv) Two hours later a second ECG was recorded.



Comment on the changes that have occurred in the activity of the heart as shown in these two ECG traces.

(3)

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(Total for question = 10 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The mammalian retina is a tissue that detects light.

(a) Rod cells and bipolar neurones are cell types found in the retina.

(i) What is the name of the chemical formed when light is absorbed by the photopigment found in rod cells?

(1)

- ☐ **A** cis-retinal
- ☐ **B** phytochrome
- ☐ **C** rhodopsin
- ☐ **D** trans-retinal

(ii) Which row in the table shows the effects of light absorption on cells in the retina?

(1)

		Rod cell	Bipolar neurone
☐	**A**	hyperpolarised	hyperpolarised
☐	**B**	hyperpolarised	depolarised
☐	**C**	depolarised	hyperpolarised
☐	**D**	depolarised	depolarised

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(b) An investigation studied the effects of light intensity on the mammalian retina.

Samples of retina were exposed to different light intensities and the mean potential difference of the bipolar neurones was recorded.
The results are shown in the table.

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Light intensity / a.u.	Mean potential difference / mV
2	11
4	17
8	19
12	20
16	20
20	20

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(i) Describe the effect of light intensity on the mean potential difference of bipolar neurones.

(2)

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(ii) Calculate the percentage change in the mean potential difference when the light intensity increases from 4 to 12 a.u.

(2)

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Answer .....

(iii) Give **two** reasons why some people might have objections to the use of mammalian retinas in this investigation.

(2)

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(c) Which statement describes the response of the muscles in the iris to increasing light intensity?

(1)

- ☒ **A** circular and radial muscles contract
- ☒ **B** circular and radial muscles relax
- ☒ **C** circular muscles contract and radial muscles relax
- ☒ **D** circular muscles relax and radial muscles contract

(Total for question = 9 marks)

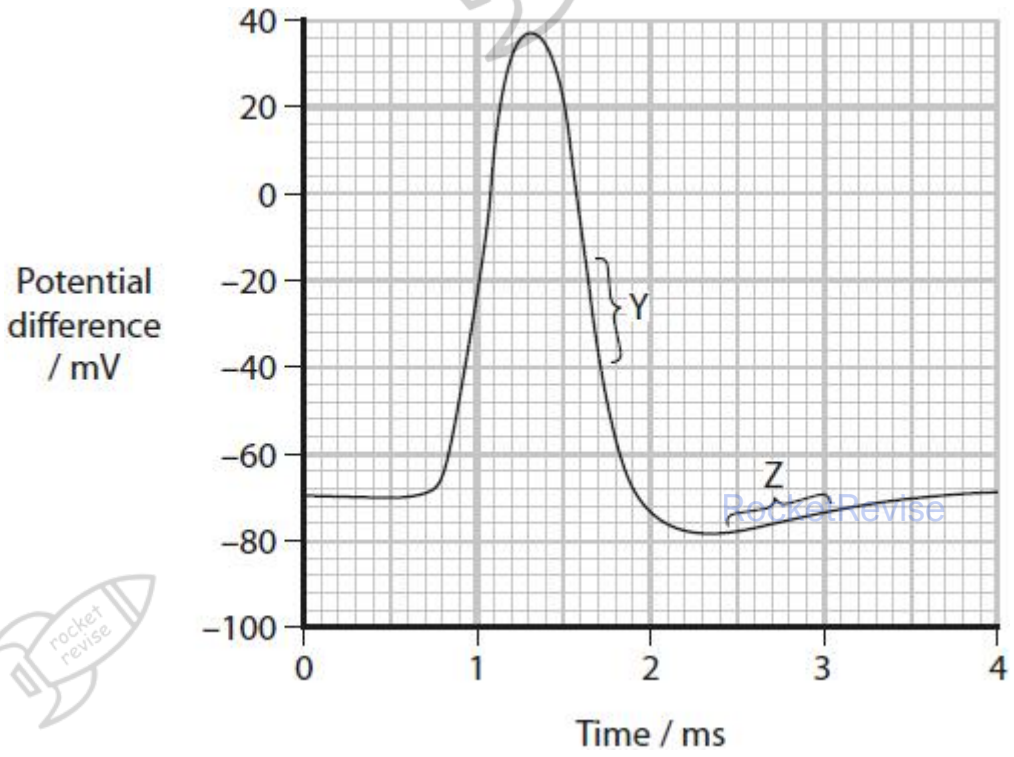
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Q10.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Nerve impulses travel along axons as action potentials.

(a) The graph shows an oscilloscope trace for an action potential in an axon.



(i) Which is the resting potential for this axon?

(1)

- ☒ **A** –70 mV
- ☒ **B** –78 mV
- ☒ **C** 38 mV
- ☒ **D** 108 mV

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(ii) Both voltage-gated sodium and voltage-gated potassium ion channels are involved in generating an action potential.
Which of these voltage-gated channels are open at Y?

(1)

- **A** no voltage-gated ion channels are open
- **B** voltage-gated sodium ion channels only
- **C** voltage-gated potassium ion channels only
- **D** both sodium and potassium voltage-gated ion channels

(iii) Which is the state of polarisation of the membrane at Z?

(1)

- **A** depolarised
- **B** hyperpolarised
- **C** hypopolarised
- **D** unpolarised

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(iv) How many of the following statements are correct?

- the magnitude of the action potential is proportional to the strength of stimulus that generates the action potential
- action potentials spread out in both directions along the axon
- following an action potential, there is a refractory period during which it is not possible to generate a new action potential

(1)

- **A** none
- **B** one
- **C** two
- **D** three

(b) Explain why maintaining a resting potential requires ATP.

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(Total for question = 7 marks)

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Q11.

Parkinson's disease is a nervous system disorder that affects movement.

(a) In people with Parkinson's disease, some neurones in the brain die.

Many of the symptoms of Parkinson's disease are due to a loss of neurones that produce a neurotransmitter called dopamine.

(i) Describe how a neurotransmitter transmits a nerve impulse across a synapse.

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(ii) One treatment for Parkinson's disease is a drug called L-DOPA.
Describe how L-DOPA reduces some of the symptoms of Parkinson's disease.

(3)

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(b) Scientists can induce pluripotent stem cells to produce transcription factors involved in the synthesis of dopamine.

(i) Explain how transcription factors are involved in the synthesis of dopamine.

(2)

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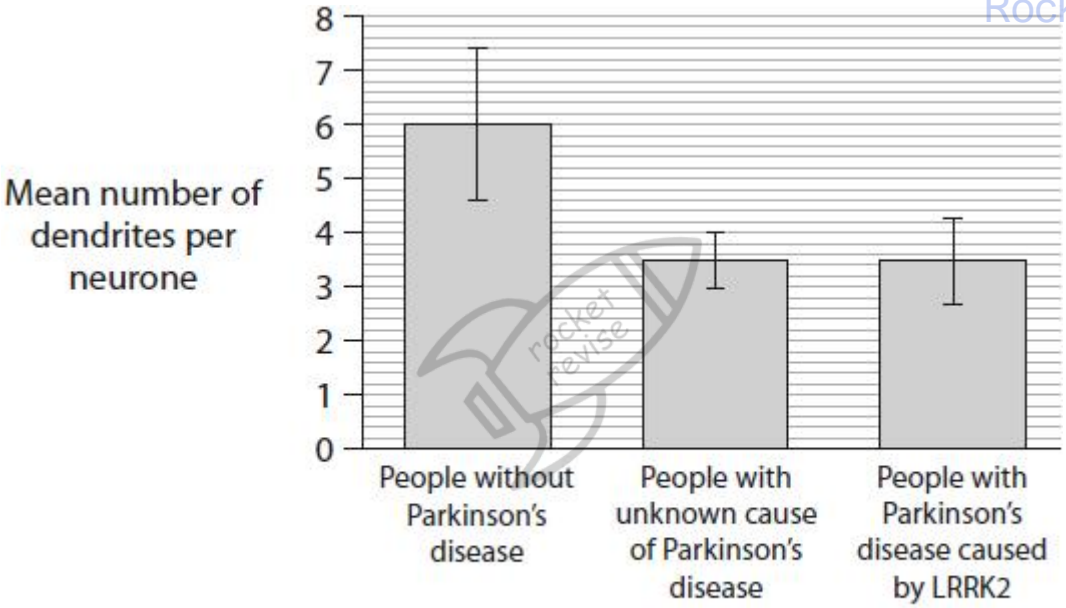
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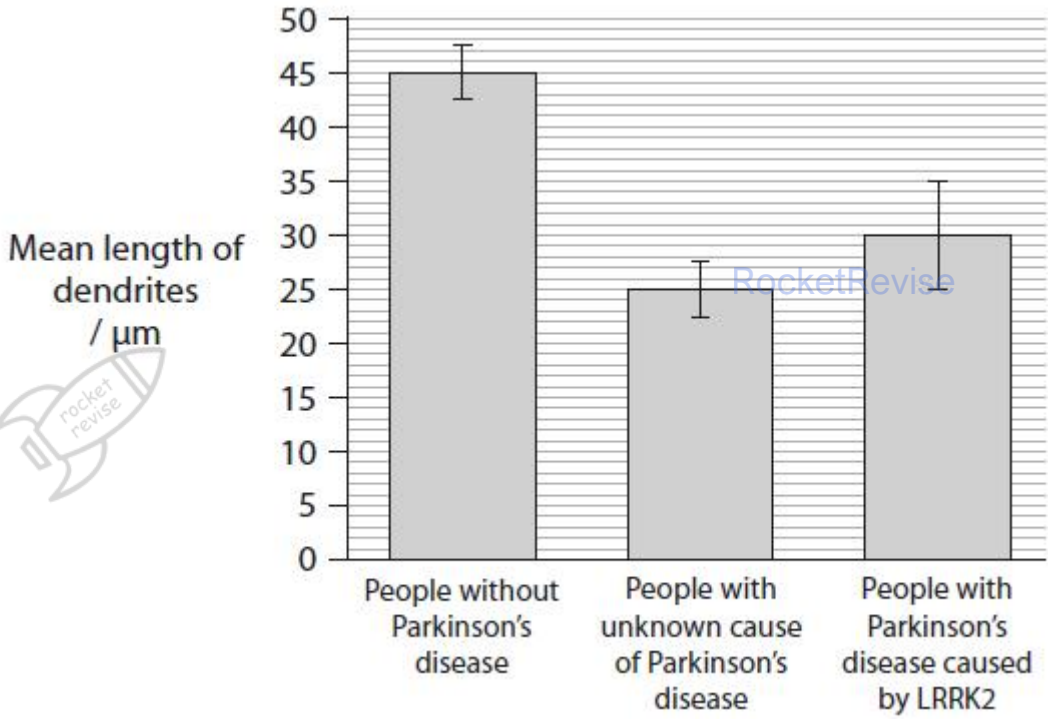
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(ii) Some people with Parkinson's disease have a DNA mutation called LRRK2.
Scientists studied the effect of this mutation on the dendrites of dopamine neurones produced from the stem cells of three groups of people.
The graphs show some of the results of their study.

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Comment on these results.

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(Total for question = 12 marks)

Q12.

Answer the question with a cross in the box you think is correct . If you change your mind about an answer, put a line through the box and then mark your new answer with a cross .

The brain is responsible for processing information and coordinating responses.

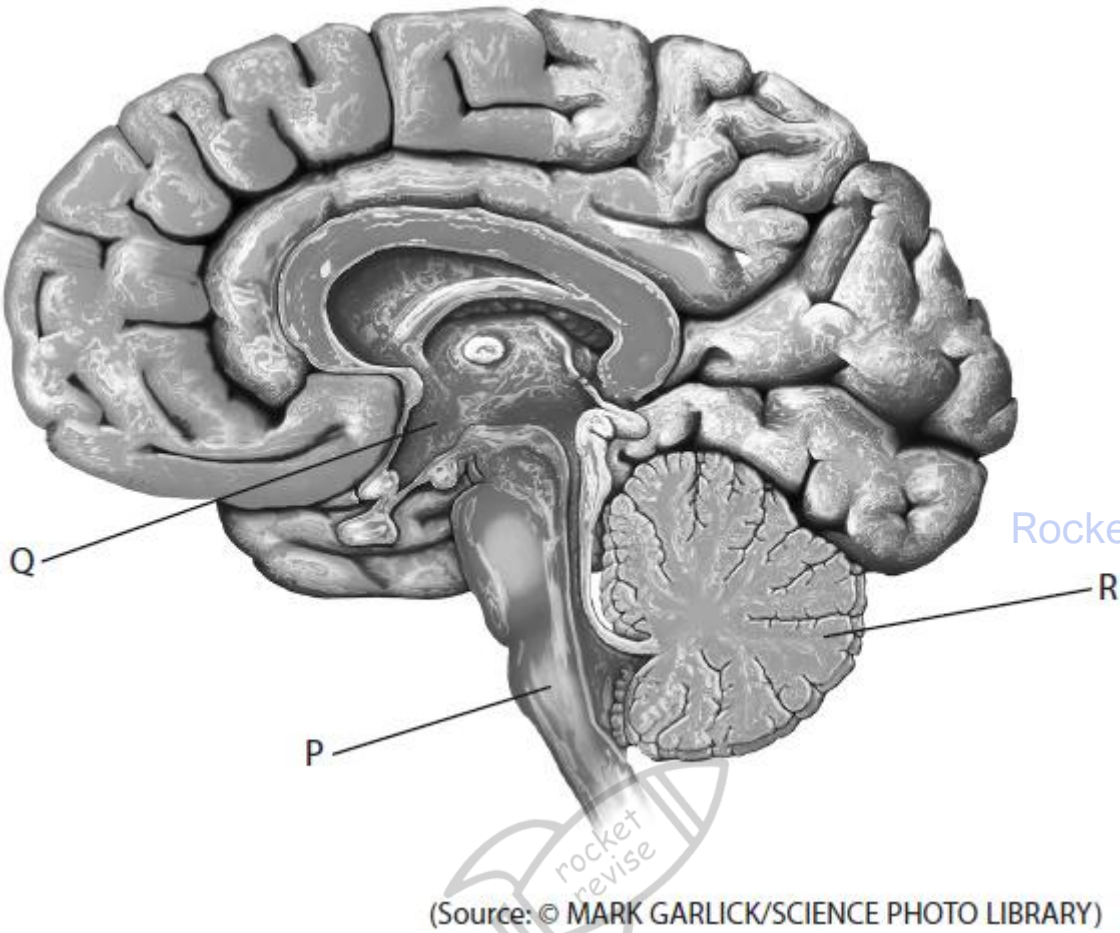
(a) The diagram shows a human brain.

Three regions of the brain have been labelled.

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(Source: © MARK GARLICK/SCIENCE PHOTO LIBRARY)

Complete the table with the name and function of each region.

(2)

Label	Name of the region	Function of the region
P	medulla oblongata	
Q		
R		balance and coordinating movement

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(b) Seizures in the brain can be investigated using different imaging techniques.

(i) Which one of the following statements is correct?

(1)

- ☒ **A** CT scans are produced by the absorbance of magnetic fields
- ☒ **B** CT scans are produced by the absorbance of radio waves by deoxyhaemoglobin
- ☒ **C** CT scans produce a cross-sectional image of a thin slice through the body
- ☒ **D** CT scans can be used to study brain activity while a person is undertaking tasks

(ii) During a seizure, part of the brain will be overactive.
Describe how positron emission tomography (PET) could be used to identify the part of the brain involved.

(3)

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(c) Some seizures can result in an increase in body temperature.

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Suggest how a seizure in part of the brain could lead to an increase in body temperature.

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(Total for question = 9 marks)

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Q13.

The scientific document you have studied is adapted from an article in Journal of Cachexia, Sarcopenia and Muscle entitled *Skeletal muscle performance and ageing*. (Tieland, 2018).

Use the information from the scientific document and your own knowledge to answer the following questions.

(a) Describe the structure of a skeletal muscle fibre (paragraph 2).

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(b) Explain how myokines may exert endocrine effects on organs such as the liver (paragraph 3).

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(c) Explain why the changes that occur to muscle structure and composition in the elderly can affect their ability to rise from a chair (paragraph 4).

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(d) Explain why Ca^{2+} binding to troponin C allows muscle fibres to contract (paragraph 5).

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(e) Explain how the 'high-energy phosphates' provide the energy necessary for muscle activity (paragraph 11).

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(f) Suggest explanations for the effects that a 'reduction in tendon stiffness' resulting from ageing could have on the movement at a joint (paragraph 12).

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(g) Suggest how myostatin might act as a negative regulator of muscle growth (paragraph 13).

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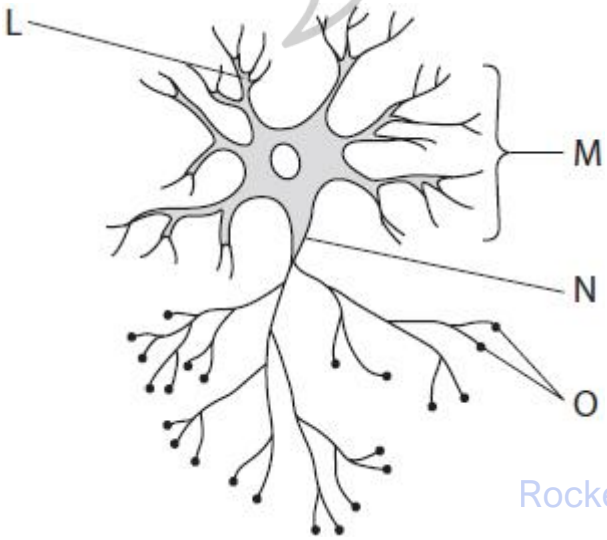
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(Total for question = 20 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows a relay neurone.



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(a) (i) Complete the table by filling in the names of each labelled part.

(2)

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Label	Part
L	
M	
N	
O	

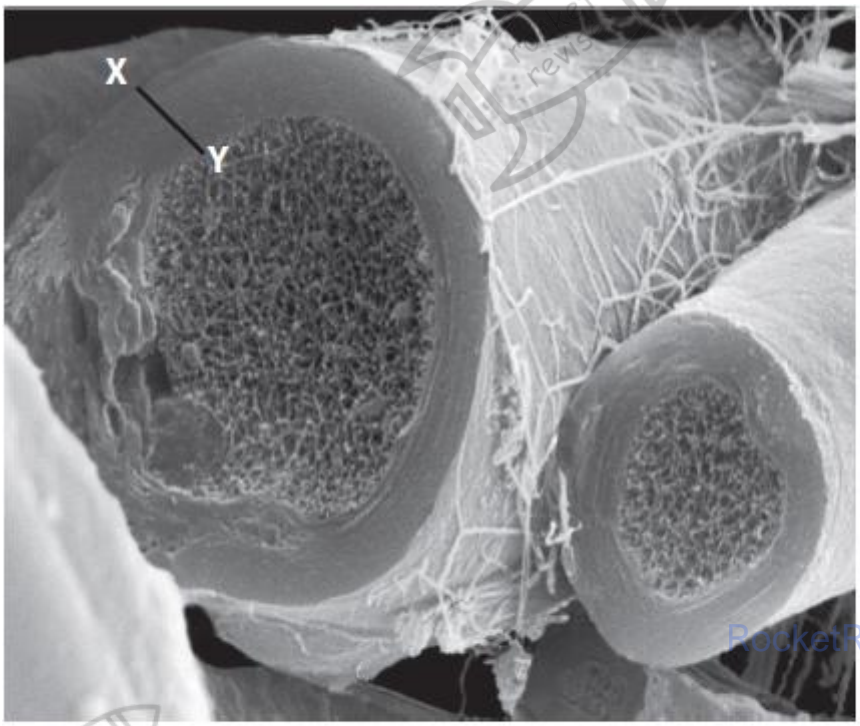
(ii) Lidocaine is a drug that is used as a local anaesthetic.
Lidocaine works by

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(1)

- ☐ **A** blocking the binding of acetylcholine to its receptor
- ☐ **B** blocking the release of acetylcholine
- ☐ **C** inhibiting potassium ion channels
- ☐ **D** inhibiting sodium ion channels

(b) The photograph shows a nerve with an outer layer of connective tissue.



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(Source: © Science Photo Library/Alamy Stock Photo)

Magnification × 3300

(i) State the type of microscope used to produce this image.

(1)

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(ii) Calculate the thickness of the connective tissue layer between the points X and Y.
Give your answer in micrometres (μm).

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Answer ..... µm

(iii) The nerve shown in the photograph contains many myelinated neurones.
Explain why a myelinated neurone conducts an impulse faster than a non-myelinated neurone of the same diameter.

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(Total for question = 9 marks)

Q15.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Animals and plants can detect and respond to light.

(a) Rod cells respond to light by producing action potentials in the optic nerve.

(i) In which structure are rod cells located?

(1)

- ☐ **A** iris
- ☐ **B** pupil
- ☐ **C** retina
- ☐ **D** spinal cord

(ii) Describe the role of rhodopsin in rod cells.

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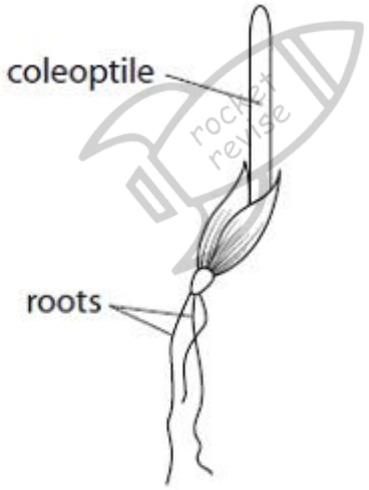
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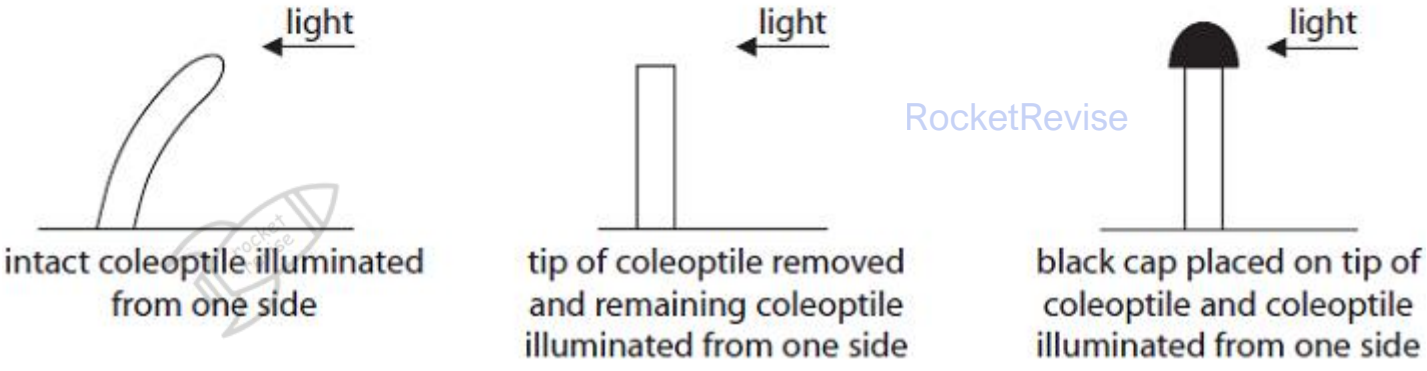
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*(b) The drawing shows the results of some experiments investigating phototropisms.

The diagram shows a coleoptile of a seedling.



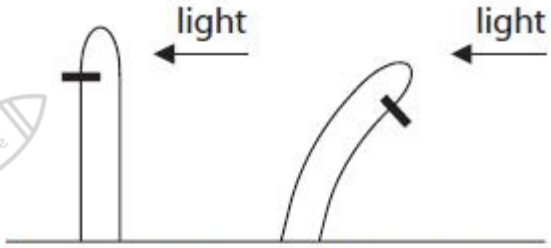
Experiment 1:



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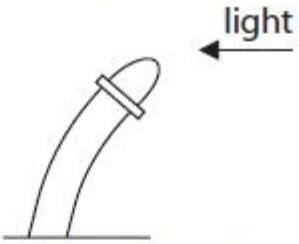
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Experiment 2a:



plastic sheet inserted below the tip of the coleoptile and the coleoptile illuminated from one side

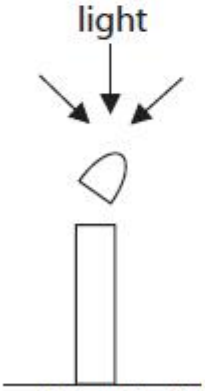
Experiment 2b:



gel layer placed below the tip of the coleoptile and the coleoptile illuminated from one side

Experiment 3:

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coleoptile illuminated and the tip removed



tip replaced on one side of the remaining coleoptile that is then kept in the dark

Scientists concluded that these results suggest a plant growth substance is involved in regulating the phototropic response.
Discuss how the results of these experiments support the role of a plant growth substance in the phototropic response of plants.
Use the information for these experiments and your own knowledge to support your answer.

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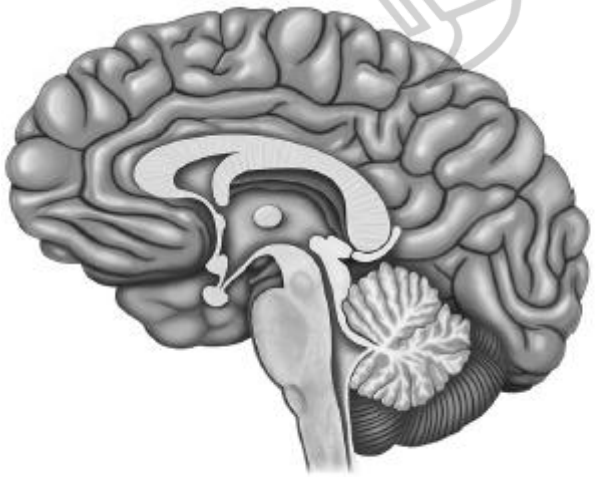
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(Total for question = 10 marks)

Q16.

The brain is made up of more than 1×10^{11} neurones.

(a) The diagram shows a section through the human brain.



(Source: © ilusmedical/Shutterstock)

Label the diagram to show the location of the following:

- pituitary gland
- medulla oblongata
- cerebral hemisphere (cerebrum)

(3)

(b) Complete the table to show the parts of the brain and their functions.

(2)

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Part of the brain	Function
cerebellum	
	control of heart rate
cerebral hemisphere (cerebrum)	

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(Total for question = 5 marks)

Q17.

The scientific document you have studied is adapted from articles in *StatPearls: Anatomy, Autonomic Nervous System, British Journal of Cardiology: Postural Orthostatic Tachycardia Syndrome (POTS) and Cellular and Molecular Life Sciences: The enteric nervous system in gastrointestinal disease etiology*.

Use the information from the scientific document and your own knowledge to answer the following questions.

(a) The autonomic nervous system regulates aspects of respiration (paragraph 1).

Describe how the autonomic nervous system regulates the breathing rate when a person is resting.

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(b) Explain how neurones can respond to different neurotransmitter substances (paragraph 4).

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(c) Suggest how the activity of the sympathetic nervous system in immune organs could regulate inflammation (paragraph 7).

(2)

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(d) Explain how an electrocardiogram could be used to show that a person has postural orthostatic tachycardia syndrome (POTS) (Box 1).

(2)

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(e) Explain how reflex circuits in the gut cause a bolus to move (paragraphs 10 and 11).

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(f) State what is meant by the term ischemia (paragraph 12).

(1)

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(g) Suggest why embryological ENS cells may fail to colonise the bowel in a developing embryo (paragraph 13).

(1)

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(h) Explain how the interaction of microbial and other factors could result in gastrointestinal inflammation such as IBD (Box 2).

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(Total for question = 20 marks)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The nervous system enables an organism to respond to environmental stimuli.

(a) Several types of neurone are found in the nervous system.

(i) Which term describes the space between a sensory neurone and a relay neurone?

(1)

- ☐ **A** axon
- ☐ **B** cell body
- ☐ **C** postsynaptic membrane
- ☐ **D** synaptic cleft

(ii) Compare and contrast the function of a sensory neurone and a motor neurone.

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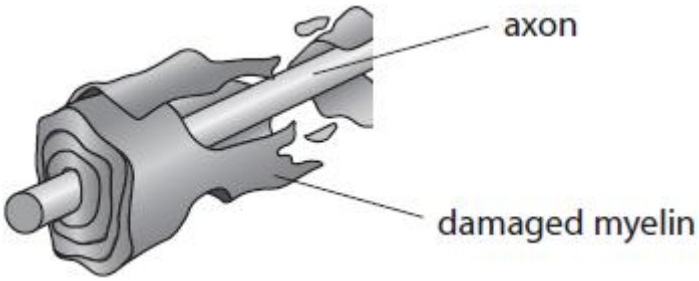
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(b) Guillain-Barré syndrome is a disorder of the nervous system.

The immune system damages the myelin sheath surrounding the axons of individuals with Guillain-Barré syndrome.



Symptoms of this disorder include:

- inability to maintain steadiness when walking
- inability to feel any sensations in the limbs

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- loss of reflexes such as knee jerk.

Suggest why a person with Guillain-Barré syndrome will suffer from the symptoms of this condition.

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(c) Mutations in the genes coding for voltage-gated calcium ion channels have been linked to neurological conditions.

Explain how these mutations might cause these neurological conditions.

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(Total for question = 11 marks)

Q19.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Plants respond to external stimuli.

(a) Phytochrome molecules are found in plants.

Phytochrome exists in two forms, Pr (P_{660}) and Pfr (P_{730}).
Which row shows the action of these two forms of phytochrome?

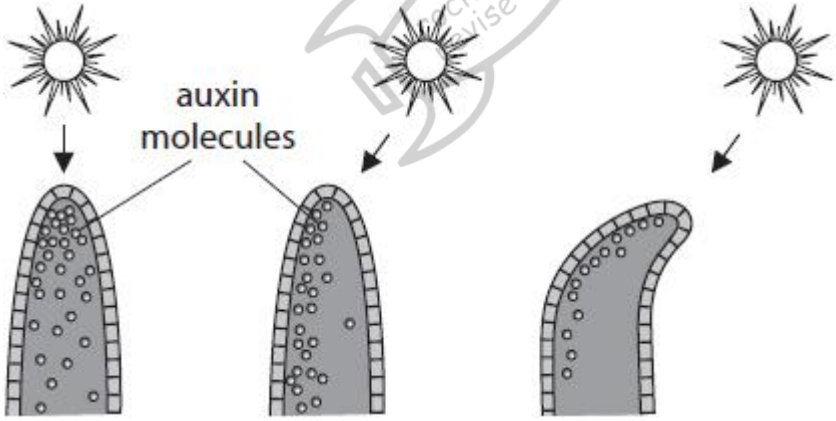
(1)

		Absorbs far red light	Breaks down in the dark
☒	**A**	Pfr	Pfr
☒	**B**	Pfr	Pr
☒	**C**	Pr	Pfr
☒	**D**	Pr	Pr

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(b) Auxins (IAA) are plant growth substances.

The diagram shows auxin molecules producing a phototropic response in a shoot.



Explain the role of auxins in this phototropic response.
Use the information in the diagram to support your answer.

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(c) Seeds contain starch. This starch is used to supply energy when the seed germinates.

Gibberellins are a group of naturally occurring plant growth substances.
Gibberellins regulate starch hydrolysis in the seed.
Explain how gibberellins can produce this effect on the seed.

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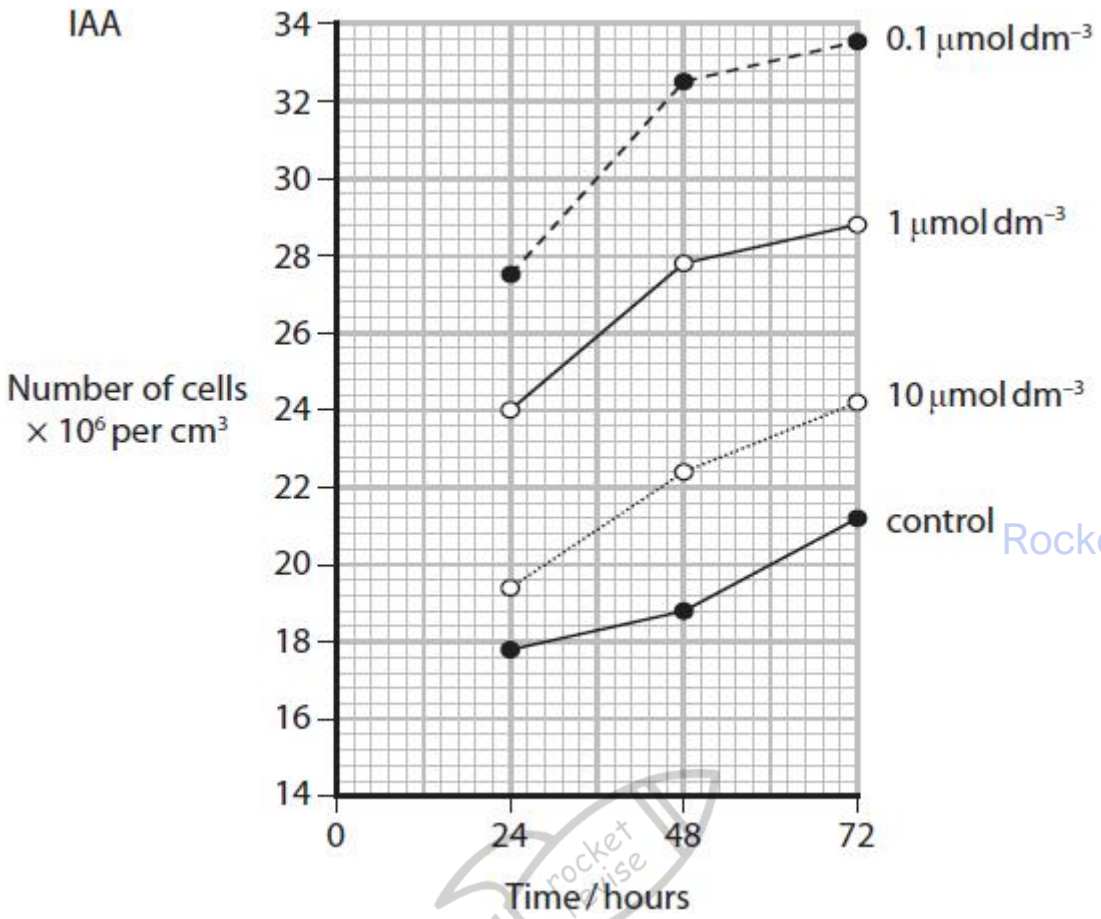
(d) Indole acetic acid (IAA) is a natural plant auxin. 1-Naphthaleneacetic acid (NAA) is a synthetic auxin.

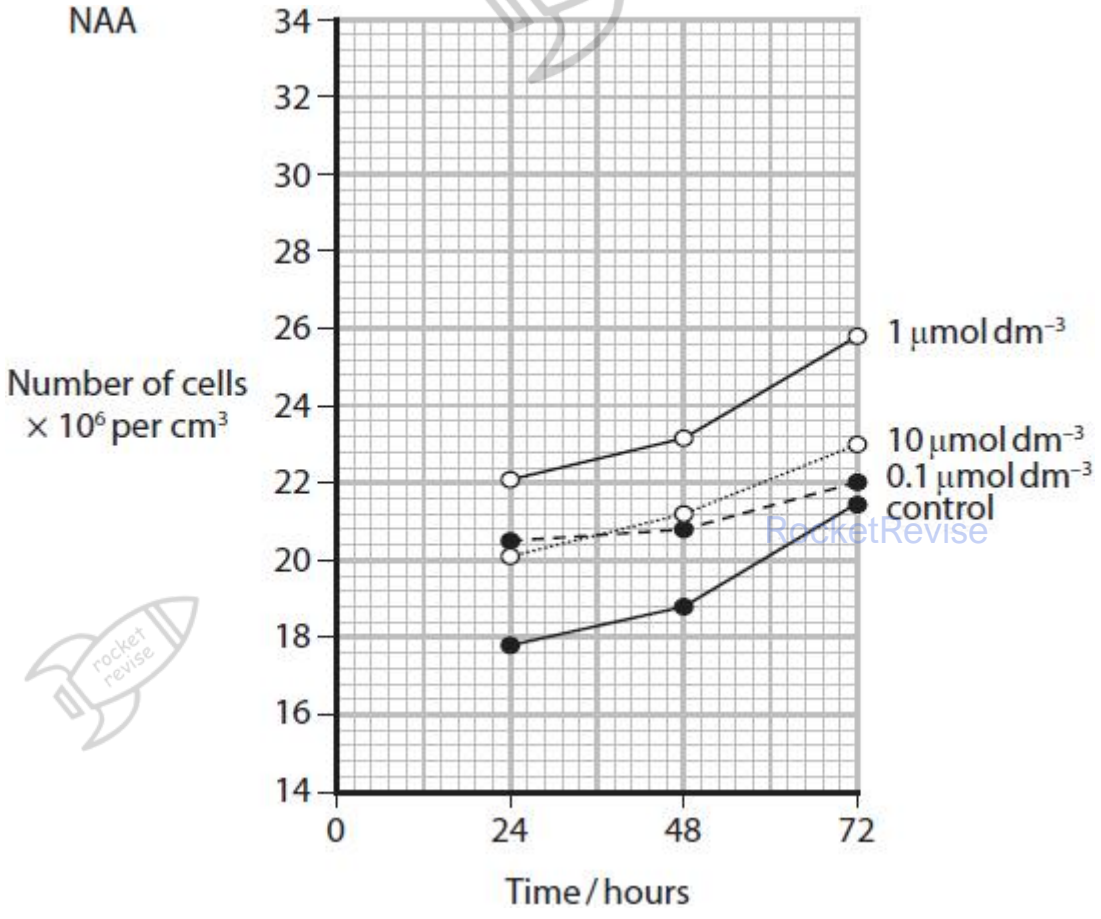
The effects of these two auxins on the growth of *Chlorella* was investigated.
Chlorella is a single-celled photosynthetic organism.
The graphs show the effect of different concentrations of the auxins on the growth of *Chlorella*.

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Comment on the effect of these auxins on the growth of *Chlorella*.

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(Total for question = 12 marks)

Q20.

Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Whales are large mammals.

The photograph shows a beluga whale and a human diver.

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(Source: © Andrey Nekrasov/Alamy Stock Photo)

(a) (i) A beluga whale can become habituated to human divers.

How many of the following statements about habituation are correct?

(1)

- habituation is an example of anatomical adaptation
- habituation is only observed in mammals
- habituation only occurs after repeated exposure to the same stimulus

☒ **A** 0
☒ **B** 1
☒ **C** 2
☒ **D** 3

(ii) A beluga whale can respond to a stimulus using a spinal reflex arc.
Which row is correct?

(1)

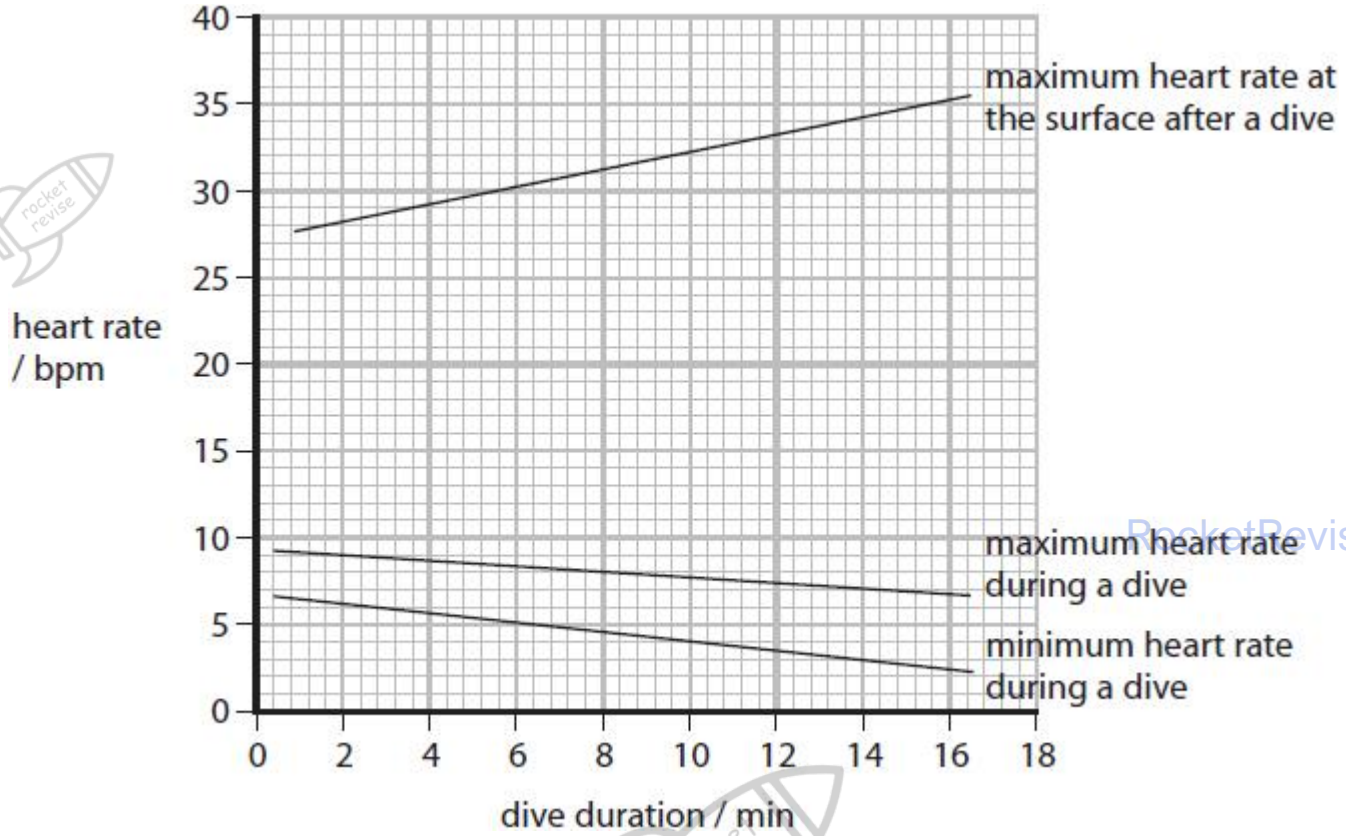
	Location of relay neurone	Location of cell body on the sensory neurone
☒ **A**	grey matter of spinal cord	at the end of the axon
☒ **B**	grey matter of spinal cord	in the middle of the axon
☒ **C**	white matter of spinal cord	at the end of the axon
☒ **D**	white matter of spinal cord	in the middle of the axon

(b) The heart rate of a whale changes when it is diving.

The graph shows the changes in heart rate that occur during dives of different durations.
The maximum heart rate at the surface is when the whale returns to the surface after a dive.

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(i) Comment on the changes in heart rate of the whale when diving.

(3)

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(ii) Explain how the changes in heart rate are controlled when the whale returns to the surface immediately after the dive.

(3)

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(Total for question = 8 marks)

Q21.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Chemicals are used to regulate responses in both plants and animals.

(a) In mammals antidiuretic hormone (ADH) is involved in regulating plasma concentration and blood volume.

(i) Which of the following produces ADH?

(1)

- ☒ **A** adrenal gland
- ☒ **B** hypothalamus
- ☒ **C** medulla oblongata
- ☒ **D** pituitary gland

(ii) Which structures does ADH act on?

(1)

- ☒ **A** capillaries
- ☒ **B** collecting ducts
- ☒ **C** proximal tubules
- ☒ **D** ureters

(iii) The control of plasma concentration and blood volume in the body is an example of which type of process?

(1)

- ☒ **A** countercurrent multiplier
- ☒ **B** habituation
- ☒ **C** negative feedback
- ☒ **D** positive feedback

(iv) Name the type of receptor in the body that detects changes in the plasma concentration.

(1)

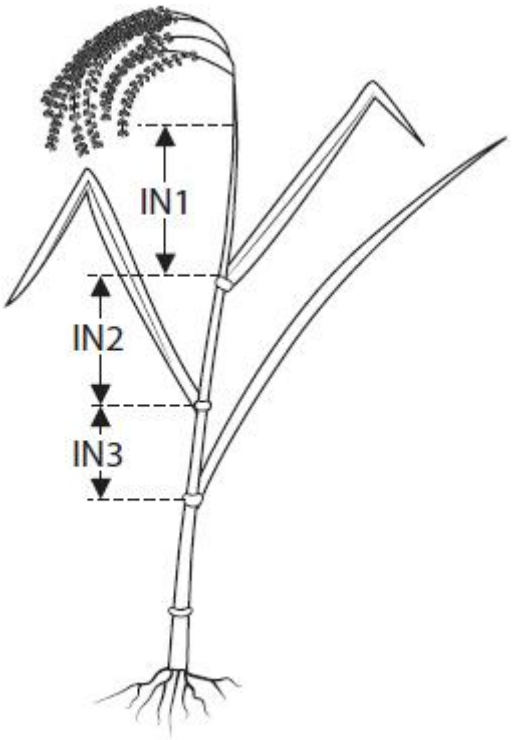
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(b) The effect of a gibberellin on the growth of cereal plants was investigated.

Different concentrations of the gibberellin were sprayed on cereal plants.
The lengths of the first, second and third internodes (IN1, IN2, IN3) were measured, as shown in the diagram.

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The table shows the effect of the gibberellin on the mean lengths of the first, second and third internodes.

Concentration of the gibberellin / ppm	Mean length / cm		
	first internode	second internode	third internode
0	36	19	14
10	36	19	14
100	41	21	14
1000	52	22	15
2500	56	22	15

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(i) Comment on the effect of the gibberellin on the growth of cereal plants.

(2)

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(ii) Suggest how the gibberellin causes this effect.

(3)

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(Total for question = 9 marks)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The nervous system is a complex collection of neurones and specialised cells that transmit impulses between different parts of the body.

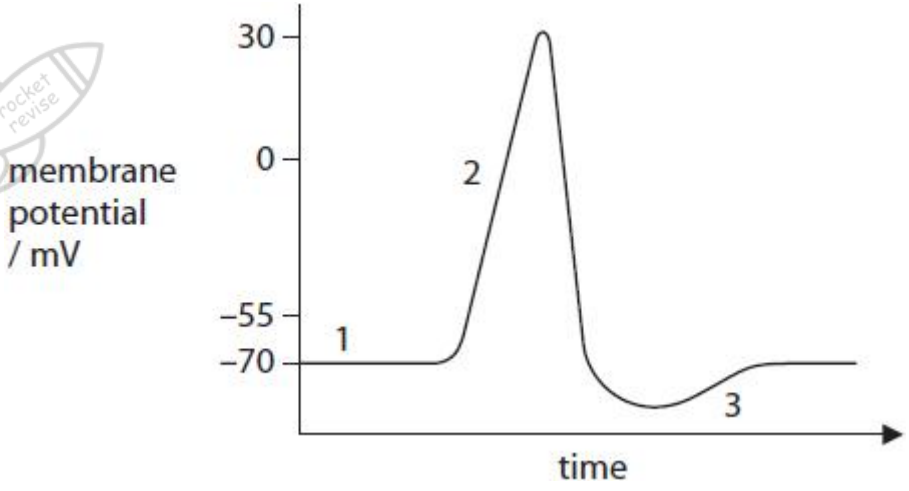
(a) Which substance can transmit a nerve impulse from one neurone to the next one?

(1)

- ☐ **A** acetylcholine
- ☐ **B** ADP
- ☐ **C** cholesterol
- ☐ **D** NADP

(b) The diagram shows the change in membrane potential as an action potential is transmitted in a motor neurone.

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Explain why the membrane potential changes between 1 and 2.

(2)

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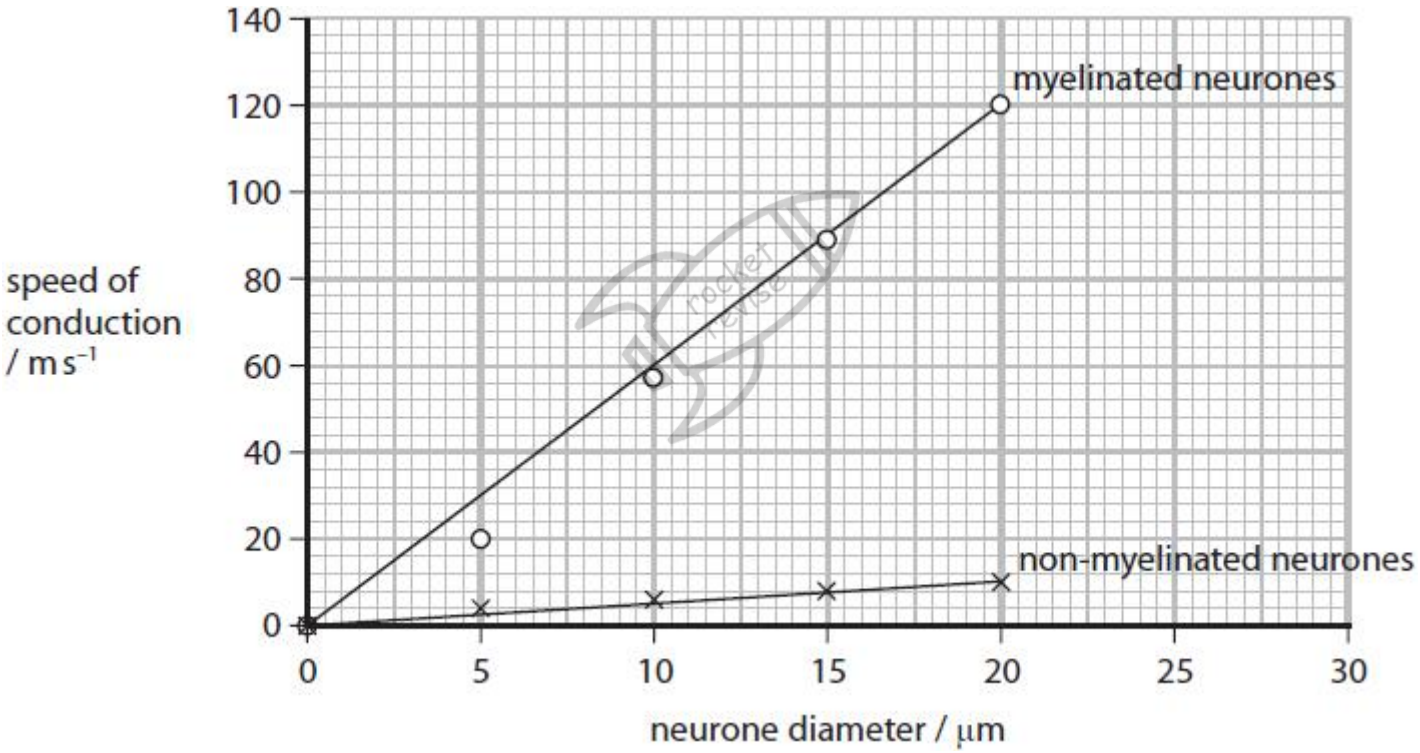
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(c) Myelinated neurones are surrounded by a fatty sheath.

The graph shows the speed of conduction in myelinated neurones and non-myelinated neurones.

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(i) Calculate the percentage difference in the gradient of the myelinated and non-myelinated neurones.

The gradient for non-myelinated neurones is 0.5.

The equation for a straight line is $y = mx + c$.

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(3)

Answer ..... %

(ii) Explain why the speed of conduction differs in myelinated and non-myelinated neurones.

(3)

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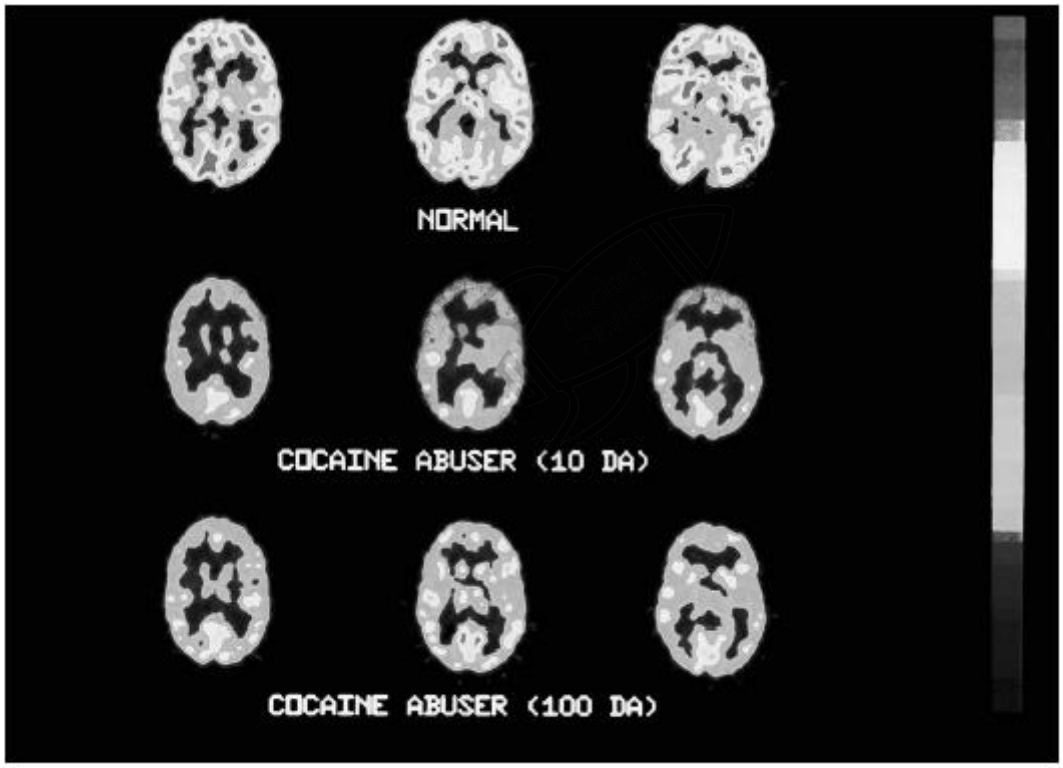
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(d) The positron emission tomography (PET) images show the brain of an individual before (control) and after taking the drug cocaine.

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(Source: © BROOKHAVEN NATIONAL LABORATORY / SCIENCE PHOTO LIBRARY)

Explain how positron emission tomography (PET) can be used to identify the change in activity in the brain after taking this drug.

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(Total for question = 13 marks)

Q23.

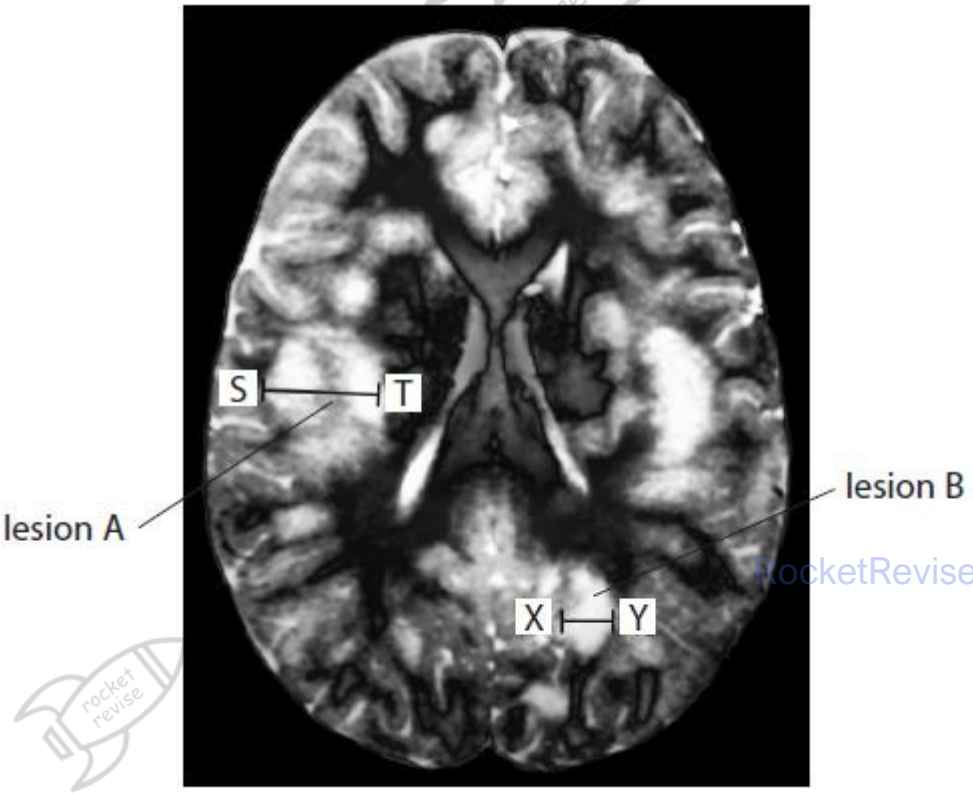
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Acute disseminated encephalomyelitis (ADEM) causes the destruction of the myelin sheath around neurones.

The destruction is the result of inflammation.

Lesions form in the brain where the myelin is destroyed.

The image is a magnetic resonance imaging (MRI) scan of a person with ADEM, showing lesions in the brain.



(Source: © ZEPHYR/SCIENCE PHOTO LIBRARY)

(a) (i) Lesion A is spherical and has a diameter (S-T) of 1.97 cm.

Calculate the volume of lesion A.
Use the formula:

$$V = \frac{4}{3}\pi r^3$$

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Give your answer to 2 decimal places.

(2)

Answer ..... cm^3

(ii) Calculate the actual width of the lesion labelled B between points X and Y.
Give your answer to 2 significant figures.

(2)

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Answer ..... cm

(b) An MRI scan can be used to diagnose a brain tumour.

Suggest why a dye might be given during the MRI scan.

(1)

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(c) A person with ADEM has problems with walking.

Explain why ADEM causes problems with walking.

(2)

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(Total for question = 7 marks)

Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some processes in plants and animals are under the control of chemicals.

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(a) (i) Which row in the table shows the effect of an increase in adrenaline in a mammal?

(1)

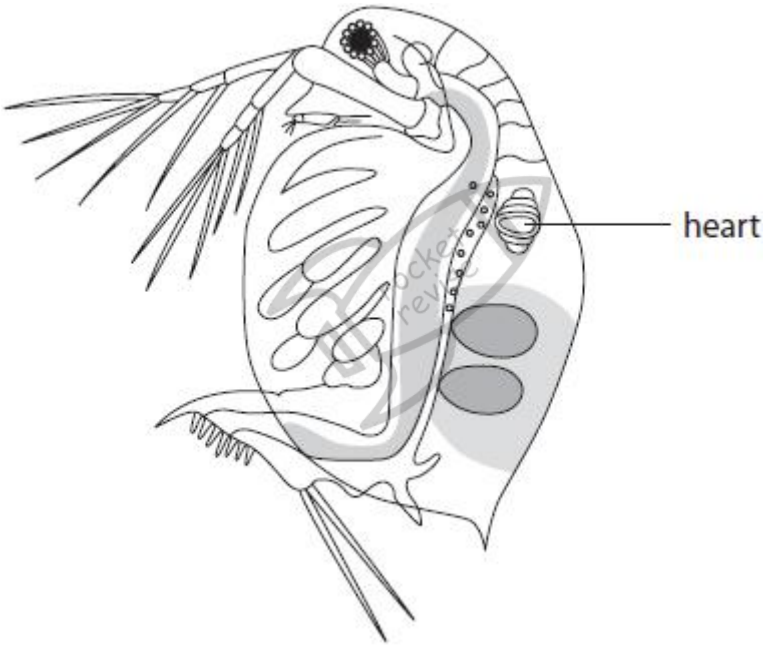
	breathing rate	heart rate
☐ A	decreases	increases
☐ B	increases	decreases
☐ C	increases	increases
☐ D	decreases	decreases

(ii) Nicotine is a drug found in tobacco products.

The effect of changing the concentration of nicotine and adrenaline on the mean heart rate of Daphnia was investigated.

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The diagram shows a Daphnia.



The results are shown in the table.

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Concentration of nicotine / mmol dm^{-3}	Mean heart rate / bpm		
	No adrenaline	Concentration of adrenaline 0.001 mmol dm^{-3}	Adrenaline 0.1 mmol dm^{-3}
1000	84	81	104
10	96	99	110
0.1	103	105	113
0.01	108	109	127
0.001	106	107	151
0.00 (control)	146	160	

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Deduce the effect of the combination of these chemicals on the heart rate of Daphnia.

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(b) Which is a description of a gibberellin?

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(1)

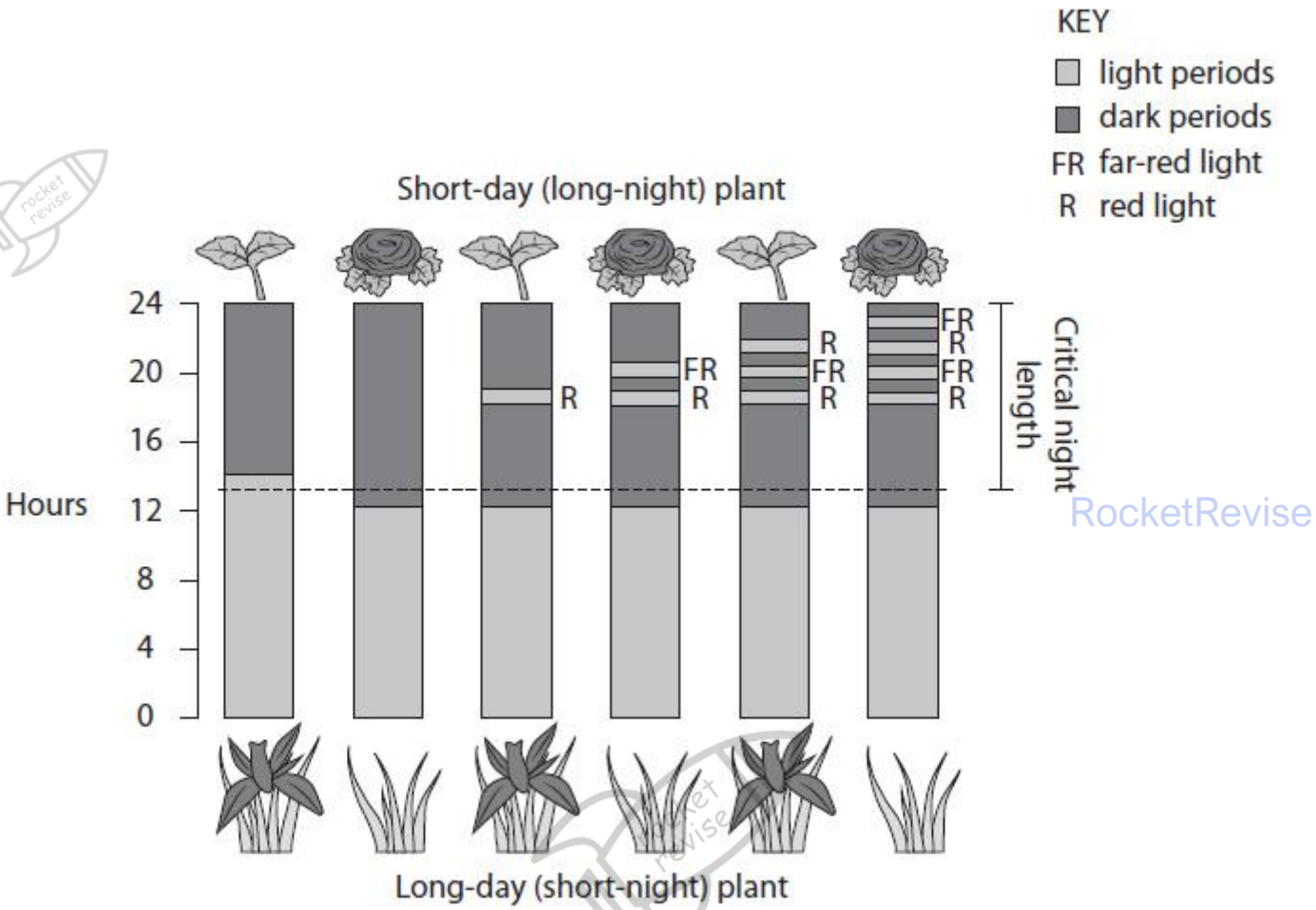
- ☒ **A** an enzyme that converts starch to glucose
- ☒ **B** a molecule that converts phytochrome far-red (Pfr) to phytochrome red (Pr)
- ☒ **C** an organic compound that stimulates the action of transcription factors
- ☒ **D** a protein that wraps around DNA

*(c) The quality and length of daylight act as signals to initiate and regulate flowering in plants. This involves phytochromes and the production of gibberellins.

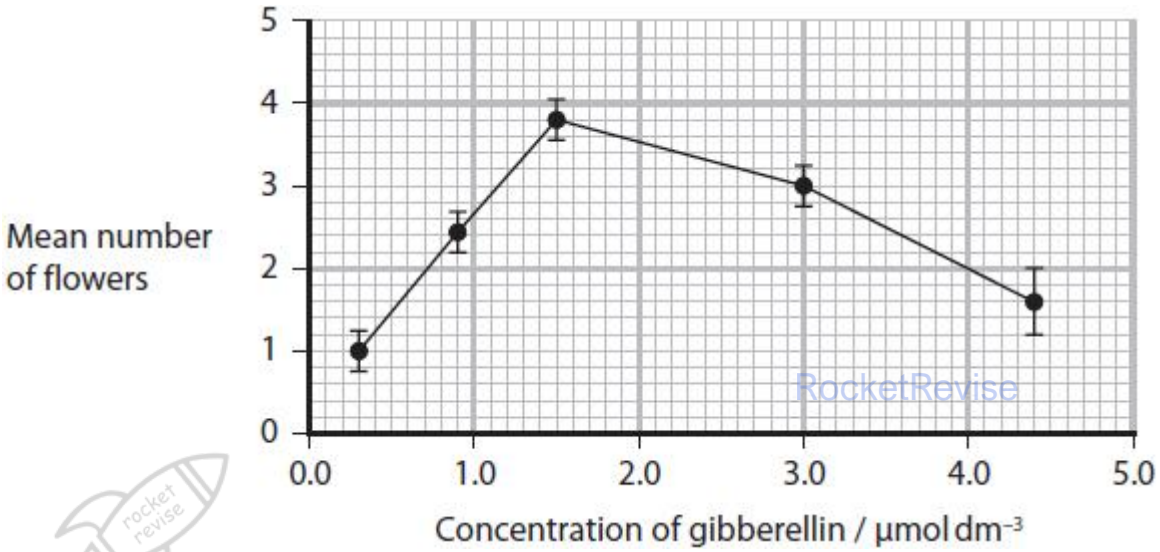
In an investigation, a long-day plant and a short-day plant were exposed to light for different lengths of time. During dark periods they were exposed to flashes of red and far-red light.
The diagram shows the results of this investigation.

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The graph shows the results of an investigation into the effect of gibberellin concentration on the flowering of another type of plant.



Discuss how phytochromes and gibberellins affect flowering in plants.

Use your own knowledge and the information in the question to support your answer.

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(Total for question = 11 marks)

Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows the eyes of a meerkat.

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(Source: Jeremy Pardoe / Alamy Stock Photo)

(a) Describe how the pupil of an eye dilates and contracts in response to changes in light intensity.

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(b) Photoreceptors stimulate nerve transmission in the eye.

(i) Which of the following pairs of molecules forms the photoreceptor in an eye?

(1)

- ☒ **A** auxin and retinol
- ☒ **B** lysozyme and ribose
- ☒ **C** opsin and retinal

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☒ **D** phytochrome and opsin

(ii) Which of these drugs would increase nerve transmission?

(1)

☒ **A** ecstasy (MDMA) and lidocaine
☒ **B** ecstasy (MDMA) and nicotine
☒ **C** lidocaine and nicotine
☒ **D** lidocaine only

(iii) In direct sunlight the diameter of the pupil was 4 mm.
In complete darkness the area of the pupil was 38.75 mm^2.
Calculate the increase in diameter of the pupil when moving from direct sunlight to complete darkness.

(2)

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Answer ..... mm

(Total for question = 8 marks)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The human eye responds to different light intensities.

(a) The table shows the mean pupil diameter in different light intensities.

Light intensity	Pupil diameter / mm
bright light	3.35
normal room light	3.86
darkness	6.41

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Calculate the percentage increase in the mean diameter of the pupil when a person moves from bright light into darkness.

(1)

Answer ..... %

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(b) During the fight or flight response the pupils of a person increase in diameter.

(i) Which is the name of the hormone involved in the fight or flight response?

(1)

- ☐ **A** adrenaline
- ☐ **B** antidiuretic hormone
- ☐ **C** thromboplastin
- ☐ **D** uracil

(ii) Describe how this change in diameter is brought about by the muscles of the iris.

(3)

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(Total for question = 5 marks)

Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Different techniques can be used in the investigation of the activity of the brain.

(a) Which type of scan can be used to study changes in brain function as they happen?

(1)

- ☐ **A** computed tomography (CT)
- ☐ **B** electrocardiogram (ECG)
- ☐ **C** functional magnetic resonance imaging (fMRI)
- ☐ **D** polymerase chain reaction (PCR)

(b) Describe the advantages and disadvantages of using positron emission tomography (PET) and magnetic resonance imaging (MRI) to investigate brain function.

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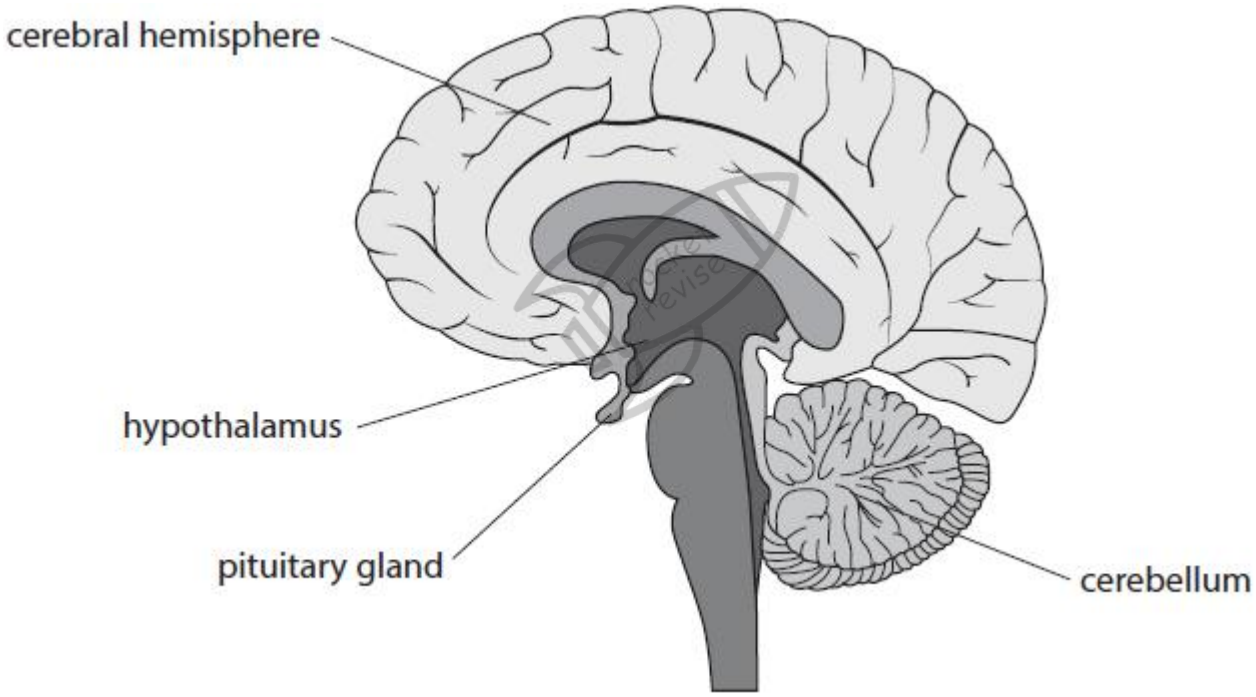
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(c) The diagram shows a section through the human brain.

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Complete the table to show **one** function for each part of the brain.

(3)

Part of brain	Function
cerebellum	
cerebral hemisphere	
hypothalamus	

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(Total for question = 7 marks)

Q28.

Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Nervous communication involves neurones and changes in ion concentrations.

(a) (i) Which description of the movement of ions will produce an action potential in an axon?

(1)

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- ☒ **A** K^+ ions move into the axon
- ☒ **B** K^+ ions move out of the axon
- ☒ **C** Na^+ ions move into the axon
- ☒ **D** Na^+ ions move out of the axon

(ii) Which row describes the events that occur during repolarisation of a neurone?

(1)

	Sodium ion channel	Membrane potential
☒ **A**	closed	decreasing
☒ **B**	open	decreasing
☒ **C**	open	increasing
☒ **D**	closed	increasing

(b) Describe the structure of a spinal reflex arc.

(3)

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(c) Explain how the nerve impulse is transmitted across a synapse.

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(Total for question = 9 marks)

Q29.

The scientific documents you have studied are adapted from articles in New Scientist entitled '*Your Second Skin*' (May 2022) and '*Anatomy Fascia*' from *Researchgate* (December 2018).

Use the information from the scientific documents and your own knowledge to answer the following questions.

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(a) Give the meaning of the term 'sensory organ' (paragraph 2).

(2)

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(b) Fascia is a type of connective tissue.

Use ticks and crosses to complete the table comparing types of connective tissue found in the human body (paragraphs 3 to 7).

(2)

Connective tissue	Contains collagen	Rich in nerves	Surrounds muscle
fascia			
ligaments			

(c) Describe how the fascia generates a 'diffuse pain' (paragraphs 8 and 9).

(3)

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(d) Explain how inflammation in the fascia is caused (paragraphs 8 and 9).

(3)

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(e) Describe how fibroblasts within the fascia could be transformed into myofibroblasts (paragraph 13).

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(f) Explain how lymph is moved in one direction in the lymphatic system (paragraphs 15 and 16).

(3)

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(g) Suggest how aging may have an adverse effect on the structure of the lymphatic system (paragraph 17).

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(Total for question = 20 marks)

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Serotonin is produced by neurones in the brain.

(a) How many of the following statements about serotonin are correct?

(1)

- L-dopa can be converted into serotonin in the brain
- Serotonin is a neurotransmitter
- Serotonin is produced by post-synaptic neurones

☐ **A** 0
☐ **B** 1
☐ **C** 2
☐ **D** 3

(b) There are 14 different serotonin receptors found in the human nervous system.

Three different genes are used to produce these receptors.
Explain how three genes can produce 14 different receptors for serotonin.

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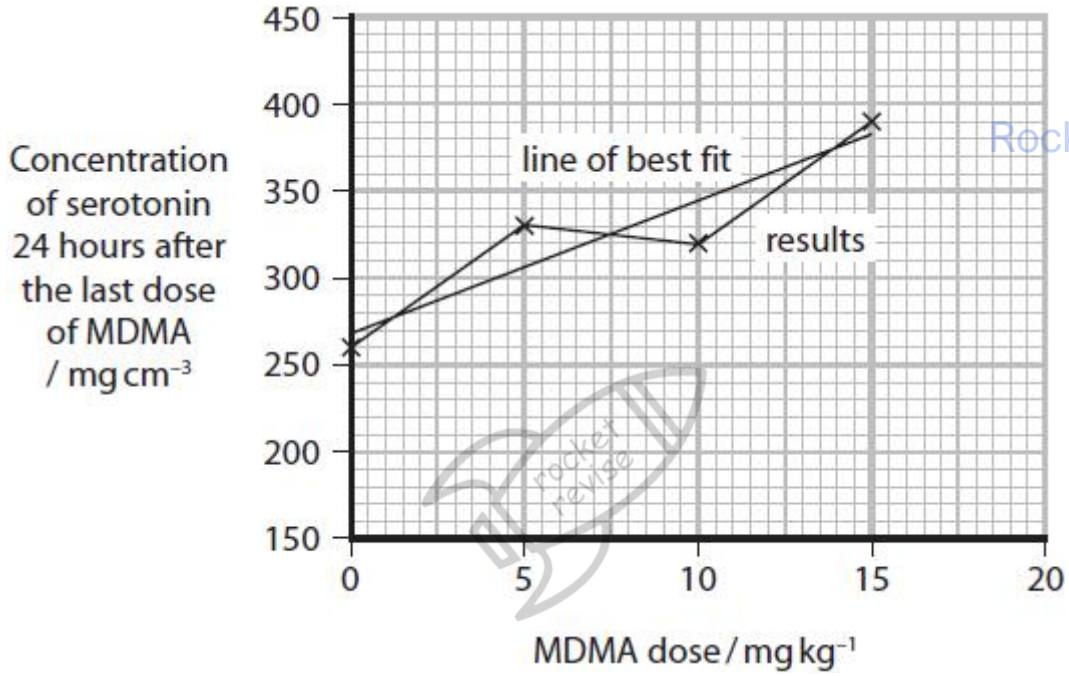
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(c) The effect of MDMA (ecstasy) on serotonin concentration in rats was investigated.

Rats were given different concentrations of MDMA by mouth twice daily for a week.
The concentration of serotonin in the cerebrospinal fluid of the rats was analysed 24 hours after the last dose of MDMA.
Cerebrospinal fluid is fluid that surrounds the brain and spinal cord.
The graph shows the effect of MDMA on the concentration of serotonin and a line of best fit is shown on the graph.



(i) Determine the effect of MDMA on the serotonin level in the cerebrospinal fluid.

(2)

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(ii) Predict the concentration of serotonin for a dose of 17.5 mg kg^{-1} of MDMA.
Use the line of best fit.
Give your answer in g cm^{-3}.

(2)

..... g cm^{-3}

(iii) Explain how the MDMA taken **by mouth** could affect the level of serotonin in the brain.
Use the information in the question to support your answer.

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(Total for question = 12 marks)

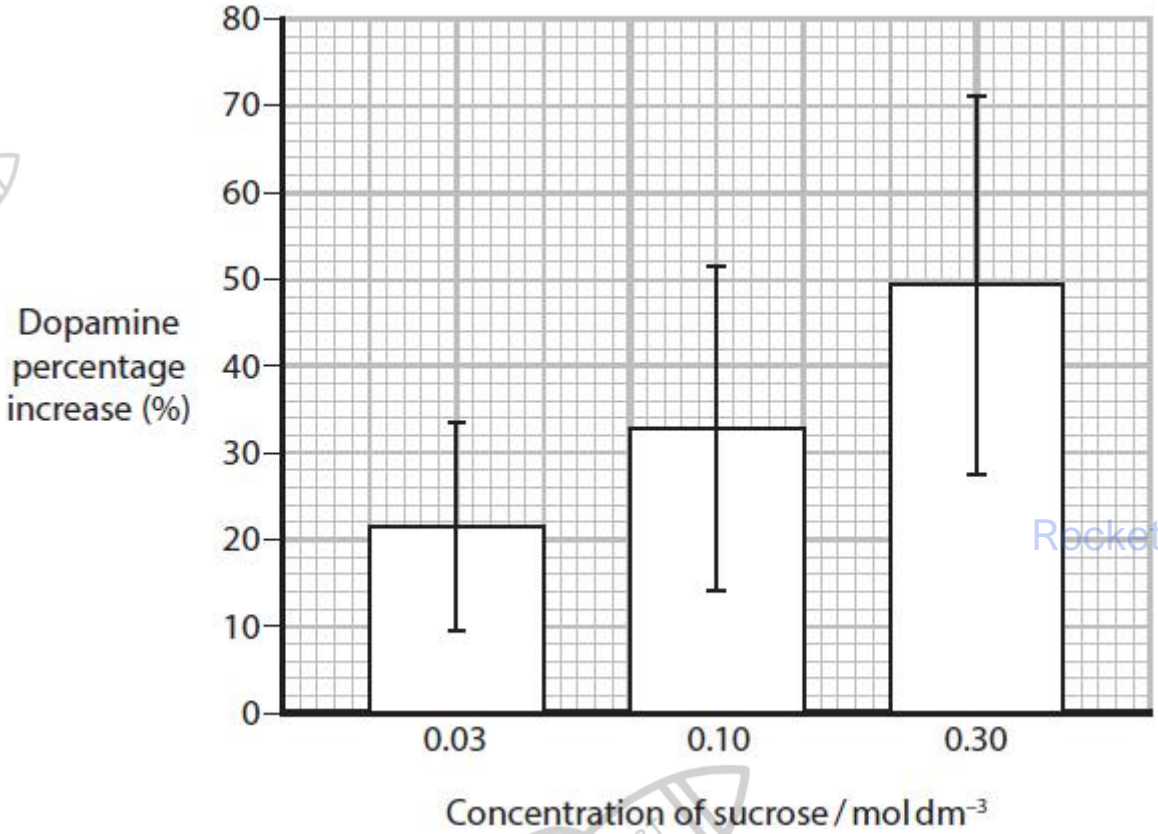
Q31.

(a) Scientists investigated the effect of sucrose solutions on the desire of rats to eat.

The release of dopamine in some areas of the brain increases the desire of the rat to eat. Scientists measured increases in the release of dopamine in the brain of rats given different concentrations of sucrose solutions to drink.

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The scientists concluded that drinking a sucrose solution had an effect on the desire of the rats to eat.

(i) Comment on the results shown in the bar chart and how they support this conclusion.

(4)

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(ii) The table gives the results for a sucrose concentration of 0.4 mol dm^{-3}.

Complete the table to calculate the standard deviation (S) for these results.

$$S = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}}$$

Give your answer to two significant figures.

(2)

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result (x)	$x - \bar{x}$	$(x - \bar{x})^2$
45	−15	225
63	3	9
74	14	196
58		
	$\sum(x - \bar{x})^2 =$	
	$n - 1 =$	
	$S =$	

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(b) When humans are in danger or under stress they respond by activating the 'fight or flight' response.

Describe the role of adrenaline in the fight or flight response.

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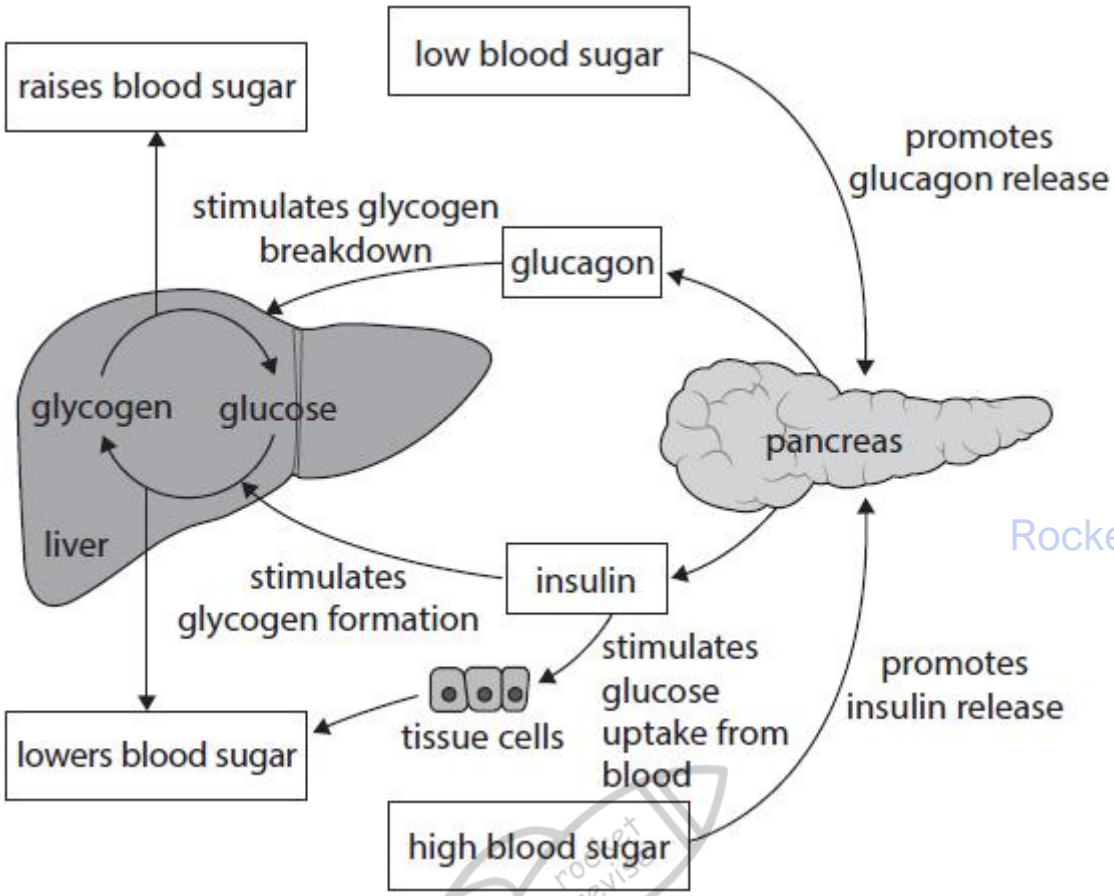
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* (c) Negative feedback is involved in maintaining systems within narrow limits.

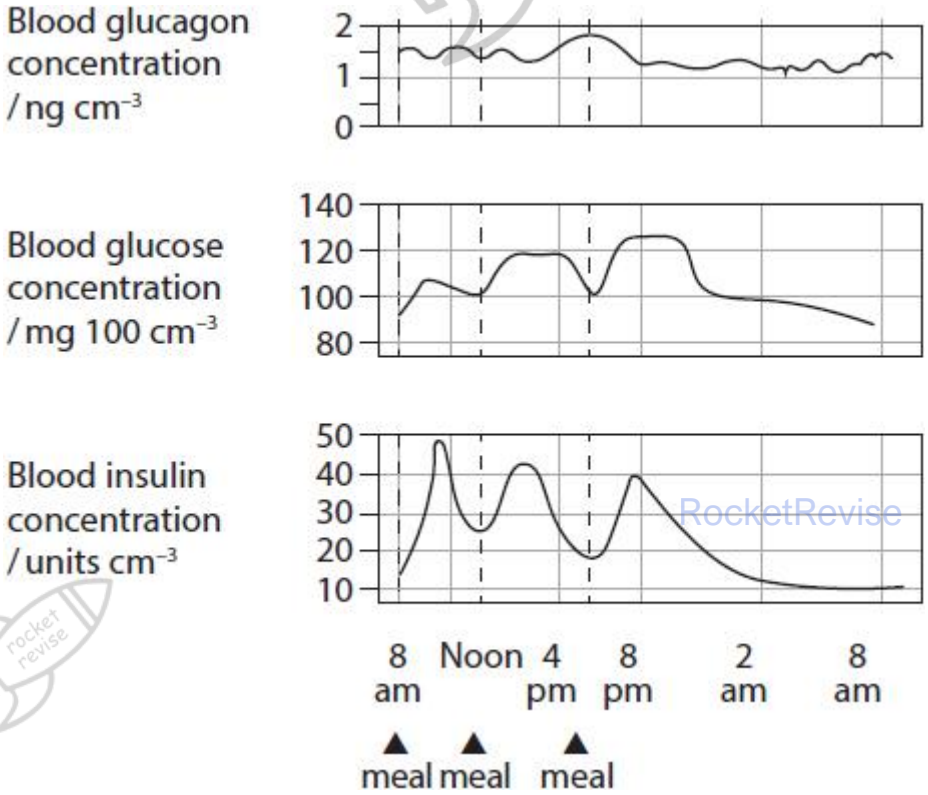
The blood glucose concentration has to be maintained within narrow limits.
Insulin and glucagon are hormones involved in the control of blood glucose levels.
The diagram shows how blood glucose concentrations are controlled in the body.

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The graphs show the daily changes in the concentrations of insulin, glucagon and glucose.



Discuss how negative feedback is involved in the control of blood glucose concentrations.
Use the information in the diagram and the graphs to support your answer.

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(Total for question = 15 marks)

Q32.

Multiple sclerosis (MS) is a disease that causes the destruction of the myelin sheaths around neurones.

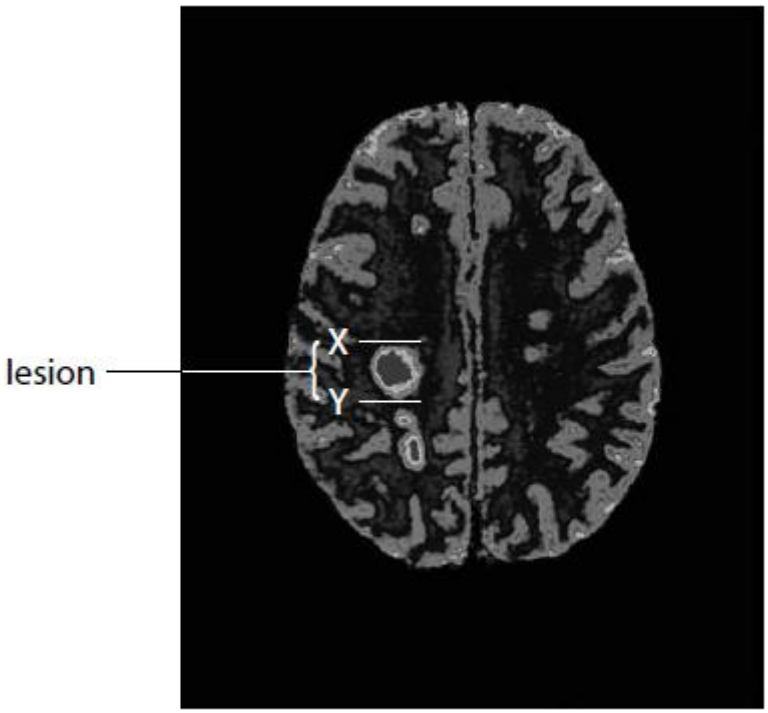
This destruction is caused by the person's own immune system.

Lesions form in the brain where myelin is destroyed.

(a) The photograph shows lesions in the brain of a person with MS, as seen using magnetic resonance imaging (MRI).

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The actual length of this brain is 15 cm.
Calculate the actual diameter, in mm, of the lesion between X and Y.

(2)

Answer ..... mm

(b) The table shows the speed of conduction of an impulse along non-myelinated and myelinated neurones with axons of different diameters.

Type of neurone	Diameter of axon / μm	Speed of conduction of impulse / m s^{-1}
non-myelinated	1	2.3
non-myelinated	2	3.1
non-myelinated	4	4.2
myelinated	1	2.6
myelinated	2	8.5

(i) Show that the effect of diameter on the speed of conduction of an impulse is six times greater in a

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myelinated neurone than in a non-myelinated neurone.

(2)

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(ii) Explain the effect of the myelin sheath and the diameter of the axon on the speed of conduction of an impulse.

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(4)

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(iii) Symptoms of MS can include blindness in one eye and lack of coordination.
Suggest why a person with MS could have these symptoms.

(3)

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(Total for question = 11 marks)

Q33.

A number of insects can transmit disease when they bite their hosts.

The photograph shows *Rhodnius prolixus*, a blood-sucking insect that transmits Chagas disease between humans.

This insect feeds on blood by extending its proboscis and pushing it through the skin of a person. It is thought to do this when it detects body warmth.



© Oxford Scientific/Getty Images

An investigation was carried out to study the effect of a heat stimulus on the proboscis extension response (PER).

Insects were kept in a temperature-controlled container with a heat pad placed in front of them. The heat pad could be touched by the proboscis when fully extended.

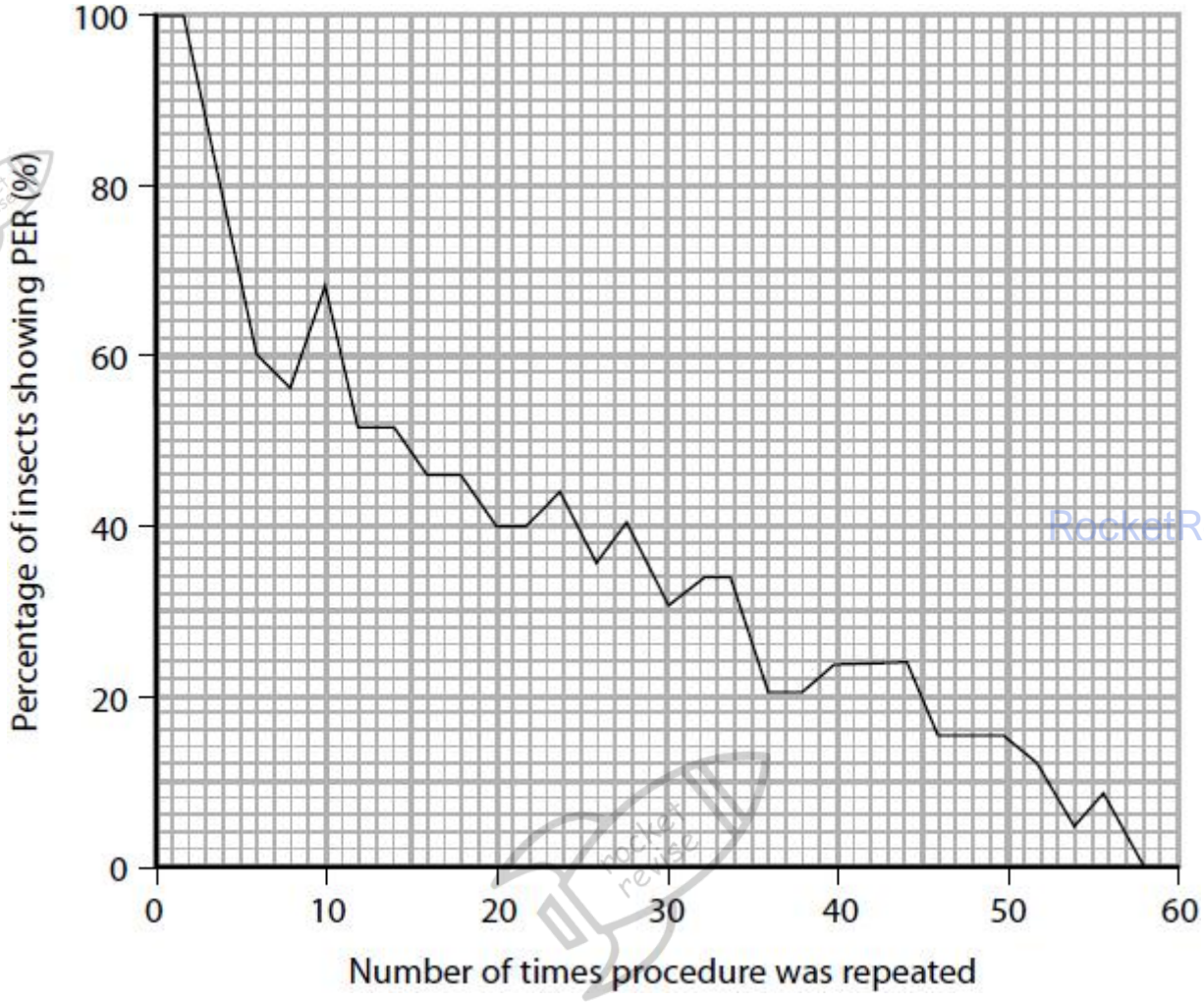
The heat pad was initially set at 25 °C. It was then heated up to 35 °C and the number of insects with a fully-extended proboscis were counted. It was then returned to a temperature of 25 °C.

This procedure was repeated a number of times.

The graph shows the results of this investigation.

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(a) In this investigation, 350 insects were used.

Calculate the mean decrease in the number of insects showing PER each time the procedure was repeated.

(3)

Answer .....

(b) Explain the results of this investigation.

(3)

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(c) Suggest why a drop in PER to a heat stimulus could be an advantage to these insects.

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(2)

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(Total for question = 8 marks)

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The fastest recorded speed by a human was set by Usain Bolt sprinting in 2009.

He achieved a speed of 12.3 metres per second.

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(a) (i) Name the main type of muscle fibre that is being used in the leg muscles of a sprinter.

(1)

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(ii) During a sprint race, leg muscles produce lactate.
Describe what happens to the lactate at the end of the race.

(3)

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(iii) Which of the following statements are features of muscle fibres in a long-distance runner?

(1)

1. have reduced blood capillary network
2. have larger number of mitochondria
3. contain a low concentration of myoglobin

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- ☐ **A** 1 only
- ☐ **B** 2 only
- ☐ **C** 1 and 3 only
- ☐ **D** 2 and 3 only

(iv) Which of the following explains how muscles contract?

(1)

- ☐ **A** Fick's law
- ☐ **B** fluid mosaic model
- ☐ **C** lock and key hypothesis
- ☐ **D** sliding filament model

*(b) Exercise has many benefits for the health and wellbeing of an individual.

An investigation was carried out on non-athletes and athletes.
All participants did a series of physical exercises and several measurements were taken.
The means of the measurements are given in the table.

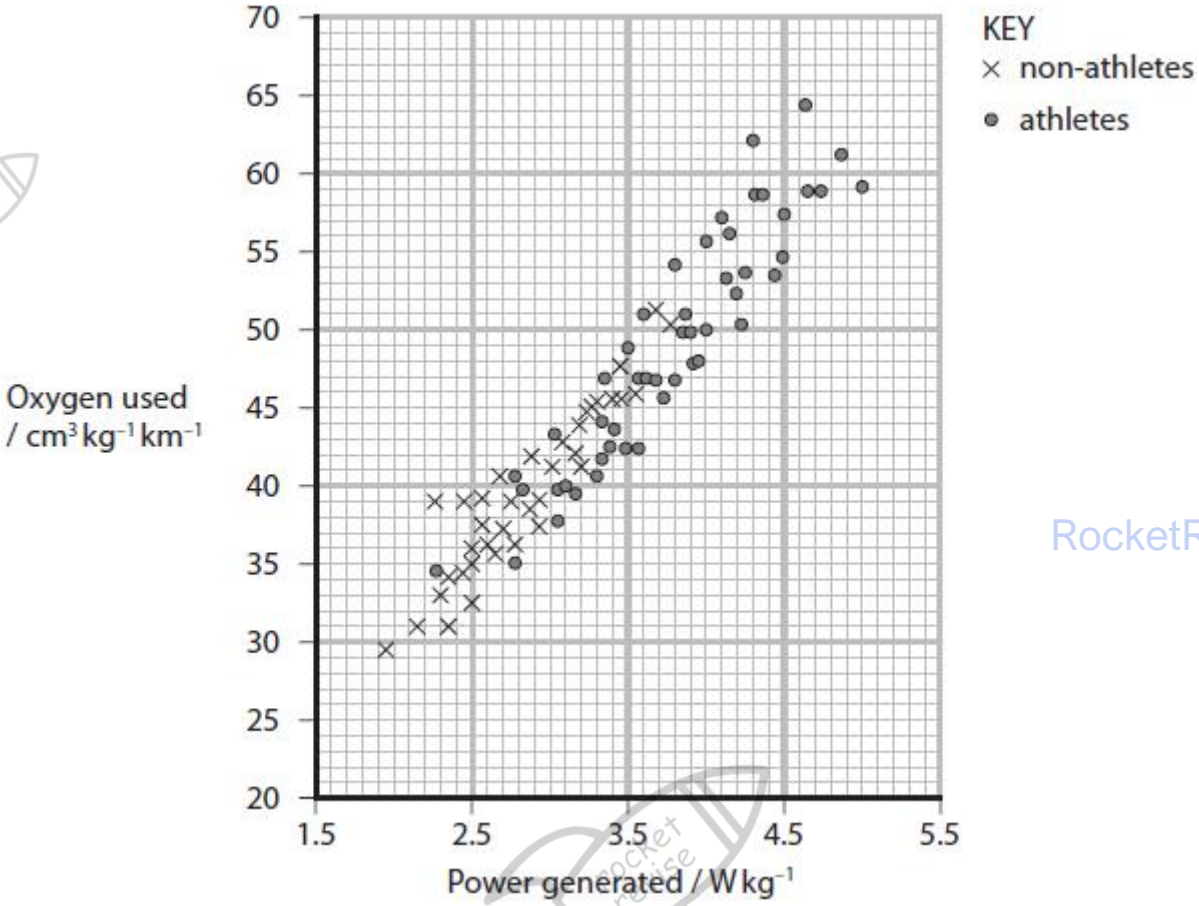
Mean measurement	Non-athletes (n=11)	Athletes (n=10)
Resting heart rate / bpm	71	66
Peak heart rate / bpm	196	190
Maximum volume of oxygen used under anaerobic conditions / $cm^3 kg^{-1} minute^{-1}$	20	35
Maximum volume of oxygen used under aerobic conditions / $cm^3 kg^{-1} minute^{-1}$	39	57

In a different investigation the relationship between power generated and oxygen used was studied in athletes and non-athletes.

The results are shown in the graph.

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Discuss the differences in the results for the athletes and non-athletes in these two investigations.

Use your knowledge and information in the table and graph to support and explain your answer.

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(Total for question = 12 marks)

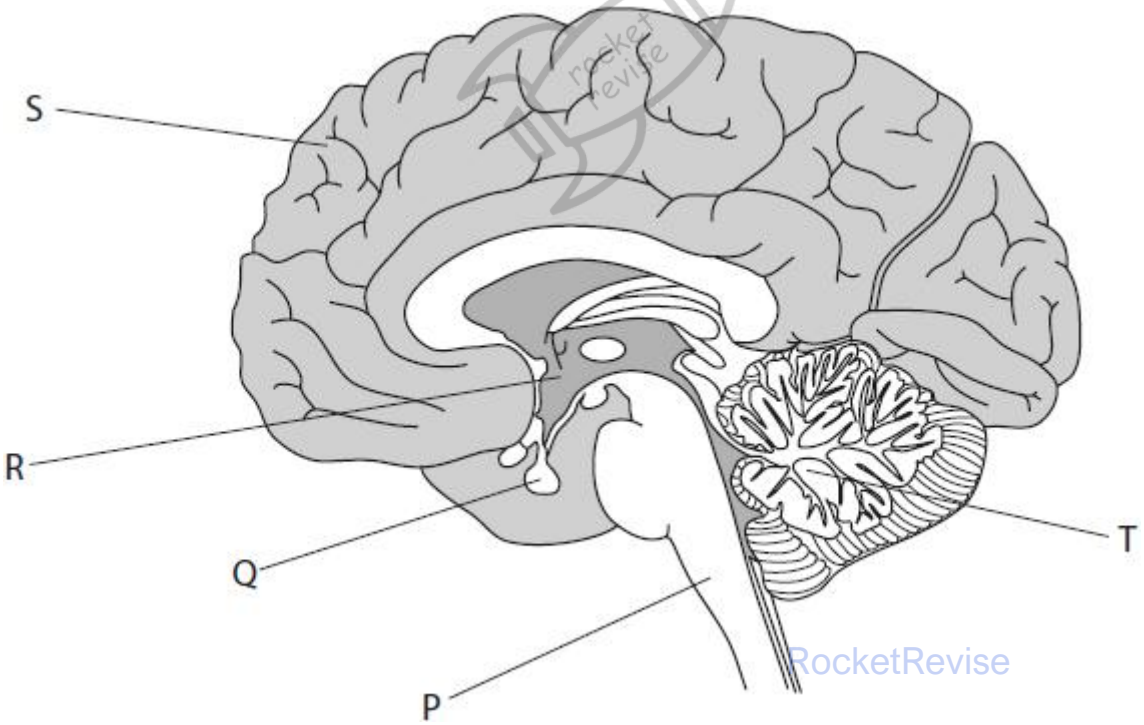
Q35.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The human brain controls many functions.

(a) The diagram shows a section through the human brain.



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(i) Which labelled part of the brain contains the cardiovascular control centre?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R
- ☒ **D** S

(ii) Which labelled part of the brain would control balance and coordinate movement?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R

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☒ **D** T

(iii) Which labelled parts would be involved in controlling the solute concentration of blood plasma?

(1)

☒ **A** P and Q
☒ **B** Q and R
☒ **C** P and R
☒ **D** S and T

(b) Dopamine and serotonin are chemicals produced in the brain that are essential for good health.

(i) Which of the following statements about dopamine is correct?

(1)

☒ **A** dopamine can pass from the bloodstream into neurones in the brain
☒ **B** dopamine is a neurotransmitter
☒ **C** high levels of dopamine are linked to depression
☒ **D** low levels of dopamine cause the breakdown of myelin

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(ii) Drugs called selective serotonin re-uptake inhibitors (SSRIs) affect the reabsorption of serotonin at synapses in the brain.
This may help to reduce depression.
Explain why SSRIs can be used to treat depression.

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(Total for question = 7 marks)

Q36.

The scientific article you have studied is adapted from an article in Clinical Nutrition entitled *Muscle tissue changes with aging* (July 2004).

Use the information from the scientific article and your own knowledge to answer the following questions.

(a) Give a definition for the term **sarcopenia** (paragraph 1).

(1)

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(b) Satellite cells are stem cells (paragraph 2).

Explain how a specialised cell such as a muscle cell is produced from a stem cell.

(3)

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(c) Testosterone is a steroid hormone.

Describe how reduced production of testosterone in the male testes can result in reduced muscle protein (paragraph 5).

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(d) Insulin control of blood glucose is an example of a negative feedback loop (paragraph 7).

Explain how negative feedback can maintain a system within a narrow limit.

(2)

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(e) Suggest how insulin increases the uptake of glucose by muscle cells (paragraph 7).

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(f) Aerobic exercise may increase muscle size (hypertrophy) in older individuals.

Describe how computed tomography (CT) scans can be used to measure these increases in muscle size in older individuals (paragraph 9).

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(g) Explain why aerobic exercise increases mitochondrial density and activity (paragraph 9).

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(h) Whey and egg protein are often used to supplement protein-poor diets.

Explain why diets rich in these proteins may not be ideal for the prolonged treatment of sarcopenia (paragraph 11).

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(Total for question = 20 marks)

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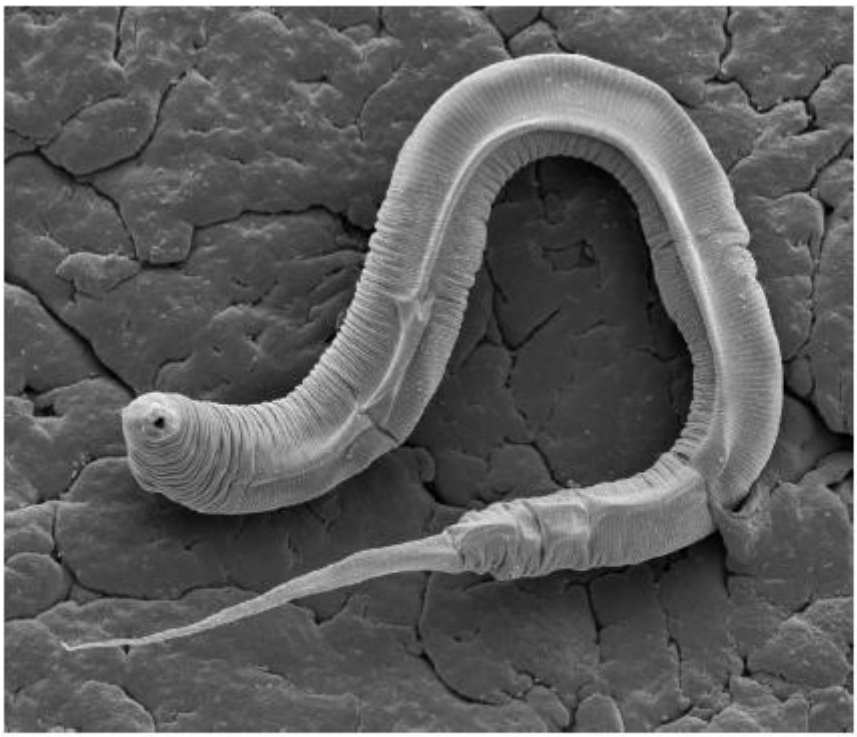
Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a roundworm, *C. elegans*.

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Magnification × 200

(Source: © STEVE GSCHMEISSNER / SCIENCE PHOTO LIBRARY)

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These roundworms are simple animals with a nervous system consisting of 302 neurones.

(a) The roundworm can respond to stimuli with reflex actions.

A reflex action is a rapid involuntary movement in response to a stimulus that involves receptors.

(i) Which receptors detect a change in light intensity and stimulate an action potential?

(1)

- ☒ **A** baroreceptors
- ☒ **B** chemoreceptors
- ☒ **C** osmoreceptors
- ☒ **D** photoreceptors

(ii) A reflex arc carries a nerve impulse from receptors in the skin to the central nervous system in a human.

Describe the structures that a nerve impulse passes through from the skin to the central nervous system.

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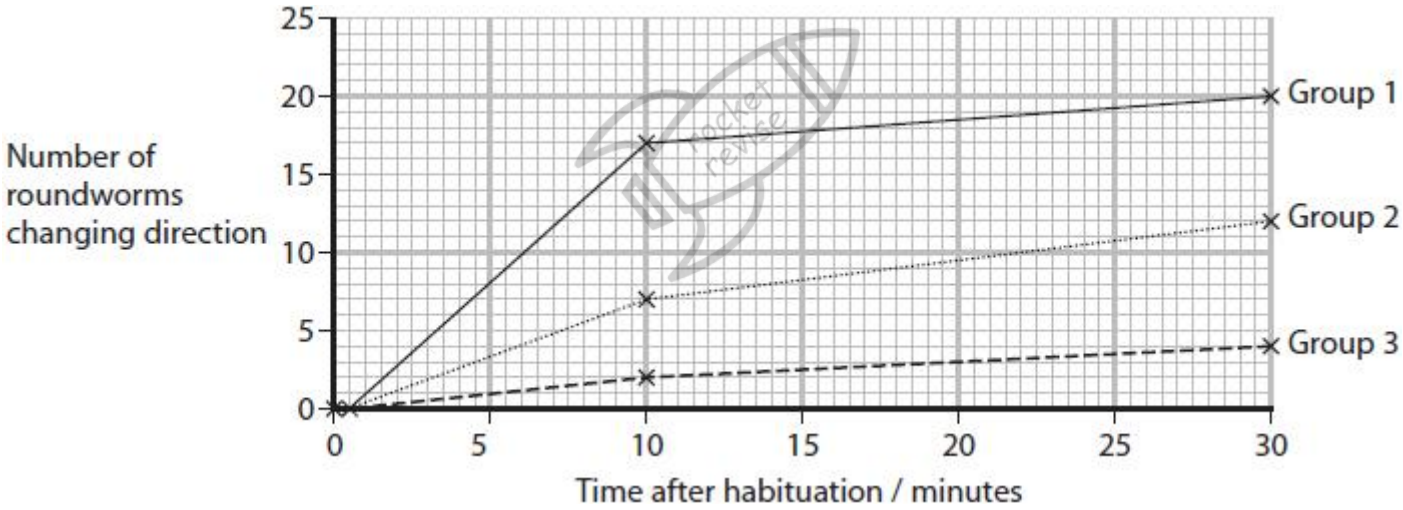
(b) Roundworms respond to physical stimuli by changing the direction in which they swim.

The recovery time of roundworms following habituation was investigated.
Three groups of 25 roundworms were kept in separate Petri dishes.
The roundworms were habituated to a physical stimulus by tapping the side of the Petri dish at different time intervals.

- Group 1 were habituated with 60 taps at 2-second intervals.
- Group 2 were habituated with 60 taps at 10-second intervals.
- Group 3 were habituated with 60 taps at 60-second intervals.

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At 0, 0.5, 10 and 30 minutes after this habituation, each Petri dish was tapped once and the number of roundworms whose swimming direction changed was recorded.
The graph shows the results of this investigation.



(i) Calculate the percentage difference between the number of roundworms in group 1 and group 2 that changed direction 30 minutes after habituation.

Use the equation:

$$\text{percentage difference} = \frac{\text{difference between two values}}{\text{mean of the two values}} \times 100$$

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(2)

Answer ..... %

(ii) Describe **three** conclusions about recovery times after habituation.

(3)

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(Total for question = 9 marks)

Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Pronghorns are similar to antelopes and are native to North America.

They live in open areas of grasslands.

The photograph shows a pronghorn.



(Source: © Paul Tessier/Shutterstock)

(a) These pronghorns can run long distances at speeds of 90 km per hour.

(i) Which of the following features of pronghorns is an adaptation for running long distances at high speeds?

(1)

- ☒ **A** decreased glomerular filtration rate
- ☒ **B** increased number of slow twitch muscle fibres
- ☒ **C** narrower airways
- ☒ **D** smaller cardiac output

(ii) Explain how the heart rate and ventilation rate of a pronghorn are increased to enable it to run long

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distances at high speeds.

(4)

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(b) Scientists measured the mass, resting metabolic rate and pulse rate of five mammals.

The table shows the results.

Mammal	Mass / kg	Resting metabolic rate / joules second^{-1}	Pulse rate / beats per minute
rat	0.260	1.45	420
dog	16.0	20.0	100
sheep	45.0	50.1	70
human	70.0	87.0	72
cow	400	267	40

(i) Describe the relationships between mass, resting metabolic rate and pulse rate shown in the table.

(3)

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(ii) When a human exercises, the metabolic rate is increased and heat energy is released. Describe how the body temperature of a human is maintained at 37°C during exercise.

(3)

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(Total for question = 11 marks)

Q39.

Animals and plants respond to changes in their environment.

(a) The mammalian eye contains photoreceptor cells which respond to light.

The photographs show a grey-bellied night monkey and a squirrel monkey.



Grey-bellied night monkey

(Source: © rod williams/Alamy Stock Photo)



Squirrel monkey

(Source: © Jesse Kraft/Alamy Stock Photo)

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The grey-bellied night monkey is active at night and the squirrel monkey is active during the day.
Rod cells are photoreceptors located in the retina. Rod cells detect low light intensities.
Cone cells are also photoreceptors located in the retina. They are responsible for colour vision and require higher light intensities.
The numbers of different photoreceptors in the retina of one eye of each of these two species of monkey were compared.
The results are shown in the table.

Species	Number of photoreceptors in the retina	
	Cone cells	Rod cells
Grey-bellied night monkey	6.0×10^6	1.4×10^8
Squirrel monkey	7.0×10^6	3.5×10^7

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(i) A section of squirrel monkey retinal tissue contained a total of 5 000 photoreceptors.
Calculate how many of these photoreceptors would be rod cells.
Assume an even distribution of rod cells and cone cells in the retina.

(2)

Answer .....

(ii) Explain why there is a difference in the numbers of rod cells and cone cells in the retinas of the grey-bellied night monkey and the squirrel monkey.

(3)

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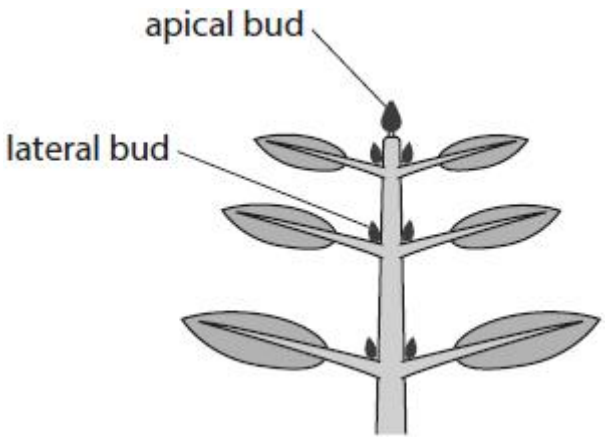
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(b) Plants also respond to their environment.

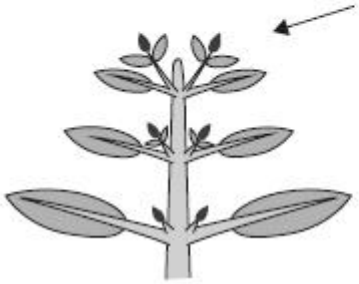
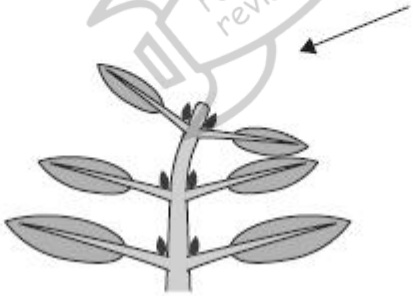
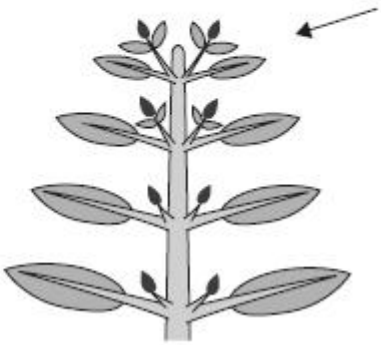
The diagram shows a seedling with the apical bud and a lateral bud labelled.

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In an investigation, three genetically identical seedlings, with the same shoot length, had their apical buds removed.
The table shows the treatments for each seedling and their appearance after being grown next to a source of light for five days.
The arrows show the direction of the light.

Seedling 1	Seedling 2	Seedling 3
no further treatment	auxin applied to the cut top of the seedling	gibberellins applied to the cut top of the seedling
		

Deduce the effect of each treatment on seedlings that have had their apical buds removed.
Use the information in the table to support your answer.

(3)

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(Total for question = 8 marks)

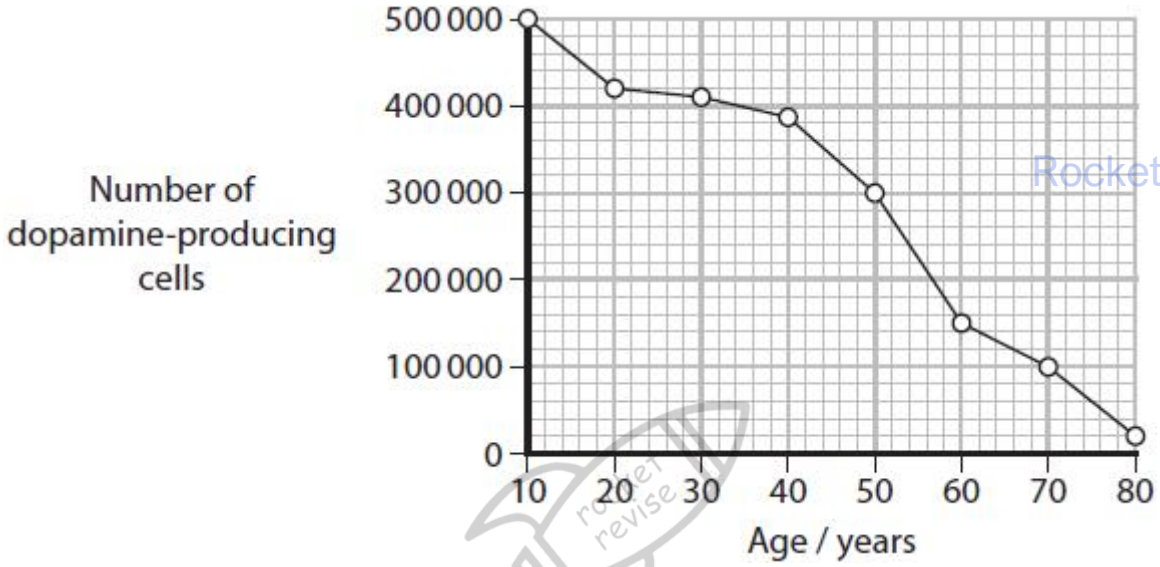
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Q40.

Parkinson's disease is caused by a lack of specific neurotransmitters.

The graph shows the effect of age on the number of dopamine-producing cells in people who develop Parkinson's disease.



(a) In people without Parkinson's disease, the number of dopamine-producing cells rarely falls below 350 000.

Explain why the symptoms of Parkinson's disease increase with age.
Use the graph to support your answer.

(4)

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(b) The drug L-DOPA is used in tablet form to treat Parkinson's disease.

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This drug is converted to dopamine by an enzyme found in the digestive system and the brain.
Carbidopa is an inhibitor of the enzyme found in the digestive system.
Suggest why carbidopa is given to patients when they are treated using L-DOPA in tablet form.

(2)

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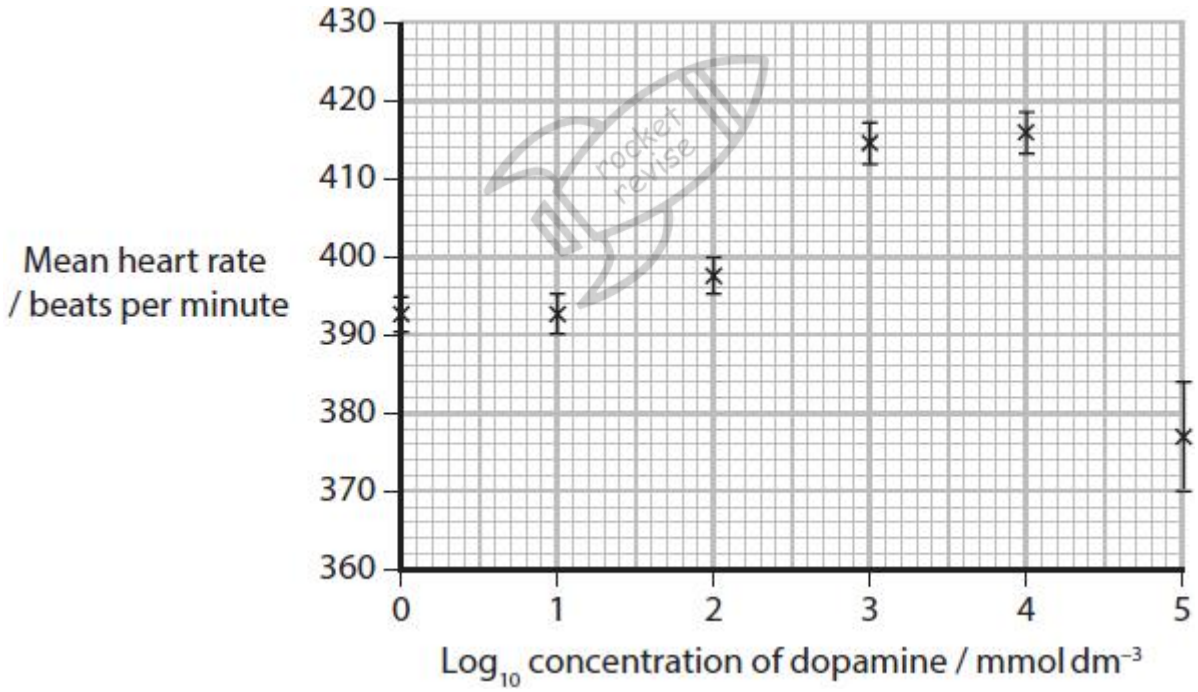
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(c) In an investigation, different concentrations of dopamine were given to *Drosophila* (fruit flies).

The mean heart rates of groups of 10 *Drosophila* were recorded at different concentrations of dopamine.
The results are shown in the graph.



Comment on the results of this investigation.

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(Total for question = 10 marks)

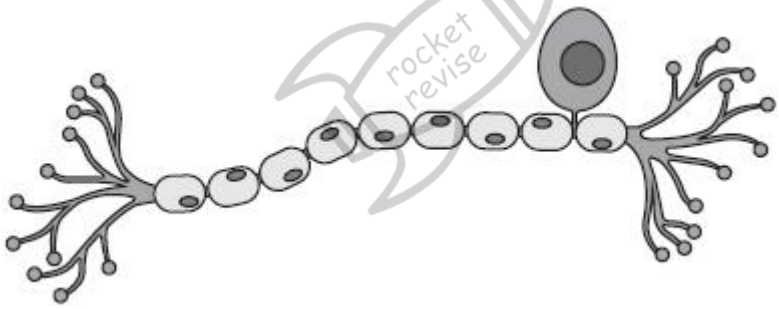
Q41.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Neurones transmit impulses between receptors and effectors.

The diagram shows a neurone.



(a) Draw an arrow on the diagram to show the direction of a nerve impulse.

(1)

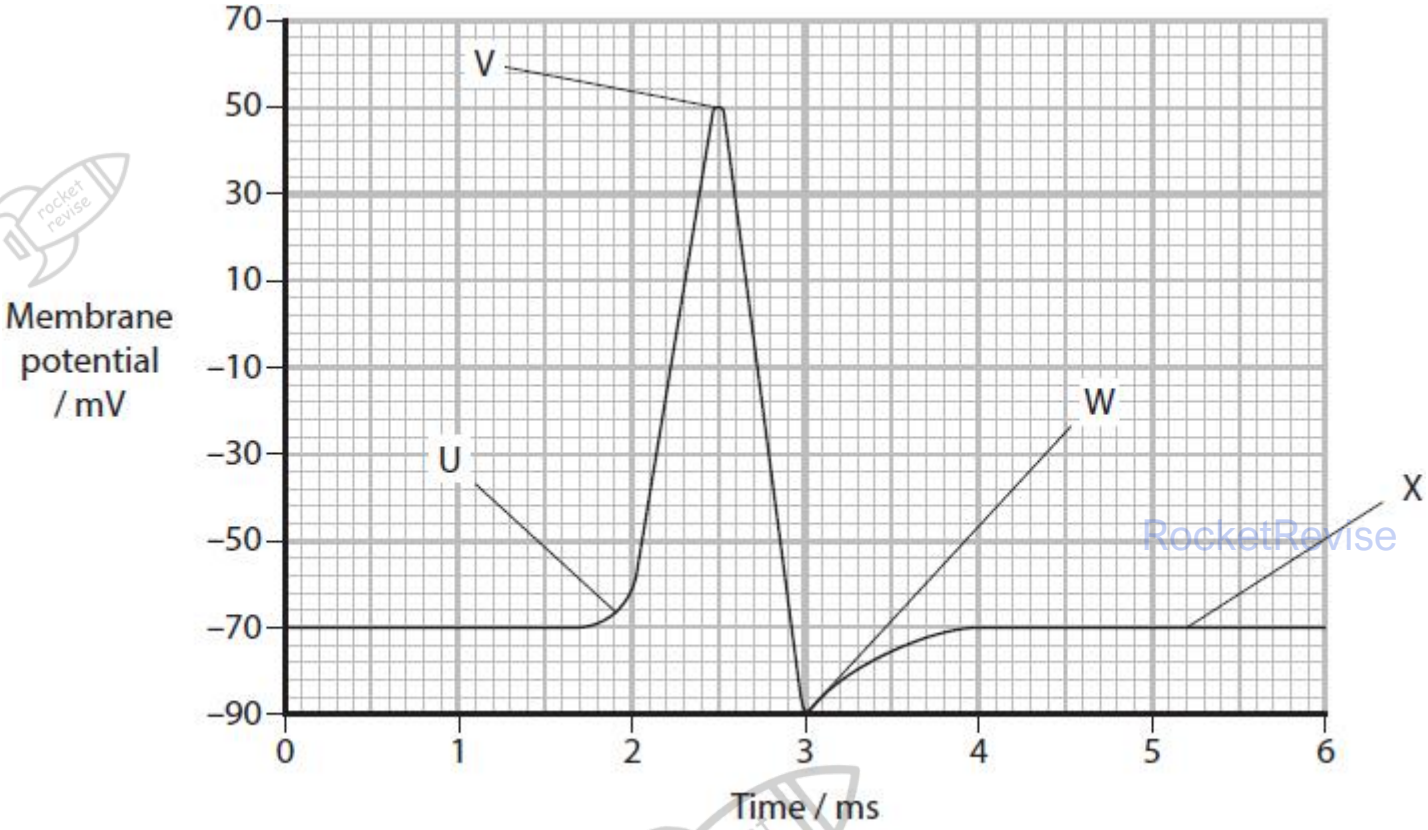
(b) Nerve impulses travel along axons due to changes in membrane potentials.

The graph shows an action potential in a neurone.

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(i) Which is the resting potential for this neurone?

(1)

- ☒ **A** –90 mV
- ☒ **B** –70 mV
- ☒ **C** +48 mV
- ☒ **D** +70 mV

(ii) Which letter on the graph shows where the neurone is hyperpolarised?

(1)

- ☒ **A** U
- ☒ **B** V
- ☒ **C** W
- ☒ **D** X

(iii) Which is the difference in membrane potential between points V and W on the graph?

(1)

- ☒ **A** 40 mV
- ☒ **B** 50 mV
- ☒ **C** 90 mV
- ☒ **D** 140 mV

(c) Give **two** differences between the **functions** of a motor neurone and a sensory neurone.

(2)

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(Total for question = 6 marks)

Q42.

The Chamorro people of Guam eat flying foxes, a type of bat, as part of their diet.

The photograph shows a flying fox.

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(Source: © H Lansdown/Alamy Stock Photo)

Scientists in Guam suggested there was a link between the diet of these people and the high incidence of the neuro-degenerative disease amyotrophic lateral sclerosis (ALS) in their population.

(a) In one investigation, scientists looked at the accumulation of the neurotoxin BMAA in the food chain of the Chamorro people.

The table shows some of the data.

Tissue	Concentration of BMAA / μg per gram
Seeds of cycad trees	37
Fat of flying fox	3 556

Calculate the percentage increase in BMAA per gram from the seeds of the cycad trees to the flying foxes.
Give your answer to **two** significant figures.

(1)

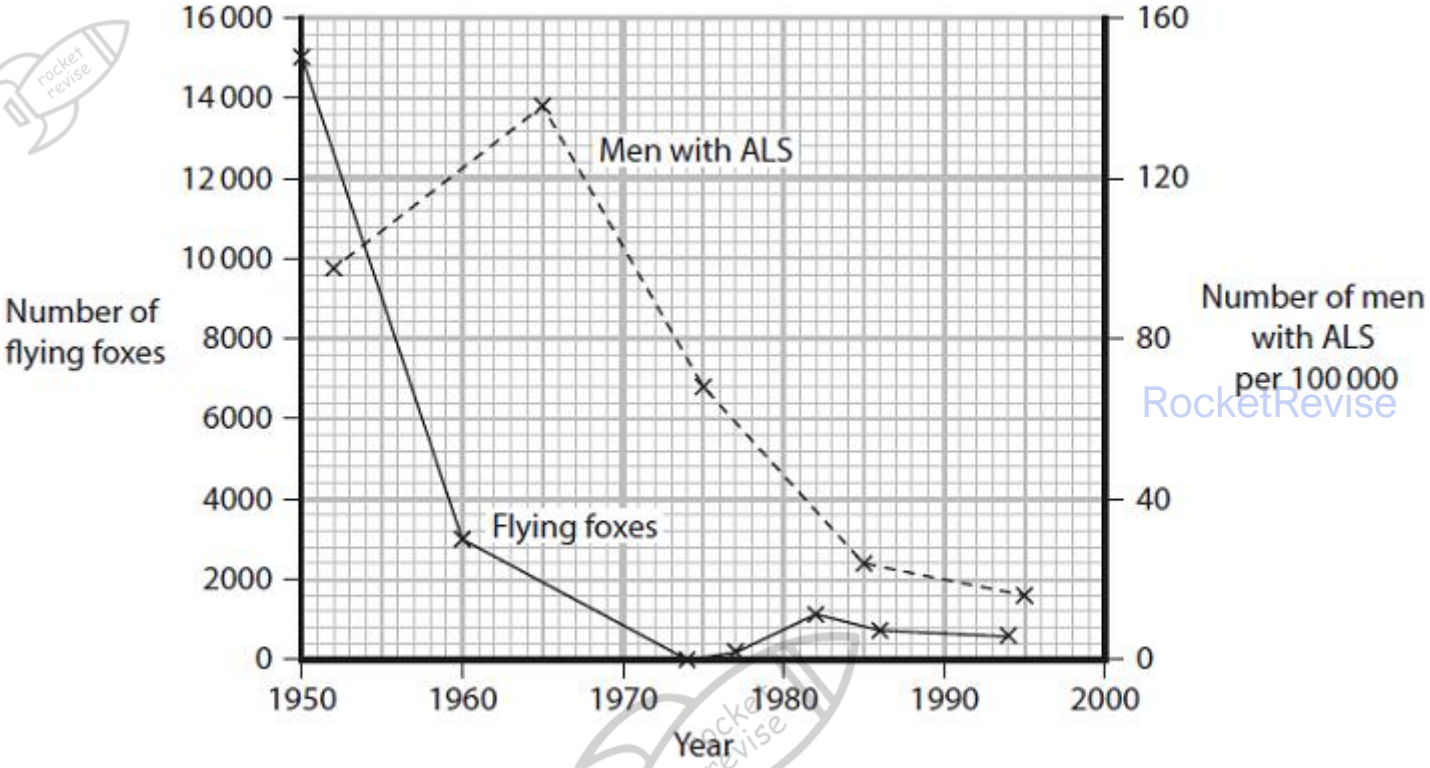
Answer ..... %

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(b) In a second investigation, the scientists looked at the flying fox population and the number of men diagnosed with ALS.

The graph shows the results of this investigation.



(i) Explain the results of this investigation.
Use information from the question to support your answer.

(4)

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(ii) The diagnosis of ALS is made using positron emission tomography (PET) scans and computed tomography (CT) scans of the brain and nervous tissue.
Compare and contrast these two techniques.

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(c) Neurotoxins can affect the potential differences generated across the membranes of motor neurones.

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(i) Draw a myelinated motor neurone and label **two** structures.

(3)

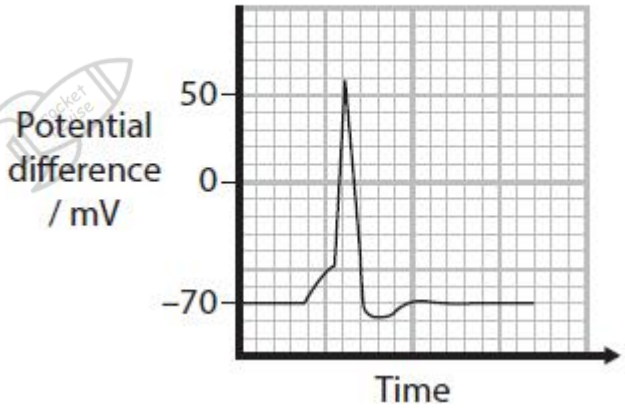
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(ii) A different neurotoxin called tetrodotoxin inhibits the transmission of nerve impulses along axons.
In an experiment, axons of motor neurones were stimulated in the absence of tetrodotoxin and in the presence of tetrodotoxin.
The graphs show the potential differences obtained in this experiment.

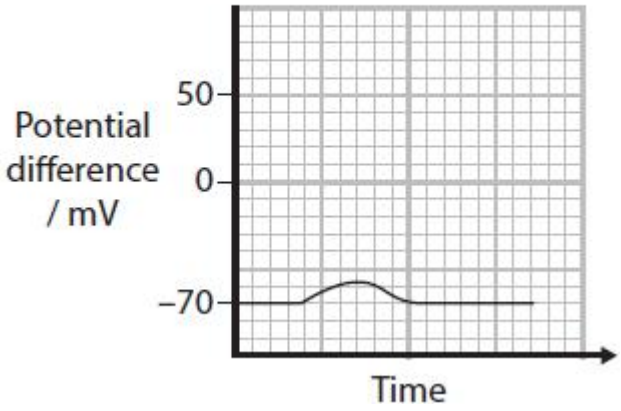
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Absence of tetrodotoxin



Presence of tetrodotoxin



Explain how tetrodotoxin inhibits the transmission of a nerve impulse along the axon.

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(Total for question = 14 marks)

Q43.

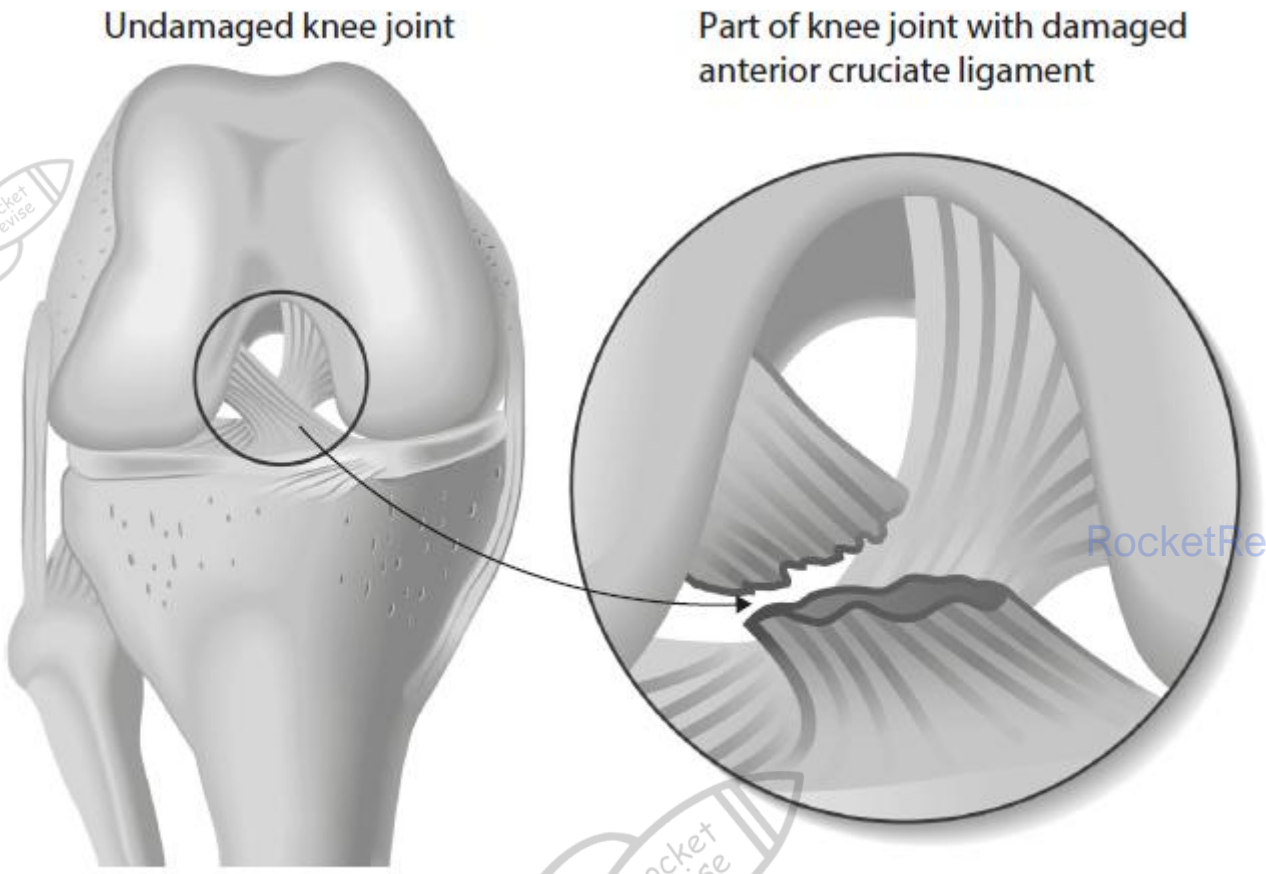
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Muscles, tendons and ligaments interact to enable movement.

(a) The diagram shows an undamaged knee joint and part of a knee joint with a damaged anterior cruciate ligament.

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(Source: © rob3000/Alamy Stock Vector)

(i) Which structure would the anterior cruciate ligament be attached to?

(1)

- ☒ **A** bone
- ☒ **B** cartilage
- ☒ **C** muscle
- ☒ **D** tendon

(ii) Damage to anterior cruciate ligaments can be detected using magnetic resonance imaging (MRI). Describe how MRI can show damage to the anterior cruciate ligament.

(2)

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(iii) In one country, there were 77 anterior cruciate ligament repairs for every 100 000 males in one year.

The population of this country in that year was 67 000 000 and 56% of the population were female. Calculate the number of anterior cruciate ligament repairs in males that year.

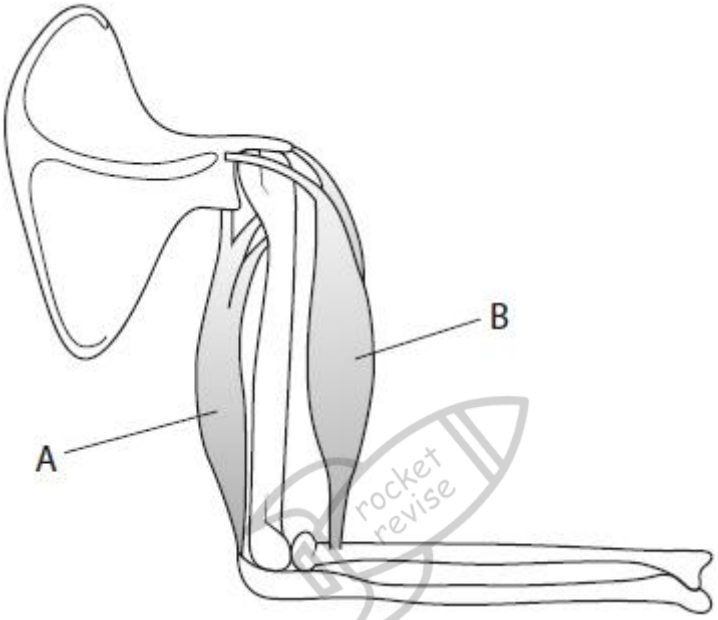
(2)

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Answer .....

(b) The diagram shows some muscles in an arm.



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(i) Name the **two** types of skeletal muscle that are contracted when the arm is held in this position.

(1)

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(ii) Describe how the arm could be straightened by the muscles labelled A and B.

(2)

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(Total for question = 8 marks)

Q44.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Plants and animals use chemicals to co-ordinate metabolic activities and development.

(a) (i) Which is a function of gibberellins in plants?

(1)

- ☐ **A** hydrolyse starch to release glucose
- ☐ **B** inhibit elongation of cells
- ☐ **C** regulate the translation of DNA
- ☐ **D** regulate the transcription of genes

(ii) Which is produced by an animal in a 'fight or flight' response?

(1)

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- ☐ **A** adenine
- ☐ **B** adrenaline
- ☐ **C** amylopectin
- ☐ **D** antihypertensive

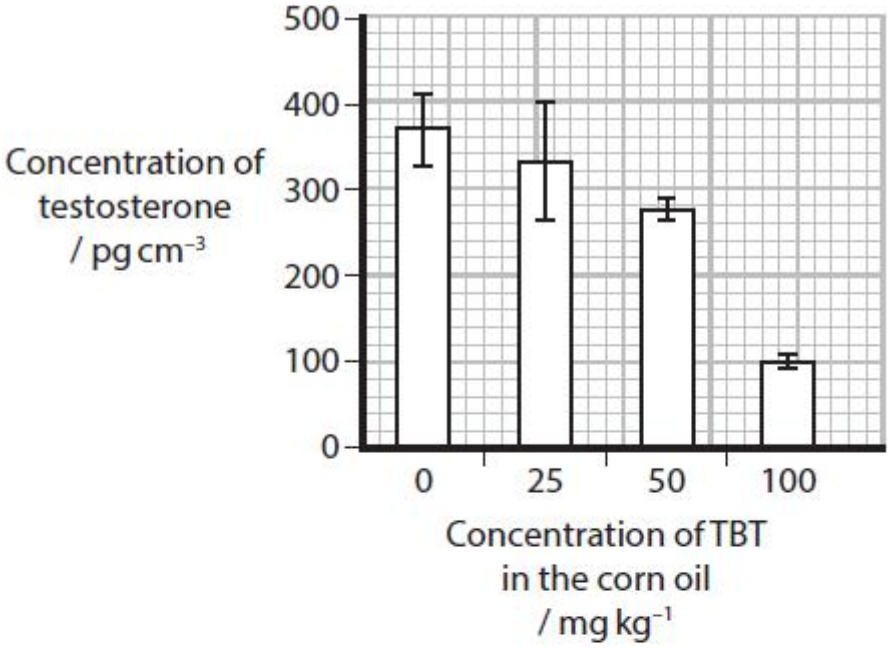
(b) Tributyltin (TBT) is a commonly used chemical in boat building.

The effect of TBT on the blood levels of two steroid hormones, testosterone and oestradiol, was investigated.
Groups of male mice were fed corn oil alone, or corn oil containing different concentrations of TBT.
The concentrations of testosterone and oestradiol in the blood of the mice was measured.
The results are shown in the graphs.

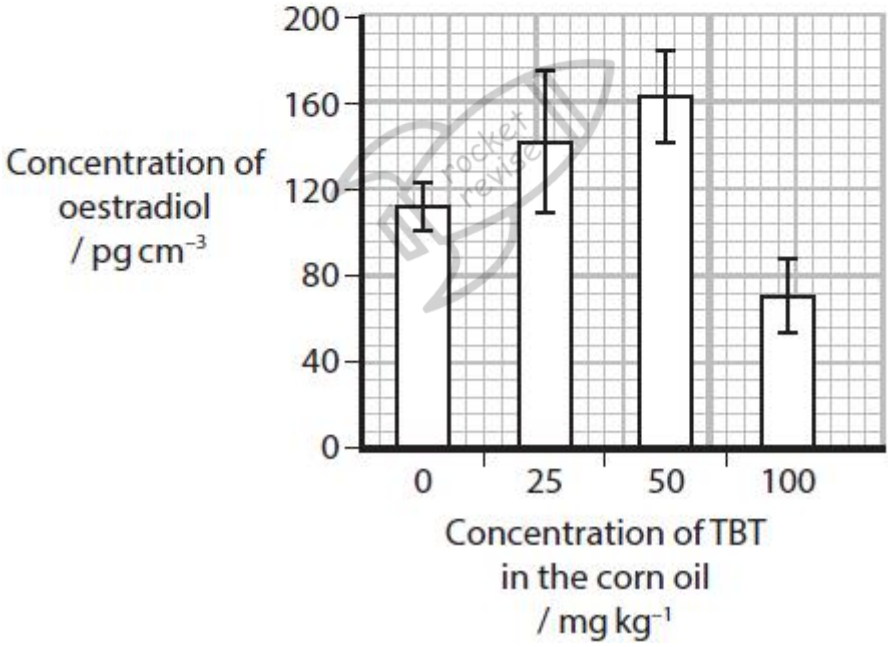
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(i) Describe **three** conclusions that can be drawn from this investigation.

(3)

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(ii) Describe how steroid hormones regulate gene transcription.

(3)

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(Total for question = 8 marks)

Q45.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Impulses are transmitted along nerves as a series of action potentials.

An action potential can be split into four main stages:

- depolarisation
- repolarisation
- hyperpolarisation
- resting state.

(a) (i) Which row shows the events that happen during depolarisation of a neurone?

(1)

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	Sodium channels	Membrane potential
☒ A	closed	decreasing
☒ B	closed	increasing
☒ C	open	decreasing
☒ D	open	increasing

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(ii) Explain what happens during repolarisation in a neurone.

(3)

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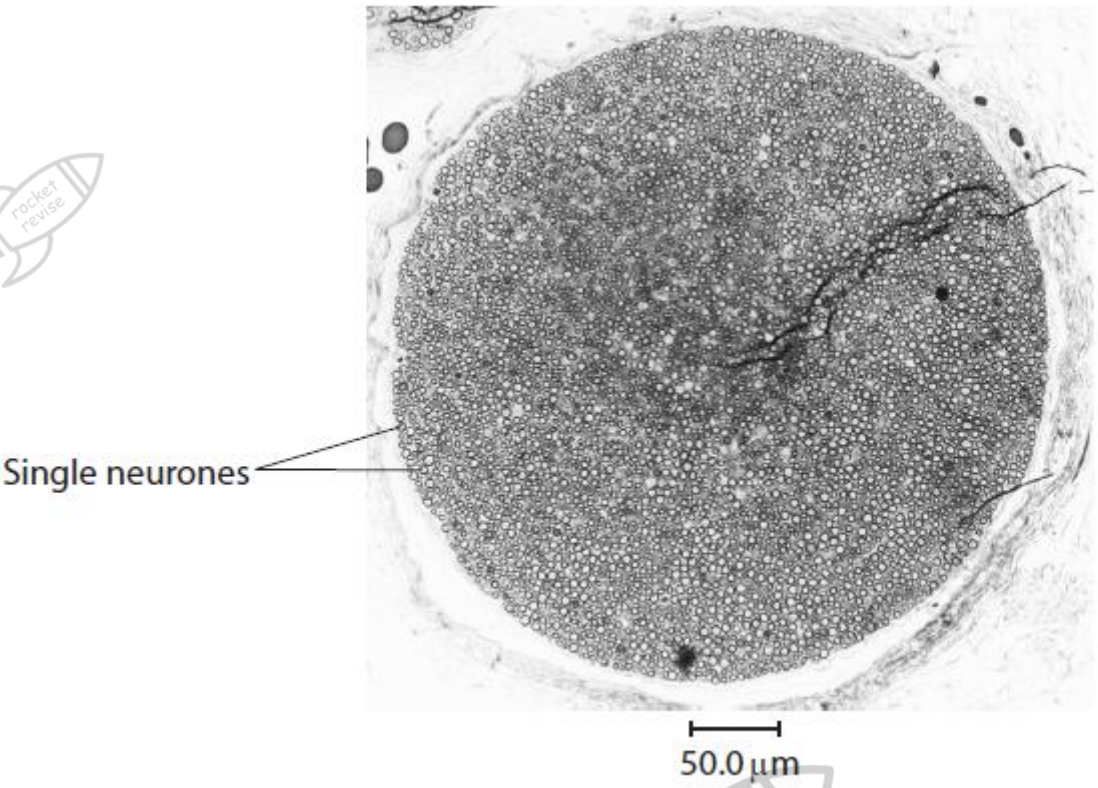
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(b) The photograph is a cross-section of part of a sciatic nerve showing the neurones.

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© Malcolm Park/Alamy Stock Photo

(i) Calculate the magnification of this photograph.
Give your answer in standard form.

(2)

Answer .....

(ii) Name a type of microscope that would be used to obtain this image.

(1)

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(iii) Explain how myelinated neurones enable a greater speed of transmission in a neurone.

(3)

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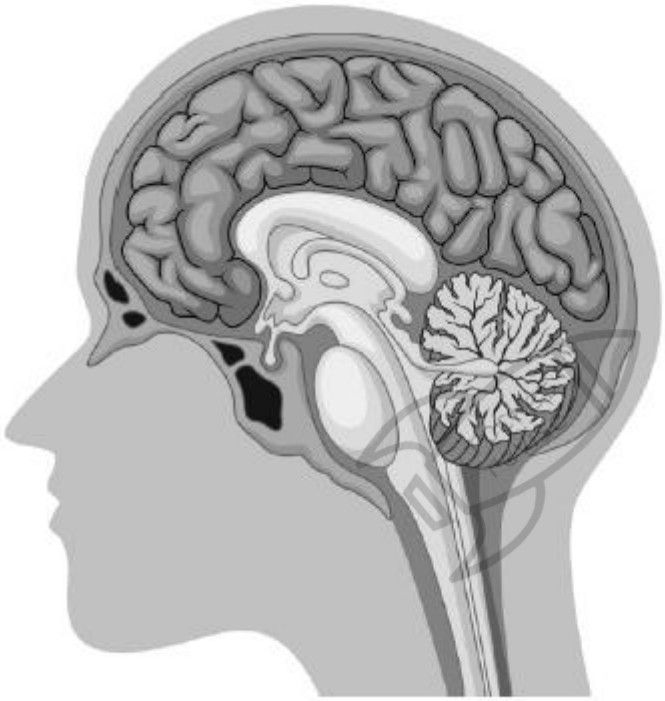
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(Total for question = 10 marks)

Q46.

The diagram shows a cross-section of the human brain.



(Source: © Rendix Alextian / Shutterstock)

(a) **Label the diagram** to show the locations of the pituitary gland (**P**), cerebellum (**C**) and hypothalamus (**H**).

(3)

(b) Complete the table to show **one** function of each part of the brain.

(3)

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Part of the brain	Function
hypothalamus	
medulla oblongata	
pituitary gland	

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(Total for question = 6 marks)

Q47.

Sodium and potassium ions are involved in the transmission of nerve impulses along an axon.

(a) Describe how an action potential is produced in an axon.

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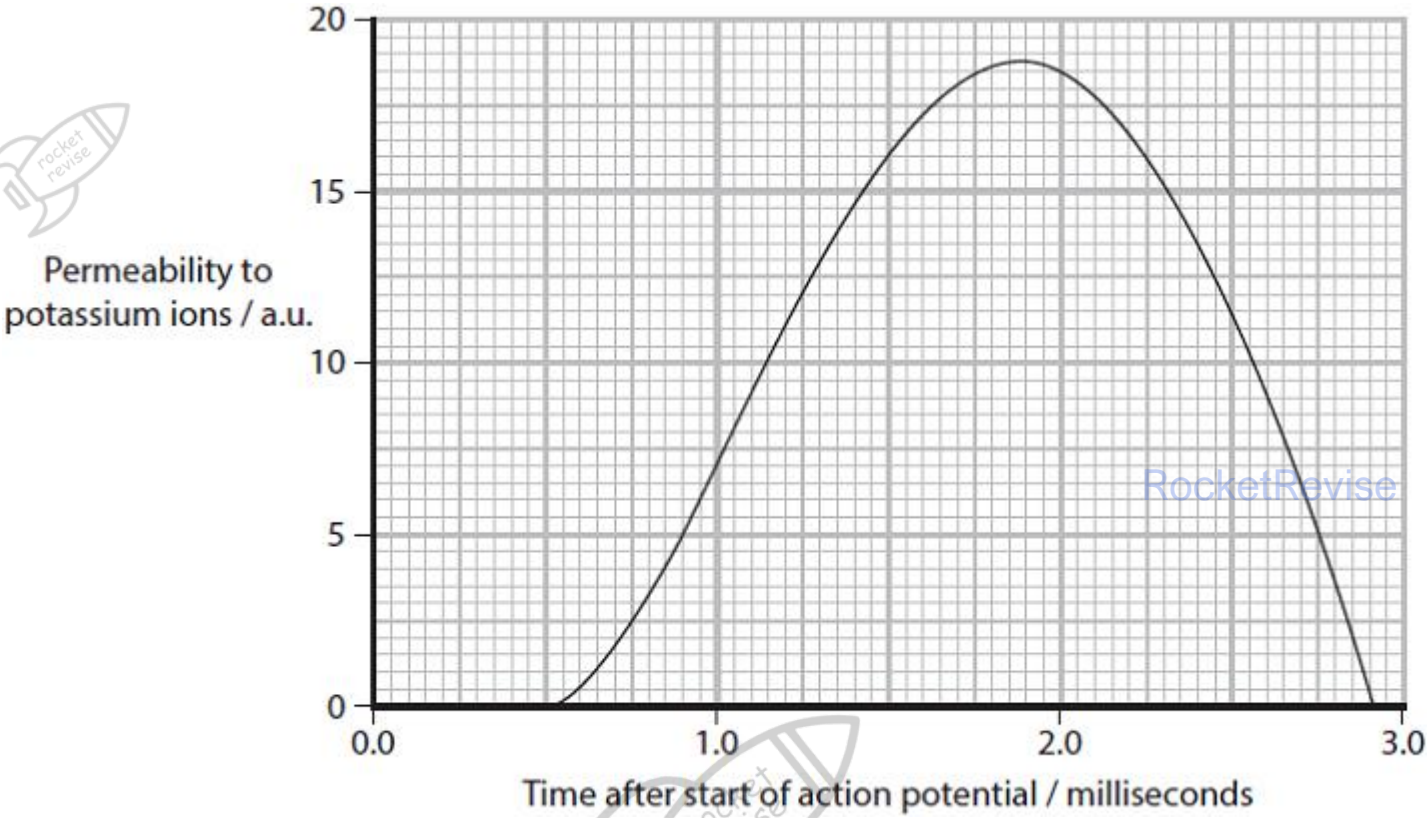
(b) Axon membranes are permeable to potassium ions.

The graph shows the permeability of an axon membrane to potassium ions after the start of an action

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potential.



Calculate the maximum rate of **increase** in permeability to potassium ions.
Give your answer to **two** significant figures with appropriate units.

(2)

Answer .....

(c) The resting potential in a neurone is the result of the difference in concentration of ions on the outside and inside of the axon when no impulse is being transmitted.

The table shows the concentration of ions outside and inside an axon.

(1)

Ion	Outside concentration / mol dm^{-3}	Inside concentration / mol dm^{-3}
Sodium (Na^+)	145	12
Potassium (K^+)	4	155
Chloride (Cl^-)	120	4

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Which is the ratio of potassium ion concentration outside and inside an axon?

- ☒ **A** 0.026 : 1
- ☒ **B** 0.3 : 1
- ☒ **C** 0.083 : 1
- ☒ **D** 30 : 1

(d) (i) Explain the effect of an increase in temperature on the speed of a nerve impulse along an axon.

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(ii) The effect of axon diameter and myelination on the speed of an impulse in four neurones was investigated.

The table shows the results of this investigation.

Neurone	Axon diameter /μm	Myelination	Speed of impulse /ms^{-1}
A	2.5	yes	7.0
B	2.5	no	3.8
C	1.5	yes	4.0
D	1.5	no	2.5

Describe **two** conclusions that can be made from this data.

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(Total for question = 11 marks)

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Q48.

The scientific document you have studied is adapted from an article in the New Scientist: Thirsty work.

Use the information from the scientific document and your own knowledge to answer the following questions.

(a) Give **one** reason why 'optimal hydration' is beneficial (paragraph 1).

(1)

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(b) Caffeine is a naturally occurring stimulant of the central nervous system.

Explain how caffeine present in an energy drink can affect the cells of the brain (paragraphs 3 and 32).

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(c) Explain **one** reason why a 'typical man' has a larger daily requirement for water compared with a 'typical woman' (paragraph 9).

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(d) Explain how the sodium ion concentration in the blood is kept within a narrow range (paragraph 17).

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(e) Explain why 'cells shrink a bit' when blood electrolyte concentration increases (paragraph 21).

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(f) Suggest how 'a high nitrate content' in beetroot juice can lead to an increased blood flow to muscles (paragraph 29).

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(g) Describe how energy drinks could be shown to cause irregular heart rhythms. (paragraph 34).

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(h) Explain **one** reason why some people are not able to tolerate cow's milk (paragraphs 30 and 31).

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(Total for question = 20 marks)

Q49.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Neurotransmitters are involved in the transmission of a nerve impulse across a synapse.

(a) Which chemical would increase the likelihood of a nerve impulse being initiated at a post-synaptic membrane?

(1)

- ☒ **A** auxin
- ☒ **B** dopamine
- ☒ **C** gibberellin
- ☒ **D** statin

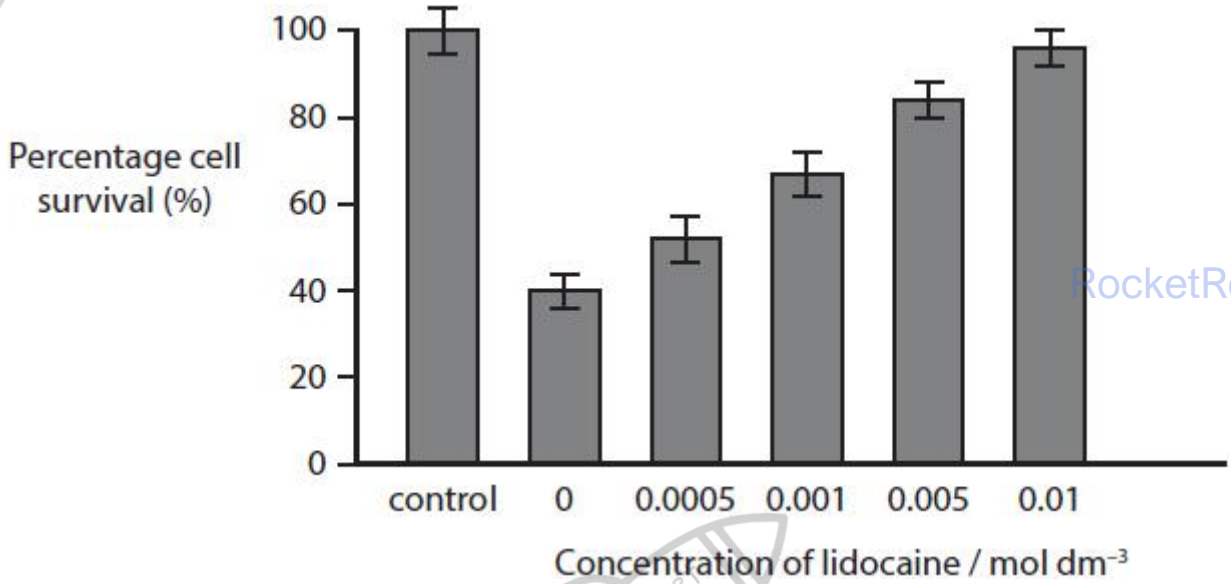
(b) Heart muscle cells obtain over 95% of their energy from oxidative phosphorylation.

When heart muscle cells are cultured in a low oxygen concentration they die.
It has been suggested that lidocaine can be used to protect heart muscle cells from this cell death.

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Lidocaine works by blocking the movement of sodium ions through the membranes surrounding the neurones. This prevents the initiation and conduction of impulses along the neurone.
In an investigation the effect of lidocaine on the cell survival of heart muscle cells cultured in a low oxygen concentration was investigated.
The control culture was given an excess of oxygen and no lidocaine. The other cultures had a low oxygen concentration and different concentrations of lidocaine.
The results are shown in the graph.



(i) Assess the effects of lidocaine on cell survival.

(3)

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(ii) Heart muscle cells are myogenic.
Which describes the term **myogenic**?

(1)

- ☒ **A** a hormone that initiates differentiation
- ☒ **B** cardiac muscle that is able to contract with no external stimulus
- ☒ **C** how a heart beat is initiated in the atrioventricular node
- ☒ **D** how muscles work in pairs

(iii) Suggest how lidocaine protects heart muscle cells from effects of low oxygen.

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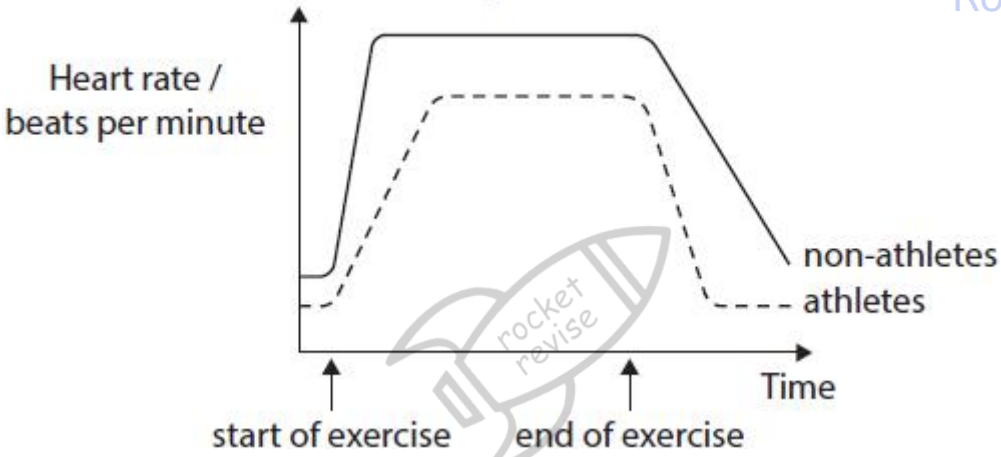
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*(c) Cardiovascular disease (CVD) is one of the biggest causes of death in the world. In 2021, nearly 25% of all deaths were attributed to CVD.

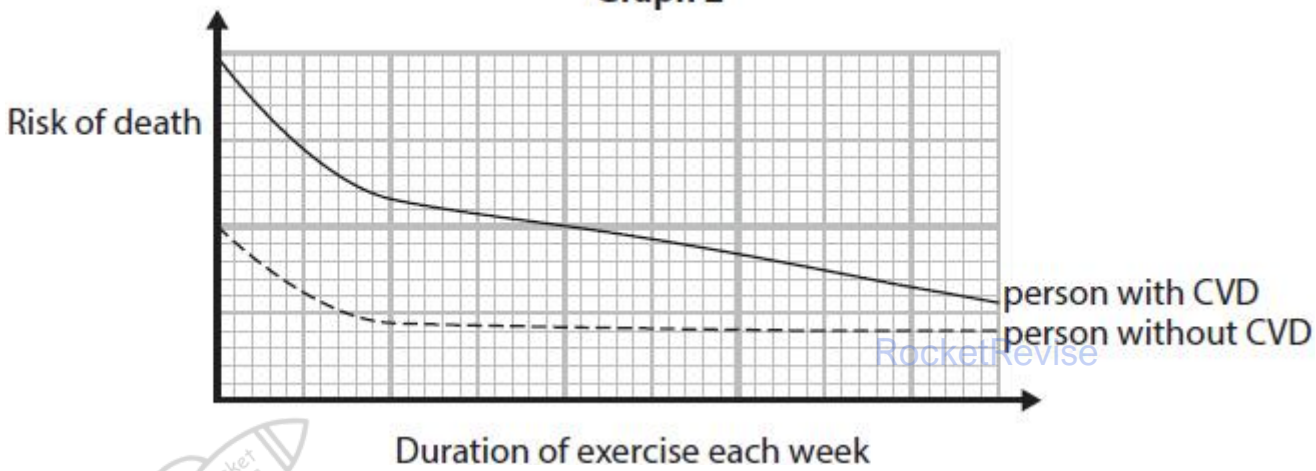
Exercise can help prevent the onset and development of CVD.
In an investigation, the heart rates of athletes and non-athletes were measured before, during and after exercise.
Graph 1 shows the relationship between heart rate and exercise for non-athletes and athletes.

Graph 1



Graph 2 shows how exercise can reduce the risk of death depending on whether the person has CVD or not.

Graph 2



Discuss how exercise reduces the risk of death from CVD.
Use the information in the graphs and your own knowledge to support your answer.

(6)

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(Total for question = 13 marks)

Q50.

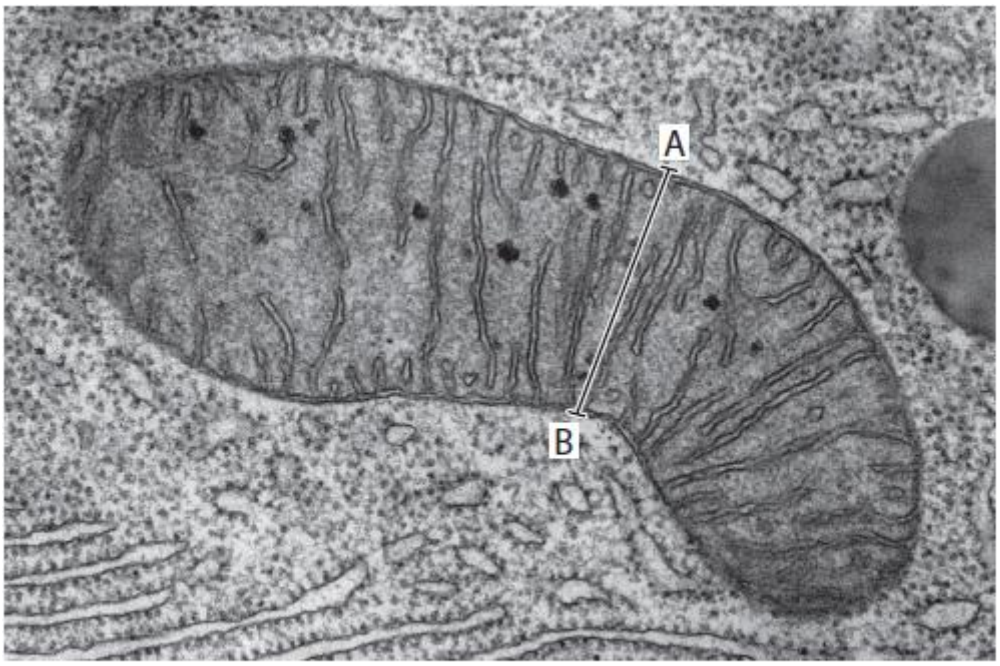
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a mitochondrion.

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(Source: © Science History Images/Alamy Stock Photo)

(a) The magnification of this photograph is ×23 500.

(i) Name the type of microscope that would be used to produce this image.

(1)

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(ii) Calculate the actual width of this mitochondrion between points A and B.
Give your answer in μm to **three** significant figures.

(2)

Answer ..... μm

(b) Respiration in a cell is composed of several processes.

(i) Which describes how ATP is synthesised by chemiosmosis?

(1)

☒ **A** hydrogen ions are pumped from the matrix to the intermembrane space through ATP synthase

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☒ **B** hydrogen ions diffuse from the intermembrane space to the matrix through ATP synthase

☒ **C** hydrogen ions diffuse from the intermembrane space to the stroma through ATP synthase

☒ **D** hydrogen ions jump from the inner membrane to the outer membrane and out through ATP synthase

(ii) Amytal is a drug that inhibits the electron transport chain in respiration.
Which row of the table shows the changes caused by amytal when it is present in a cell?

(1)

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	Oxygen used	Krebs cycle activity
☒ A	decreases	increases
☒ B	decreases	decreases
☒ C	increases	increases
☒ D	increases	decreases

(iii) Which molecule is required in the conversion of pyruvate to a 2 carbon compound in the link reaction of respiration?

(1)

- ☒ **A** oxidised NAD
- ☒ **B** oxidised NADP
- ☒ **C** reduced NAD
- ☒ **D** reduced NADP

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(c) Name **three** products of Krebs cycle.

(2)

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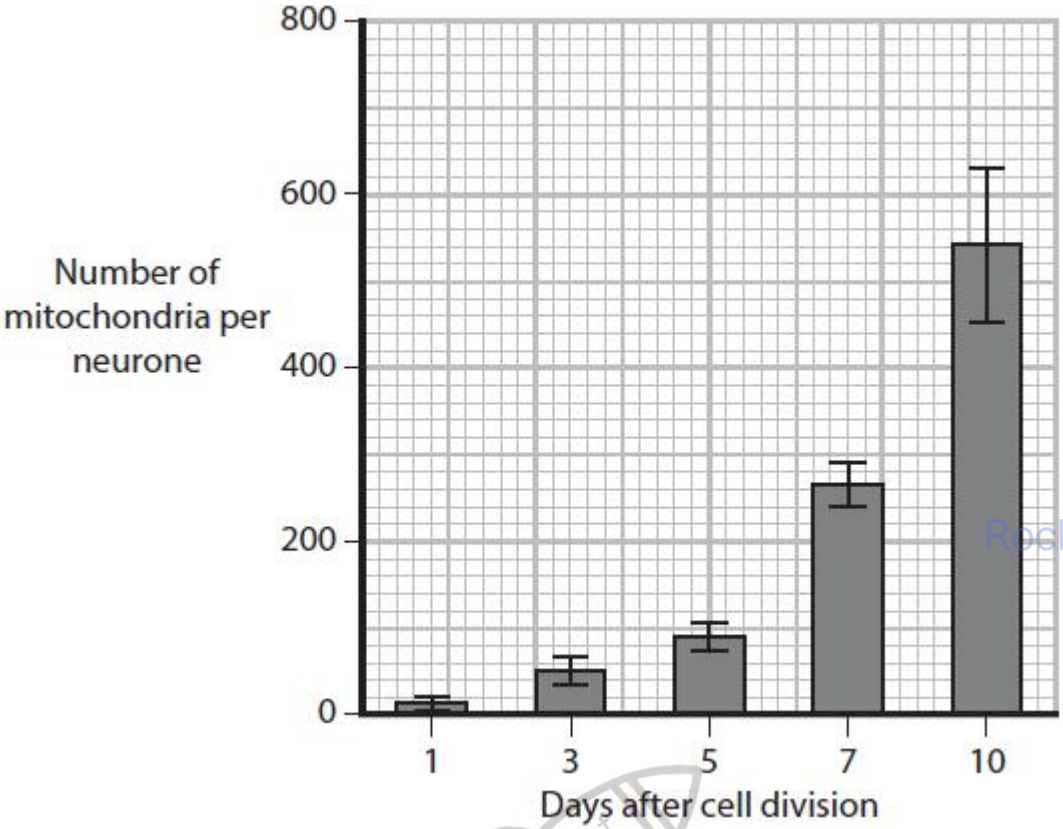
(d) As the brain develops some stem cells divide and differentiate into mature neurones, which requires ATP.

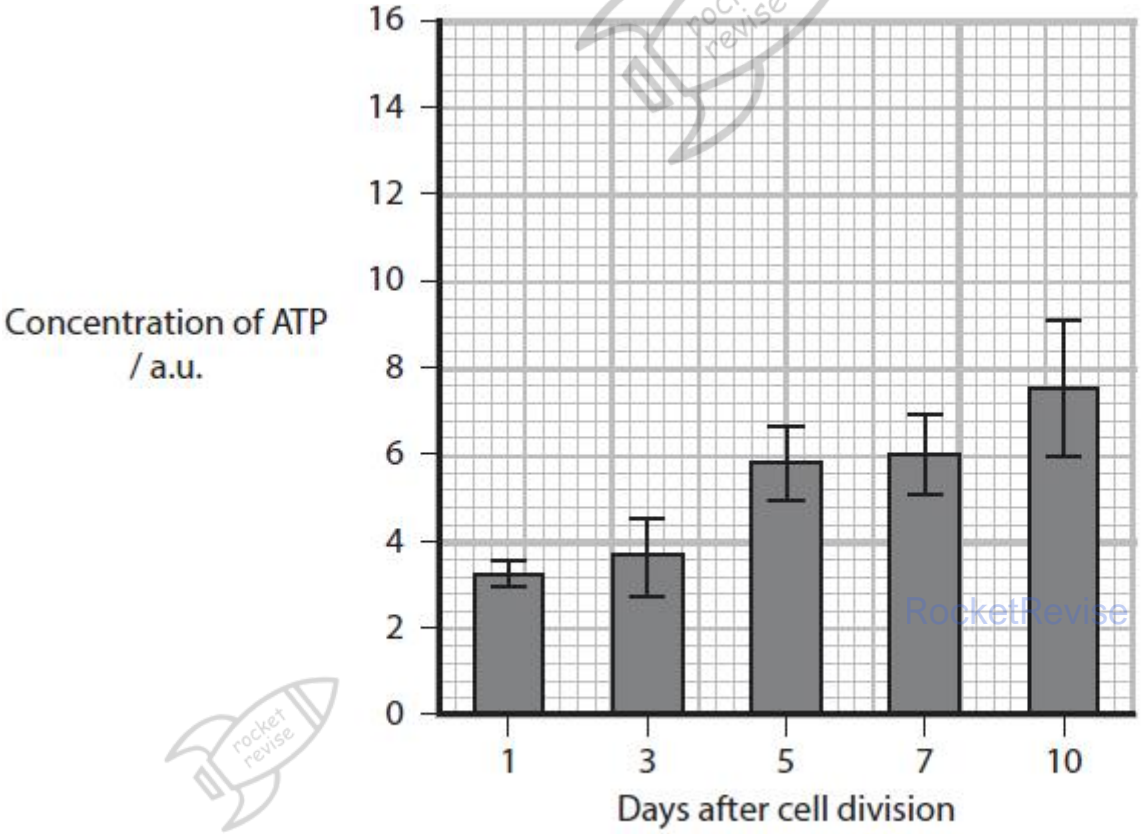
Mature neurones develop axons and form synapses with adjacent cells.
In an investigation, the number of mitochondria and the quantity of ATP made were measured after cell division and during differentiation of brain cells.
The graphs show the results of this investigation.

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(Source: https://pubmed.ncbi.nlm.nih.gov/23212379/)

Comment on the conclusions that can be made from this investigation.

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(Total for question = 12 marks)

Q51.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Hormones are chemicals that coordinate different functions in the body.

Growth hormone (GH) is one of the many hormones produced by the brain.

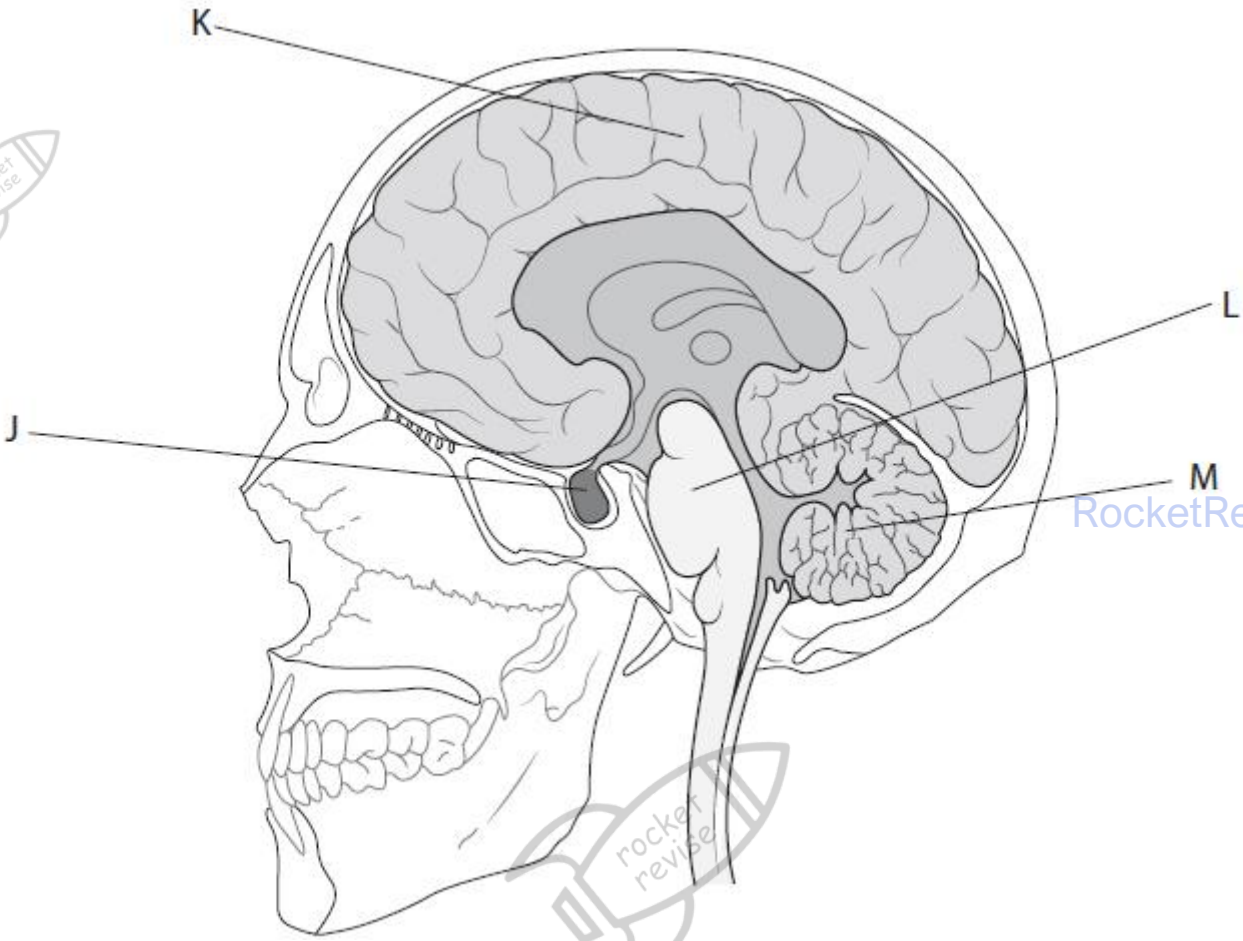
(a) Growth hormone is produced in the pituitary gland.

The diagram shows a cross section through the brain.

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(i) Which **label** identifies the pituitary gland?

(1)

- ☒ **A** J
- ☒ **B** K
- ☒ **C** L
- ☒ **D** M

(ii) Growth hormone is a peptide hormone that stimulates the synthesis of protein in muscle. Explain how GH stimulates the synthesis of protein in muscle.

(3)

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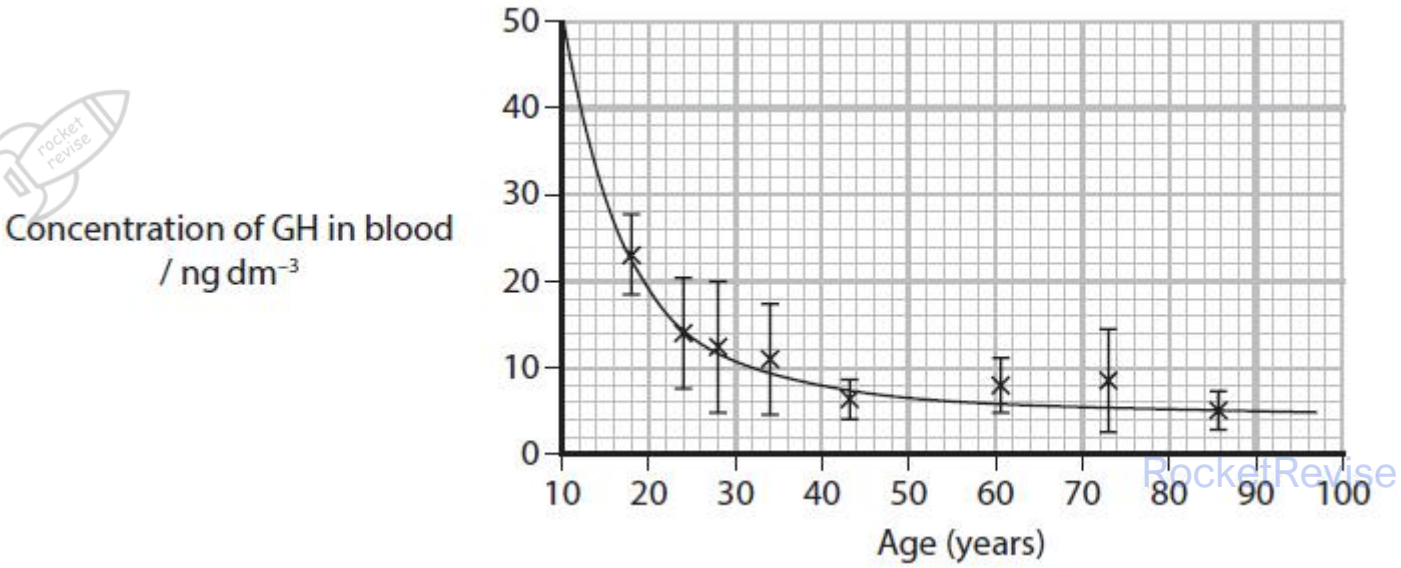
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(b) The concentration of GH in the blood changes during the life of a person.

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The graph shows the changes in the concentration of GH in the blood of males of different ages.



Comment on the changes in the concentration of GH in the blood of males.
Use the data in the graph to support your answer.

(3)

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(c) Growth hormone stimulates the development of the gland that secretes the fight or flight hormone.

Which hormone is the fight or flight hormone?

(1)

- ☒ **A** adrenaline
- ☒ **B** amylopectin
- ☒ **C** cholesterol
- ☒ **D** statin

(d) Synthetic GH can be produced from genetically modified mouse cells.

(i) Describe how mouse cells can be genetically modified to synthesise human GH.

(4)

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(ii) Suggest **one advantage** and **one disadvantage** of using genetically modified mouse cells rather than E. coli bacteria to produce human GH.

(2)

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(Total for question = 14 marks)

Q52.

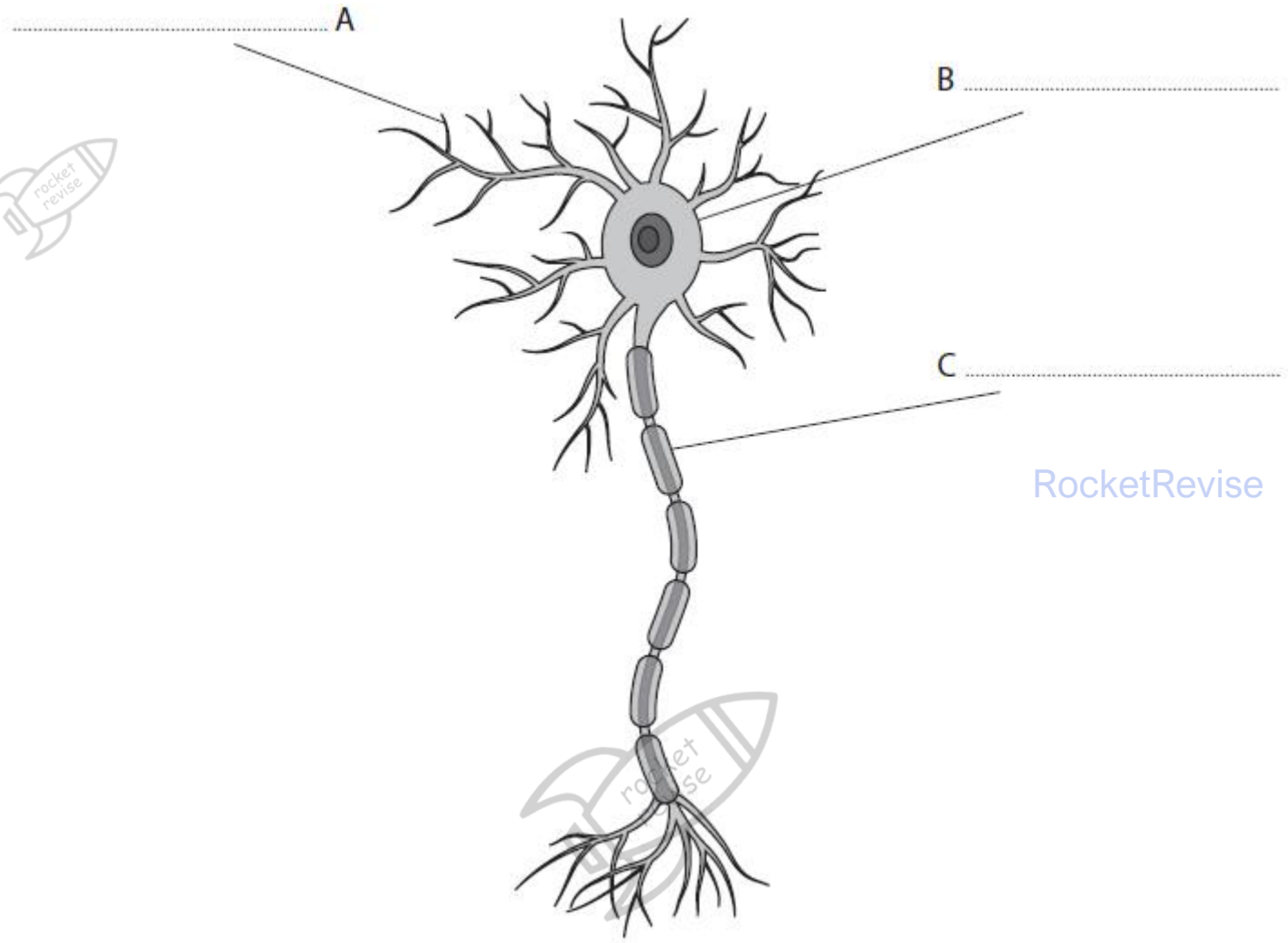
The nervous system is involved in coordination in animals.

The diagram shows a motor neurone.

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(a) Complete the diagram by naming the parts labelled A to C.

(2)

(b) Describe how a nerve impulse is conducted across the synapse between neurones.

(3)

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(Total for question = 5 marks)

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Q53.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) Parkinson's disease arises from decreased dopamine production in the brain.

Explain why L-Dopa, **given orally in tablet form**, reduces the symptoms of Parkinson's disease.

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(b) The medulla oblongata is part of the nervous system.

Which is a function of the medulla oblongata?

(1)

- ☒ **A** control of the cardiac response
- ☒ **B** long-term memory
- ☒ **C** release of ADH
- ☒ **D** voluntary movement

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*(c) Ageing affects different organ systems.

Three systems affected by ageing are the:

- cardiovascular system
- breathing and gas exchange system
- brain and nervous system.

Ageing is affected by:

- types of bacteria living on the skin and in the gut

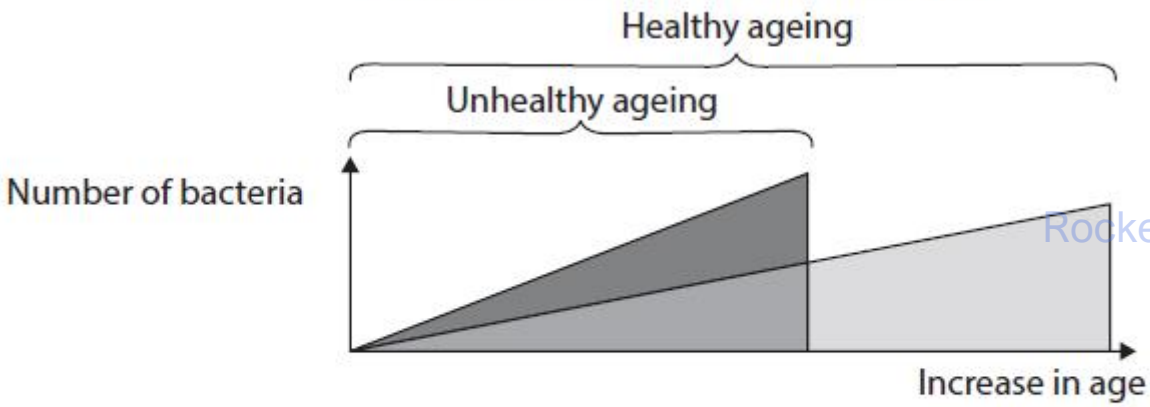
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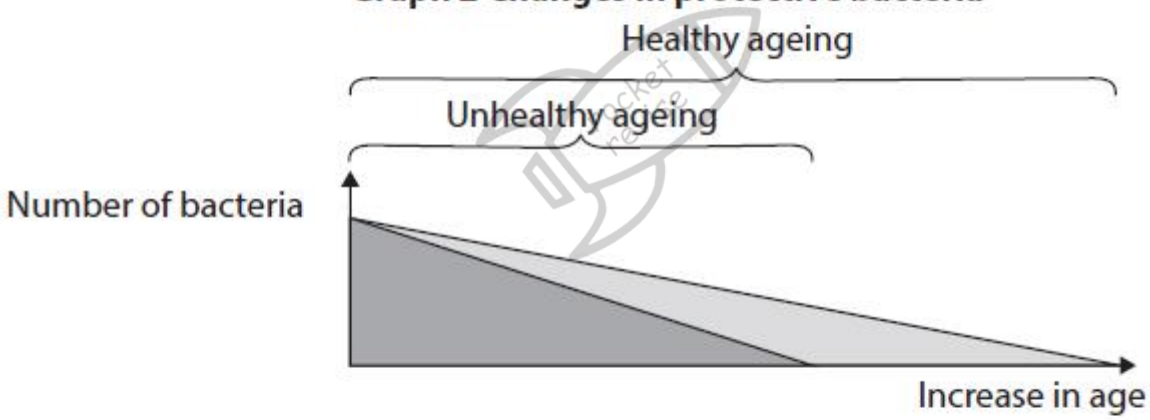
- inflammation in different parts of the body
- lifestyle.

The graphs show the changes in the types of bacteria and inflammation during healthy and unhealthy ageing.

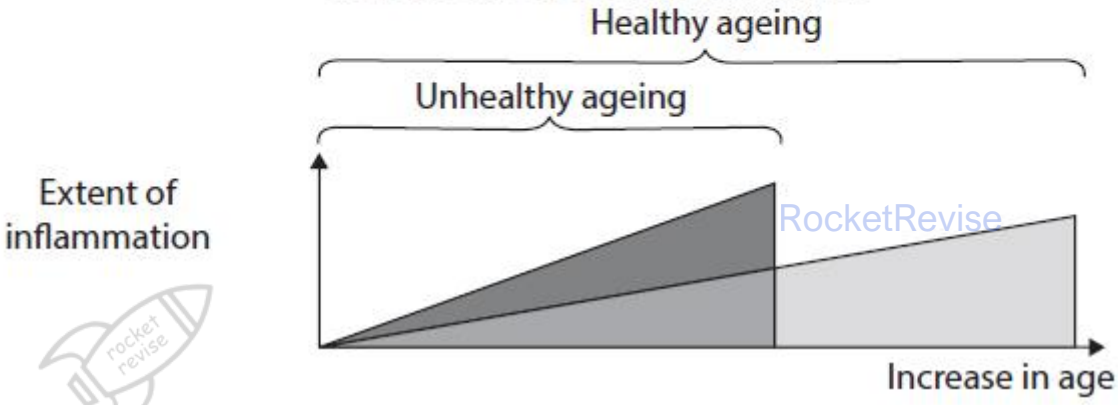
Graph 1 Changes in pathogenic bacteria



Graph 2 Changes in protective bacteria



Graph 3 Changes in inflammation



Discuss the role of bacteria, inflammation and lifestyle in ageing.
Use the information in the graphs and your own knowledge to support your answer.

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(Total for question = 11 marks)

Q54.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Specialised molecules play a role in the coordination of the response of plants to light.

(a) Which molecule detects light during germination of seeds?

(1)

- ☐ **A** chlorophyll
- ☐ **B** phytochrome
- ☐ **C** retinal
- ☐ **D** rhodopsin

(b) The effect of red, blue and white light at low and high light intensities on the growth of peanut seedlings was investigated.

The photograph shows a peanut seedling.

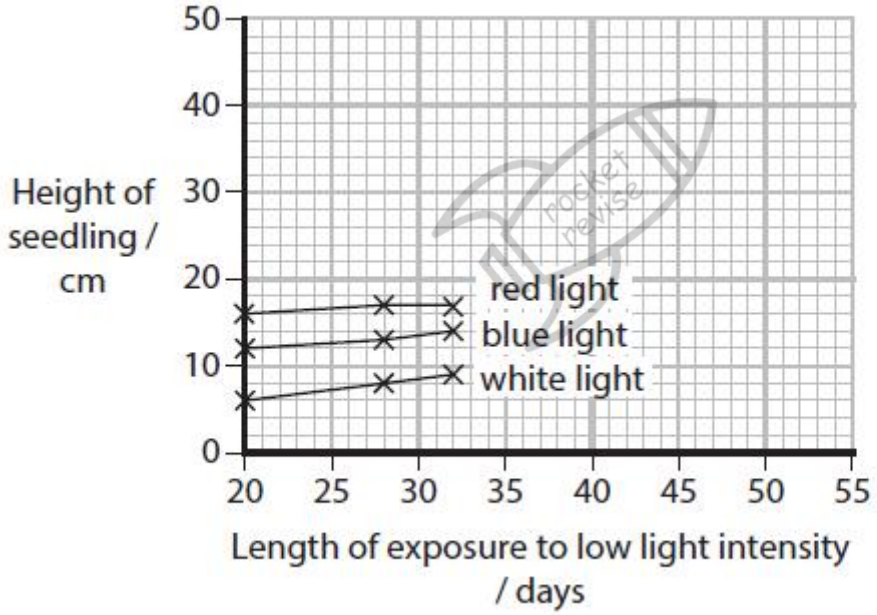
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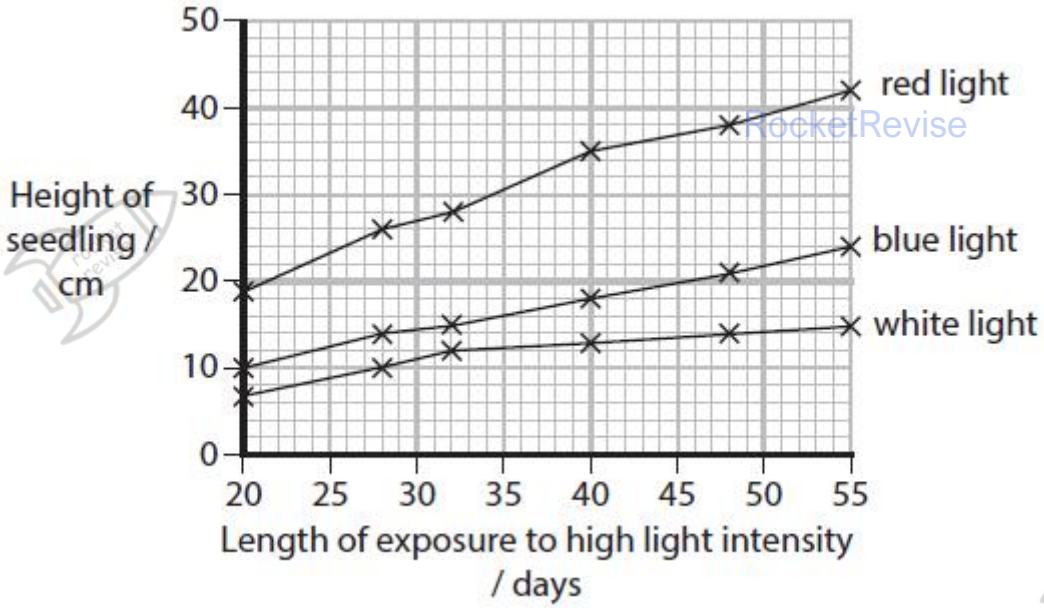
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The graphs show the results of this investigation.
The plants in low light intensities died after 32 days.





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Determine the effect of light on the growth of the seedlings in this investigation.

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(c) Auxin stimulates:

- the release of hydrogen ions that disrupt cellulose microfibrils
- the uptake of potassium ions by cells.

Explain how auxin causes cells to increase in size.

(4)

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(d) The effect of different concentrations of gibberellin on seed germination of the ashoka tree was investigated.

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The photograph shows the leaves and flowers of an ashoka tree that grows in parts of Asia.



(Source: © Roberto Pascual Gomez / Shutterstock)

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Samples containing 200 seeds were each treated with different concentrations of gibberellin. The table shows the results of this investigation.

Gibberellin concentration / ppm	Number of seeds germinating
0	26
100	92
200	114
300	66
400	46

(i) Calculate the percentage change in germination between 200 ppm and 400 ppm gibberellin concentration.

(2)

Give your answer as a whole number.

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Answer .....%

(ii) Explain how gibberellins affect germination in this investigation.

(2)

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(Total for question = 13 marks)

Q55.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A number of drugs can affect the nervous system of mammals.

(a) Which describes the effect lidocaine has on the membranes of neurones?

(1)

- ☒ **A** it closes sodium ion channels
- ☒ **B** increases the number of action potentials
- ☒ **C** it opens potassium ion channels
- ☒ **D** it opens sodium ion channels

(b) Which describes the effect of MDMA (ecstasy) on neurones?

(1)

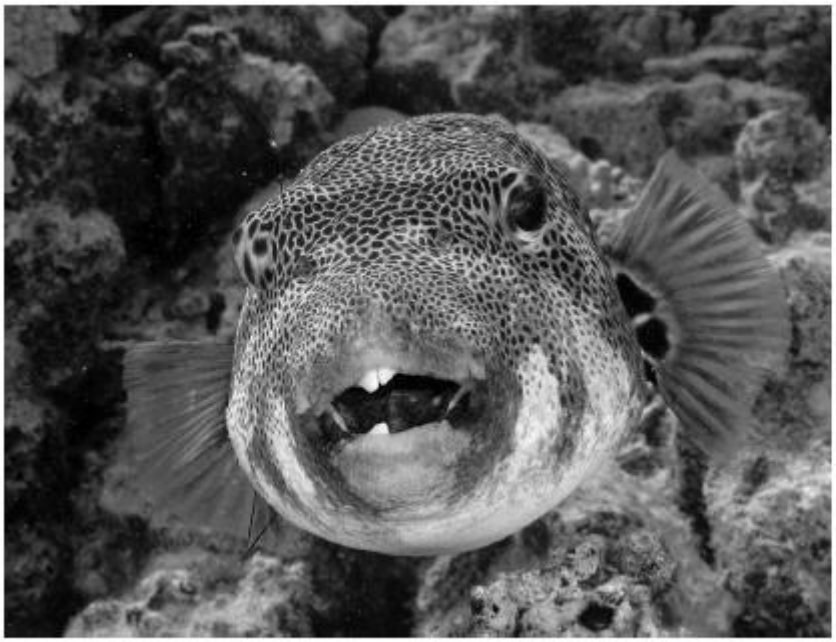
- ☒ **A** it damages the myelin sheath of neurones
- ☒ **B** it decreases the release of neurotransmitters
- ☒ **C** it decreases the speed of action potential transmission in neurones
- ☒ **D** it increases the release of neurotransmitters

(c) The photograph shows a puffer fish.

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(Source: © ImageBROKER on Offset / Shutterstock)

Tetrodotoxin (TTX) is a toxin found in puffer fish.
This toxin works by binding to ion channels in neurones.
Suggest how TTX could inhibit the nervous control of breathing.

(3)

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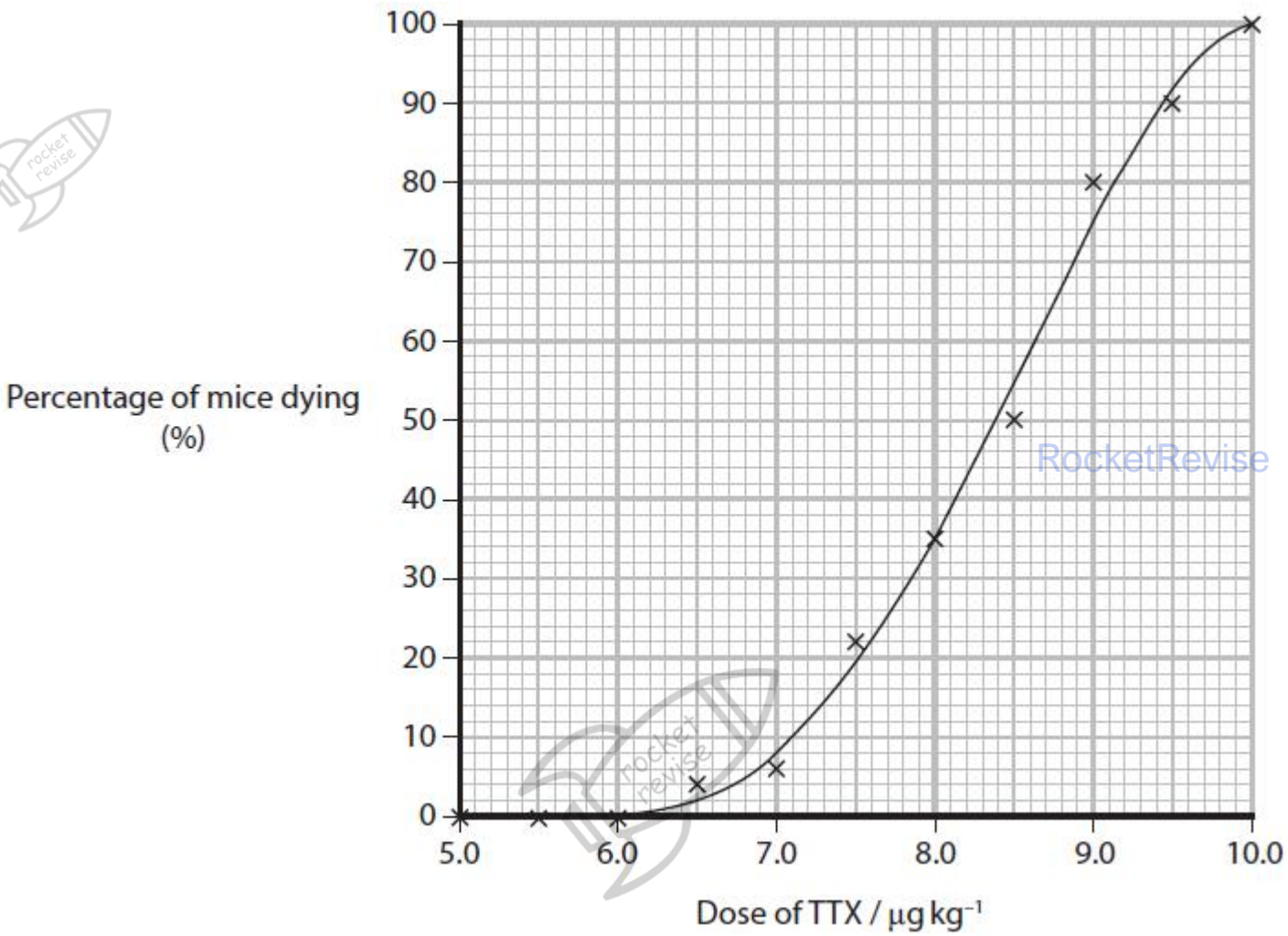
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(d) The toxicity of TTX in mice was investigated.

Groups of mice were given different doses of TTX. The percentage of mice dying at each dose was recorded.
The graph shows the results of this investigation and a line of best fit.

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(i) The LD_{50} obtained from this graph is 8.4 µg kg^{-1}.
Give the meaning of LD_{50} = 8.4 µg kg^{-1}.
Use the information in the graph to support your answer.

(1)

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(ii) Comment on the toxicity of TTX in mice.

(3)

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(Total for question = 9 marks)

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